# Control Systems

(UE23EC241B)

- Hitesh

# Unit-1 Mathematical Modelling & Analysis of Linear Systems

## System

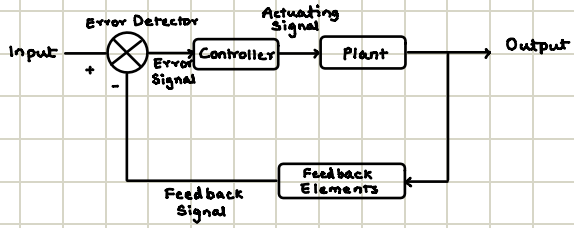
→ A set of things working together as parts of a mechanism or an interconnecting network to perform a function

ex: Bus, 2-wheeler, fan etc.,

## 2 Types of loops

1) Open Loop - Output has no influence on control action of input signal

2) Closed Loop - System considers current output & alters it to desired condition




→ Physical System

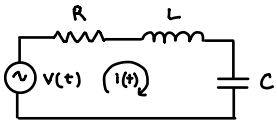
↓

Model the system

- → Mathematical Modelling
- → Transfer Function
- → State Variables

## Mathematical Models of Physical Systems

ex: RLC Circuit,




From KVL ⇒ $V(t) = Ri(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int_{-\infty}^{t} i(t)\,dt$ → ① ⇒ This is called Integro-Differential equation

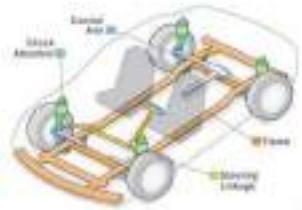
Differentiating wrt t, $0 = R\frac{di}{dt} + L\frac{d^2i}{dt^2} + \frac{i(t)}{C}$ → ②

Apply Laplace Transform to ②, $Ls^2I(s) + RsI(s) + \frac{I(s)}{C} = 0$

Apply Laplace Transform to ①, $V(s) = \left(L(s) + R + \frac{1}{Cs}\right)I(s)$

$\Rightarrow \frac{V_o(s)}{V(s)} = \frac{I(s)/Cs}{\left(L(s)+R+\frac{1}{Cs}\right)I(s)} = \frac{1}{LCs^2+RCs+1}$ → Mathematical Model ; $\frac{I(s)}{V(s)} = \frac{Cs}{LCs^2+RCs+1}$

↳ Algebraic Func$^n$



## Differential Equations of Physical Systems

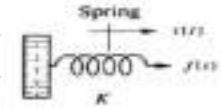
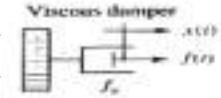
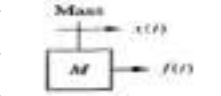
→ **Mechanical Systems**

→ Systems consisting spring, mass & damper

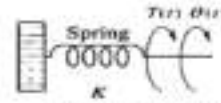
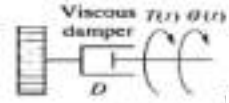
→ 2 Types:

- Translational
- Rotational

→ **Translational Mechanical Systems:**

| Component | Force - velocity | Force - Displacement |
|---|---|---|
| Spring  | $f(t) = K\int_0^t v(t)\,dt$ | $f(t) = Kx(t)$ |
| Viscous damper  | $f(t) = f_v\, v(t)$ | $f(t) = f_v \frac{dx(t)}{dt}$ |
| Mass  | $f(t) = M\frac{dv(t)}{dt}$ | $f(t) = M\frac{d^2x(t)}{dt^2}$ |

→ **Rotational Mechanical Systems:**

| Component | Torque - Angular Velocity | Torque - Angular Displacement |
|---|---|---|
| Spring  | $T(t) = K\int_0^t \omega(t)\,dt$ | $T(t) = K\theta(t)$ |
| Viscous damper  | $T(t) = D\omega(t)$ | $T(t) = D\frac{d\theta(t)}{dt}$ |
| Inertia | $T(t) = J\frac{d\omega(t)}{dt}$ | $T(t) = J\frac{d^2\theta(t)}{dt^2}$ |

## Linear Approximation of Physical Systems

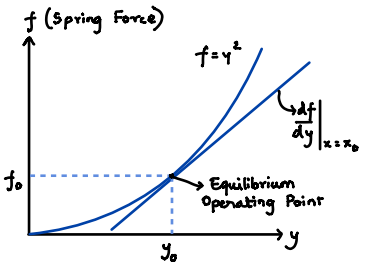
→ Most mechanical & electrical systems are linear for large range of values, and non-linear elements can be linearised assuming small signal conditions

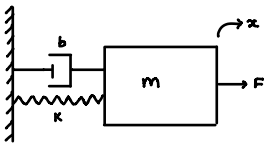
→ Consider operating point = $x_0$,

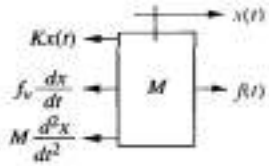
$$y = g(x) = g(x_0) + \left.\frac{dg}{dx}\right|_{x=x_0} \cdot \frac{(x-x_0)}{1!} + \left.\frac{d^2g}{dx^2}\right|_{x=x_0} \frac{(x-x_0)^2}{2!} + \dots$$

$$= g(x_0) + \left.\frac{dg}{dx}\right|_{x=x_0} \cdot (x-x_0) = y_0 + m(x-x_0)$$

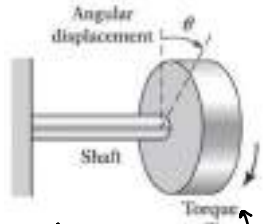
$$\Delta y = m\,\Delta x$$



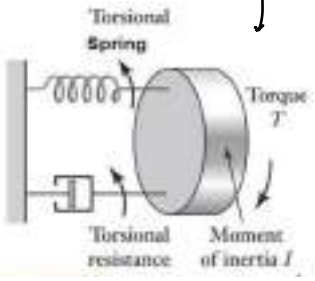


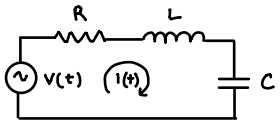




Free Body Diagram

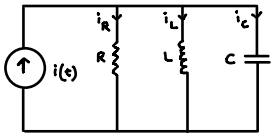



Rotating Mass on end of a shaft




Series




Parallel

→ **Translational Systems examples**

$$F = F_m + F_f + F_k$$

$$= m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + Kx$$

→ D'Alembert's Law: $\sum F = 0$

→ **Rotational Systems examples**

$$T = I\frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt} + K\theta$$

→ **Electrical System**

→ Series:

$$V = Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt$$

$$= R\frac{dq}{dt} + L\frac{d^2q}{dt^2} + \frac{q}{C} \qquad \left(i = \frac{dq}{dt}\right)$$

→ Parallel

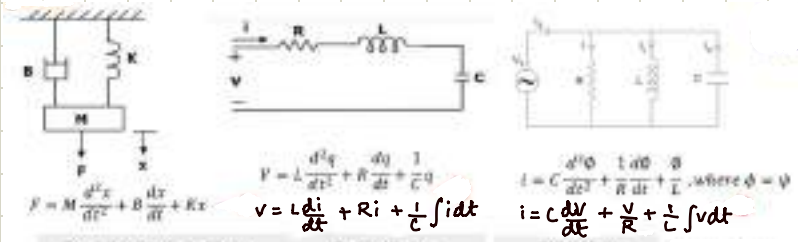
$$i = \frac{v}{R} + C\frac{dv}{dt} + \frac{1}{L}\int_{\infty}^{t} v\,dt$$

$$= \frac{1}{R}\cdot\frac{d\phi}{dt} + C\frac{d^2\phi}{dt^2} + \frac{\phi}{L} \qquad \left(v = \frac{d\phi}{dt}\right)$$

# Converting Mechanical System to Electrical System

→ 2 Ways :

- Force - Current Analogy
- Force - Voltage Analogy




$$F = M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx$$

$$V = L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q$$

$$v = L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt$$

$$i = C\frac{d^2\phi}{dt^2} + \frac{1}{R}\frac{d\phi}{dt} + \frac{\phi}{L}, \text{ where } \phi = \psi$$

$$i = C\frac{dv}{dt} + \frac{v}{R} + \frac{1}{L}\int v\,dt$$

| Translational Mechanical System | Electrical System | Electrical System |
|---|---|---|
| Force(F) | voltage(v) | Current(i) |
| Mass(M) | Inductance(L) | Capacitance(C) |
| Frictional Coefficient(B) | Resistance(R) | Reciprocal of Resistance $(\frac{1}{R})$ |
| Spring Constant(K) | Reciprocal of Capacitance $(\frac{1}{C})$ | Reciprocal of Inductance $(\frac{1}{L})$ |
| Displacement(x) | Charge(q) | Magnetic Flux(φ) |
| Velocity(v) | Current(i) | Voltage(V) |

$$T = J\frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt} + K\theta$$

$$V = L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q$$

$$= L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt$$

$$i = C\frac{d^2\phi}{dt^2} + \frac{1}{R}\frac{d\phi}{dt} + \frac{\phi}{L}$$

$$i = C\frac{dv}{dt} + \frac{v}{R} + \frac{1}{L}\int v\,dt$$

| Rotational Mechanical System | Electrical System | Electrical System |
|---|---|---|
| Torque(T) | Voltage(V) | Current(i) |
| Moment of inertia(J) | Inductance(L) | Capacitance(C) |
| Rotational friction Coefficient(B) | Resistance(R) | Reciprocal of Resistance $(\frac{1}{R})$ |
| Torsional spring constant(K) | Reciprocal of Capacitance $(\frac{1}{C})$ | Reciprocal of Inductance $(\frac{1}{L})$ |
| Angular Displacement(θ) | Charge(q) | Magnetic Flux(φ) |
| Angular velocity(ω) | Current(i) | Voltage(V) |

Input → [Process] → Output

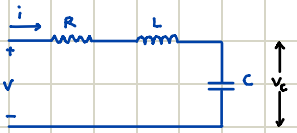
R(s) — G(s) — C(s) = R(s).G(s)

$TF = \frac{C(s)}{R(s)}$

# Transfer Function

→ The output is related to input through a function called transfer function

Q.




A. $TF = \frac{V_c(s)}{V(s)}$

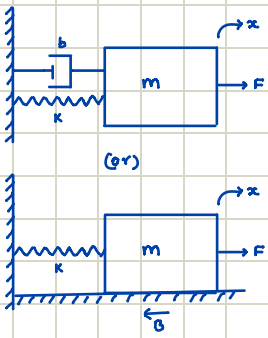
$$V_c(t) = \frac{1}{C}\int_{-\infty}^{t} i(t)dt = \frac{1}{C}\int_{-\infty}^{0} i(t)dt + \frac{1}{C}\int_{0}^{t} i(t)dt$$

$V_c(0) = 0$

$V_c(s) = \frac{I(s)}{Cs}$

$$\frac{V_c(s)}{V(s)} = \frac{I(s)/C(s)}{\left(Ls + R + \frac{1}{Cs}\right)I(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Q. Find the transfer function for the given mechanical system




A. $F = m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + Kx$

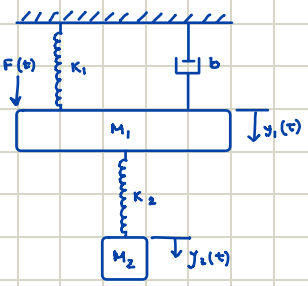
$F(s) = ms^2x(s) + bsx(s) + Kx(s)$

$= (ms^2 + bs + K)\,x(s)$

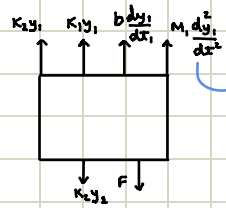
$$\frac{x(s)}{F(s)} = \frac{1}{ms^2 + bs + K} = \frac{N(s)}{D(s)} = \frac{\text{zero polynomial}}{\text{poles polynomial}}$$

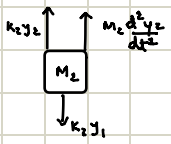
→ 2 poles

Q. Find the transfer function for the given mechanical system




A. Free body diagrams:




$$F(t) = M_1\frac{d^2y_1}{dt^2} + K_1y_1 + b\frac{dy_1}{dt} + K_2(y_1 - y_2)$$

$$0 = K_2(y_1 - y_2) + M_2\frac{d^2y_2}{dt^2}$$

$F(s) = (M_1s^2 + bs + K_1 + K_2)\,y_1(s) - K_2y_2(s)$

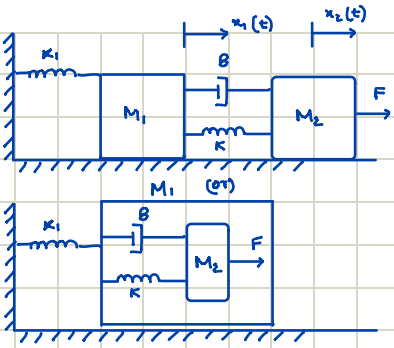
$= (M_1s^2 + bs + K_1 + K_2)\,y_1(s) - \frac{K_2^2y_1(s)}{M_2s^2 + K_2}$

$0 = (M_2s^2 + K_2)\,y_2(s) - K_2y_1(s) \Rightarrow y_2(s) = \frac{K_2y_1(s)}{M_2s^2 + K_2}$

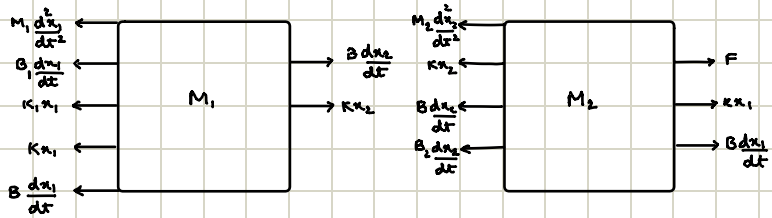
$$\Rightarrow \frac{y_1(s)}{F(s)} = \frac{1}{M_1s^2 + bs + K_1 + K_2 - \frac{K_2^2}{M_2s^2 + K_2}}$$

$B_1$ & $B_2$ are frictional forces due to contact b/w mass and ground

Q. Find the transfer function for the given mechanical system




A. Free body diagram



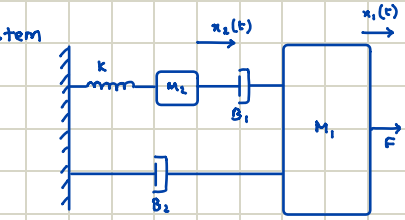

$$M_1\frac{d^2x_1}{dt^2} + B_1\frac{dx_1}{dt} + K_1x_1 + Kx_1 + B\frac{dx_1}{dt} = B\frac{dx_2}{dt} + Kx_2 \qquad F = M_2\frac{d^2x_2}{dt^2} + Kx_2 + B\frac{dx_2}{dt} + B_2\frac{dx_2}{dt} - Kx_1 - B\frac{dx_1}{dt}$$

$$(M_1s^2 + B_1s + K_1 + K + Bs)X_1(s) = (Bs + K)X_2(s) \qquad F(s) = (M_2s^2 + K + Bs + B_2s)X_2(s) - (K + Bs)X_1(s)$$

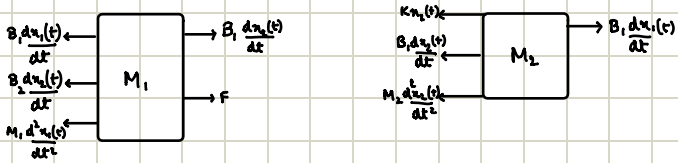
$$\Rightarrow F(s) = \left(\frac{(M_2s^2 + K + Bs + B_2s)(M_1s^2 + B_1s + K_1 + K + Bs)}{Bs + K} - (K + Bs)\right)X_1(s)$$

$$\frac{F(s)}{x_1(s)} = \left(\frac{(M_2s^2 + K + Bs + B_2s)(M_1s^2 + B_1s + K_1 + K + Bs)}{Bs + K} - (K + Bs)\right)$$

Q. Find the transfer function for the given mechanical system




A.




$$F = M_1\frac{d^2x_1}{dt^2} + (B_2 - B_1)\frac{dx_2}{dt} + B_1\frac{dx_1}{dt} \qquad 0 = M_2\frac{d^2x_2}{dt^2} + B_1\frac{dx_2}{dt} - B_1\frac{dx_1}{dt} + Kx_2$$

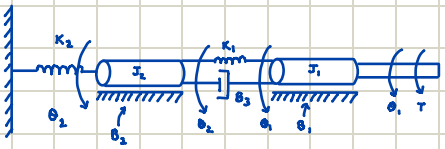
$$F(s) = (M_1s^2 + B_1s)x_1(s) + (B_2 - B_1)sx_2(s) \qquad 0 = (M_2s^2 - B_1s + K)x_2(s) - B_1sx_1(s)$$

$$= (M_1s^2 + B_1s)x_1(s) + \frac{(B_2 - B_1)s \cdot B_1sx_1(s)}{M_2s^2 - B_1s + K} \qquad x_2(s) = \frac{B_1sx_1(s)}{M_2s^2 - B_1s + K}$$

$$\frac{F(s)}{x_1(s)} = (M_1s^2 + B_1s) + \frac{(B_2 - B_1)B_1s^2}{M_2s^2 - B_1s + K}$$

## Analogous Systems

Q.



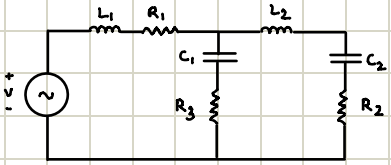

A. $$T = J_1\frac{d^2\theta_1}{dt^2} + B_1\frac{d\theta_1}{dt} + K_1(\theta_1-\theta_2) + B_3\frac{d}{dt}(\theta_1-\theta_2)$$

$$0 = J_2\frac{d^2\theta_2}{dt^2} + K_1(\theta_2-\theta_1) + K_2(\theta_2) + B_3\frac{d}{dt}(\theta_2-\theta_1) + B_2\frac{d\theta_2}{dt}$$

Converting Rotational - Mechanical System to Electrical System (T - V)

$$V = L_1\frac{d^2q_1}{dt^2} + R_1\frac{dq_1}{dt} + \frac{1}{C_1}(q_1-q_2) + R_3\frac{d}{dt}(q_1-q_2)$$

$$= L_1\frac{di_1}{dt} + R_1i_1 + \frac{1}{C_1}\int(i_1-i_2)dt + R_3(i_1-i_2)$$



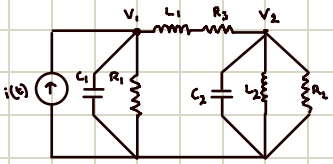

$$0 = L_2\frac{d^2q_2}{dt^2} + \frac{1}{C_1}(q_2-q_1) + \frac{1}{C_2}q_2 + R_3\frac{d}{dt}(q_2-q_1) + R_2\frac{dq_2}{dt}$$

$$= L_2\frac{di_2}{dt} + \frac{1}{C_1}\int(i_2-i_1)dt + \frac{1}{C_2}\int i_2dt + R_3(i_2-i_1) + R_2i_2$$

Converting Rotational - Mechanical System to Electrical System (T - I)

$$i = C_1\frac{d^2\phi_1}{dt^2} + \frac{1}{R_1}\frac{d\phi_1}{dt} + \frac{1}{L_1}(\phi_1-\phi_2) + \frac{1}{R_3}\frac{d}{dt}(\phi_1-\phi_2)$$

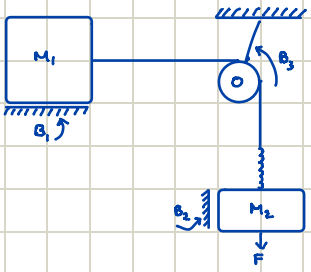
$$= C_1\frac{dV_1}{dt} + \frac{1}{R_1}V_1 + \frac{1}{L_1}\int(V_1-V_2)dt + \frac{1}{R_3}(V_1-V_2)$$



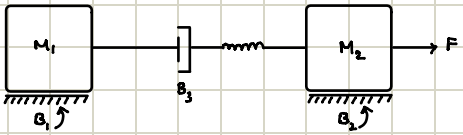

$$0 = C_2\frac{d^2\phi_2}{dt^2} + \frac{1}{L_1}(\phi_2-\phi_1) + \frac{1}{L_2}\phi_2 + \frac{1}{R_3}\frac{d}{dt}(\phi_2-\phi_1) + \frac{1}{R_2}\cdot\frac{d\phi_2}{dt}$$

$$= C_2\frac{dV_2}{dt} + \frac{1}{L_1}\int(V_2-V_1)dt + \frac{1}{L_2}\int V_2dt + \frac{1}{R_3}(V_2-V_1) + \frac{V_2}{R_2}$$

Q.



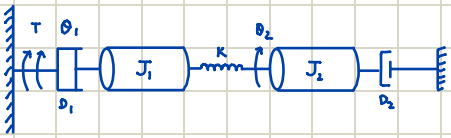

A. Equivalent Diagram




$$F = M_2\frac{dx_1^2}{dt^2} + B_2\frac{dx_2}{dt_2} + Kx_2$$

$$B_1\frac{dx_1}{dt} + B_3\frac{dx_1}{dt} + M_1\frac{dx_1^2}{dt^2} = 0$$

Q.



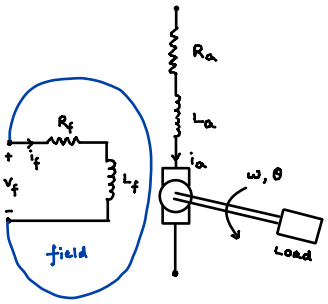

A. $$\bar{T} = D_1\frac{d\theta_1}{dt} + J_1\frac{d^2\theta_1}{dt^2} + K(\theta_1 - \theta_2)$$

$$0 = J_2\frac{d^2\theta_2}{dt^2} + D_2\frac{d\theta_2}{dt} + K(\theta_2 - \theta_1)$$

## Electromechanical System (DC Motor)

→ A DC motor converts direct current which is an electrical energy to rotational mechanical energy. The basic principle is that whenever a current carrying conductor is placed in a magnetic field, it experiences a mechanical force. The direction of this force is given by Fleming's Left Hand Rule

### Field Controlled DC Motor




→ $I_f$ is varied, $I_a$ is kept constant

$$V_f = I_f R_f + L_f \frac{di_f}{dt}$$

$$V_f(s) = I_f(s) \cdot R_f + L_f s\, I_f(s)$$
$$= I_f(s)\left[R_f + L_f s\right]$$
$$I_f(s) = \frac{V_f(s)}{R_f + L_f s} \longrightarrow ①$$

$$T_L = J\frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt}$$

$$T_m = T_L$$
$$T_m \propto I_f$$
$$T_m = K_m I_f$$
$$T_m(s) = K_m I_f(s) \longrightarrow ②$$
$$T_L(s) = Js^2\theta(s) + Bs\,\theta(s)$$
$$= \theta(s)\left(Js^2 + Bs\right)$$
$$\theta(s) = \frac{T_L(s)}{Js^2 + Bs}$$
$$= \frac{1}{Js^2 + Bs} \cdot K_m \cdot \frac{V_f(s)}{R_f + L_f s}$$

$$\frac{\theta(s)}{V_f(s)} = \frac{K_m}{(Js^2 + Bs)(R_f + L_f s)} = \boxed{\frac{K_m}{s\,(Js + B)(R_f + sL_f)}}$$

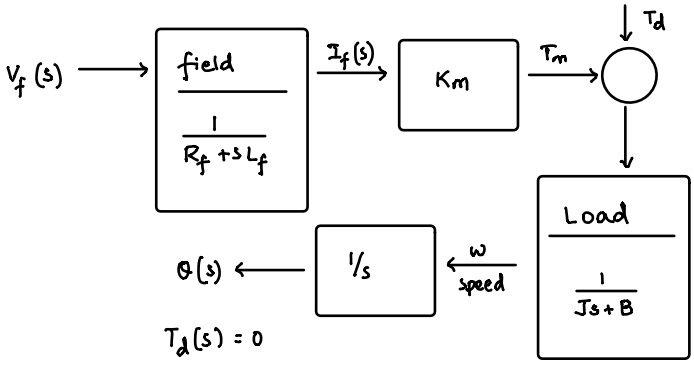
$$= \frac{\frac{K_m}{BR_f}}{s\left(\frac{Js}{B} + 1\right)\left(\frac{sL_f}{R_f} + 1\right)}$$

$$\boxed{\frac{\theta(s)}{V_f(s)} = \frac{K_m/BR_f}{s(sT_L + 1)(sT_f + 1)}}$$

$T_L$ : Load Time Constant

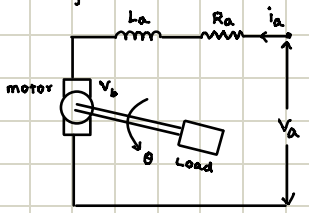
$T_f$ : Field Time Constant

### Block Diagram ⇒



## Armature Controlled DC Motor

→ Let $I_f$ = constant & $I_a$ = variable




$V_b$ : back electromotive force voltage proportional to speed of motor

$$V_a = i_a R_a + L_a \frac{di_a}{dt} + V_b$$

$$V_a(s) = R_a I_a(s) + sL_a I_a(s) + V_b(s)$$

$$V_a(s) - V_b(s) = I_a(s)\left(R_a + sL_a\right)$$

$$\frac{V_a(s) - V_b(s)}{I_a(s)} = \left(R_a + sL_a\right)$$

$T_m \propto I_a$

↳ Generated Torque

$$T_m(s) = K_m I_a(s)$$

Back emf, $V_b \propto \omega \Rightarrow V_b = K_b \omega$

$$= K_b \frac{d\theta}{dt}$$

$$V_b(s) = K_b s\,\theta(s)$$

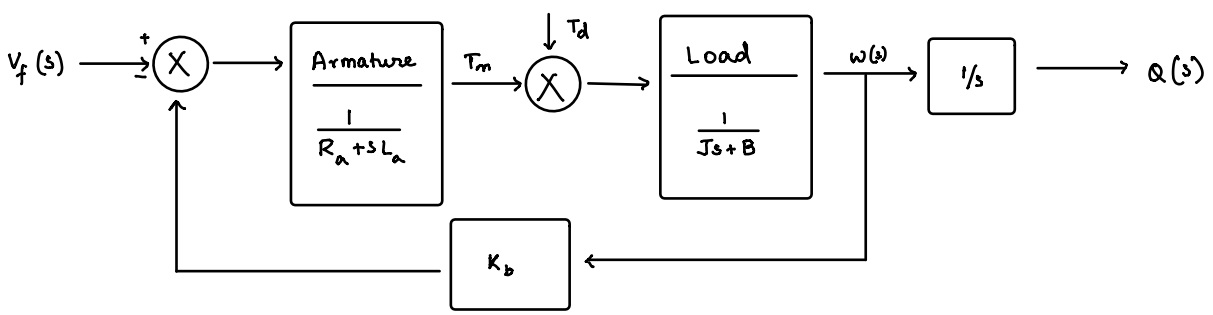
The generated motor torque $T_m(s) = \left(Js^2 + Bs\right)\theta(s)$

$$\theta(s) = \frac{T_m(s)}{Js^2 + Bs}$$

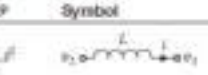
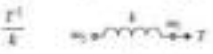
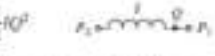
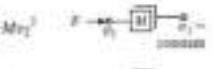
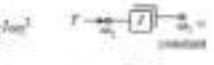
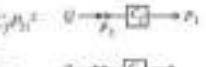
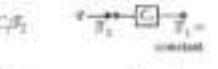
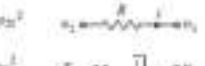
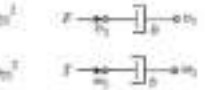
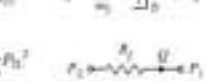
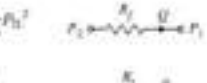
$$= \frac{K_m I_a(s)}{Js^2 + Bs}$$

$$= \frac{K_m}{Js^2 + Bs} \cdot \left(\frac{V_a(s) - K_b s\,\theta(s)}{R_a + sL_a}\right)$$

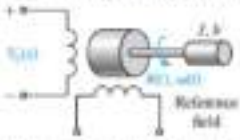
$$= \frac{K_m V_a(s)}{\left(Js^2 + Bs\right)\left(R_a + sL_a\right)} - \frac{K_m K_b s\,\theta(s)}{\left(Js^2 + Bs\right)\left(R_a + sL_a\right)}$$



# Transfer functions of Dynamic Elements & Networks

| Type of Element | Physical Element | Governing Equation | Energy $E$ or Power $\mathcal{P}$ | Symbol |
|---|---|---|---|---|
| Inductive storage | Electrical inductance | $v_{21} = L\frac{di}{dt}$ | $E = \frac{1}{2}Li^2$ |  |
| | Translational spring | $v_{21} = \frac{1}{k}\frac{dF}{dt}$ | $E = \frac{1}{2}\frac{F^2}{k}$ | |
| | Rotational spring | $\omega_{21} = \frac{1}{k}\frac{dT}{dt}$ | $E = \frac{1}{2}\frac{T^2}{k}$ |  |
| | Fluid inertia | $P_{21} = I\frac{dQ}{dt}$ | $E = \frac{1}{2}IQ^2$ |  |
| Capacitive storage | Electrical capacitance | $i = C\frac{dv_{21}}{dt}$ | $E = \frac{1}{2}Cv_{21}^2$ | |
| | Translational mass | $F = M\frac{dv_2}{dt}$ | $E = \frac{1}{2}Mv_2^2$ |  |
| | Rotational mass | $T = J\frac{d\omega_2}{dt}$ | $E = \frac{1}{2}J\omega_2^2$ |  |
| | Fluid capacitance | $Q = C_f\frac{dP_{21}}{dt}$ | $E = \frac{1}{2}C_fP_{21}^2$ |  |
| | Thermal capacitance | $q = C_t\frac{dT_2}{dt}$ | $E = C_tT_2$ |  |
| Energy dissipators | Electrical resistance | $i = \frac{1}{R}v_{21}$ | $\mathcal{P} = \frac{1}{R}v_{21}^2$ |  |
| | Translational damper | $F = bv_{21}$ | $\mathcal{P} = bv_{21}^2$ |  |
| | Rotational damper | $T = b\omega_{21}$ | $\mathcal{P} = b\omega_{21}^2$ |  |
| | Fluid resistance | $Q = \frac{1}{R_f}P_{21}$ | $\mathcal{P} = \frac{1}{R_f}P_{21}^2$ |  |
| | Thermal resistance | $q = \frac{1}{R_t}T_{21}$ | $\mathcal{P} = \frac{1}{R_t}T_{21}$ | |

7. AC motor, two-phase control field, rotational actuator

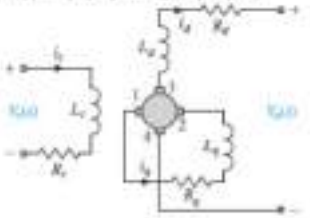



$$\frac{\theta(s)}{V_c(s)} = \frac{K_m}{s(\tau s + 1)}$$

$\tau = J/(b - m)$

$m$ = slope of linearized torque-speed curve (normally negative)

8. Rotary Amplifier (Amplidyne)



$$\frac{V_g(s)}{V_c(s)} = \frac{K/(R_cR_g)}{(s\tau_c + 1)(s\tau_g + 1)}$$

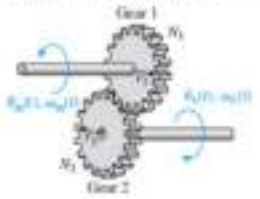
$\tau_c = L_c/R_c \qquad \tau_g = L_g/R_g$

for the unloaded case, $I_d = 0$, $\tau_c = \tau_g$

$0.05 \text{ s} < \tau_c < 0.5 \text{ s}$

$V_g, V_{bk} = V_d$

10. Gear train, rotational transformer

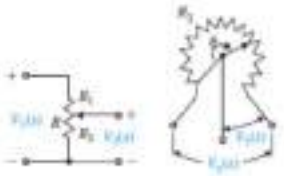



$$\text{Gear ratio} = n = \frac{N_1}{N_2}$$

$$N_2\theta_2(t) = N_1\theta_{in}(t), \qquad \theta_2(t) = n\theta_{in}(t)$$

$$\omega_2(t) = n\omega_{in}(t)$$

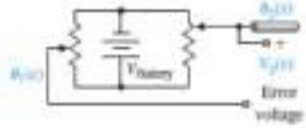
11. Potentiometer, voltage control



$$\frac{V_2(s)}{V_1(s)} = \frac{R_2}{R} = \frac{R_2}{R_1 + R_2}$$

$$\frac{R_2}{R} = \frac{\theta}{\theta_{max}}$$

12. Potentiometer, error detector bridge

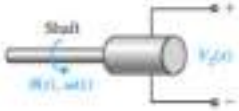



$$V_1(s) = k_p(\theta_1(s) - \theta_2(s))$$

$$V_2(s) = k_p\theta_{error}(s)$$

$$k_p = \frac{V_{Battery}}{\theta_{max}}$$

13. Tachometer, velocity sensor




$$V_2(s) = K_t\omega(s) = K_t s\theta(s)$$

$K_t$ = constant

14. DC amplifier

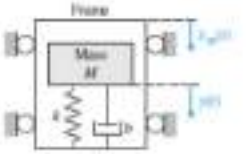
$$\frac{V_2(s)}{V_1(s)} = \frac{k_a}{s\tau + 1}$$

$R_o$ = output resistance

$C_o$ = output capacitance

$\tau = R_oC_o$, $\tau \ll 1$s and is often negligible for controller amplifier

15. Accelerometer, acceleration sensor



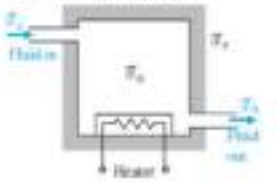

$$x_o(t) = y(t) - x_{in}(t),$$

$$\frac{X_o(s)}{X_{in}(s)} = \frac{-s^2}{s^2 + (b/M)s + k/M}$$

For low-frequency oscillations, where $\omega < \omega_n$,

$$\frac{X_o(j\omega)}{X_{in}(j\omega)} \simeq \frac{\omega^2}{k/M}$$

16. Thermal heating system




$$\frac{\Theta(s)}{q(s)} = \frac{1}{C_t s + (QS + 1/R_t)}, \text{ where}$$

$\Theta = \Theta_o - \Theta_a$ = temperature difference due to thermal process

$C_t$ = thermal capacitance

$Q$ = fluid flow rate = constant

$S$ = specific heat of water

$R_t$ = thermal resistance of insulation

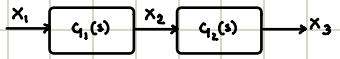
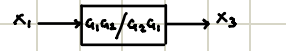
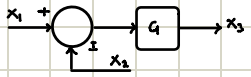
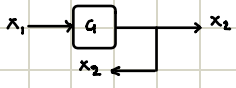
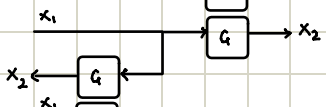
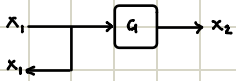
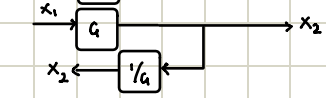
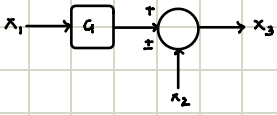
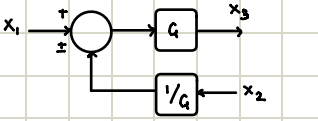
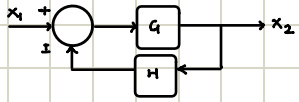
$q(s)$ = transform of rate of heat flow of heating element

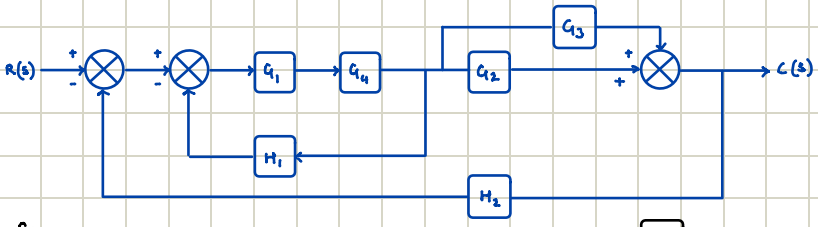
## Block Diagram Reduction Techniques

→ Block Diagram consists of unidirectional operational blocks that represent the transfer function of variables of interest

→ It gives a pictorial representation of control system

→ Types of Block diagram reduction techniques

| | Transformation | Original Diagram | Equivalent Diagram |
|---|---|---|---|
| 1) | Combining blocks in cascade |  |  |
| 2) | Moving a summing point behind block |  | |
| 3) | Moving a pickoff point ahead of block |  |  |
| 4) | Moving a pickoff point behind a block |  |  |
| 5) | Moving a summing point ahead of block |  |  |
| 6) | Eliminating a feedback loop |  | |

Q. 

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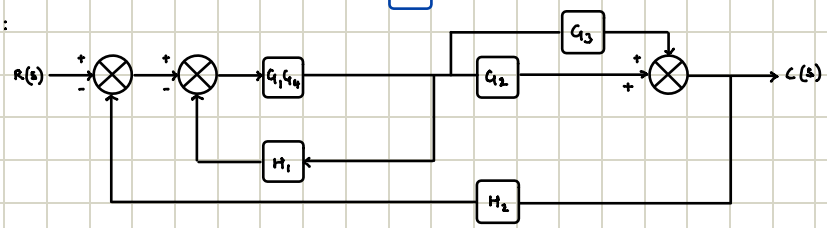

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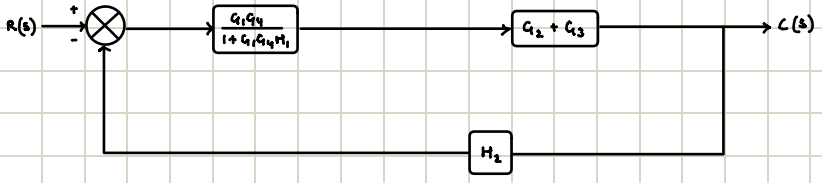

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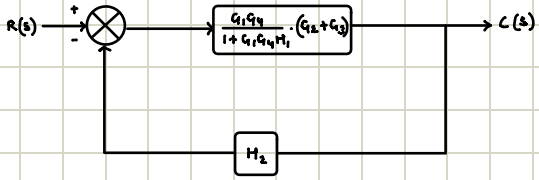


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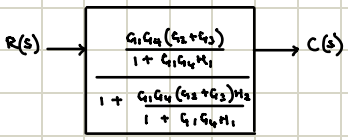


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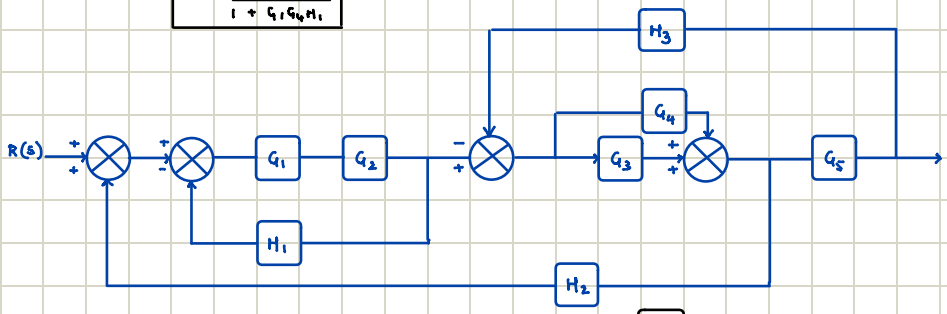
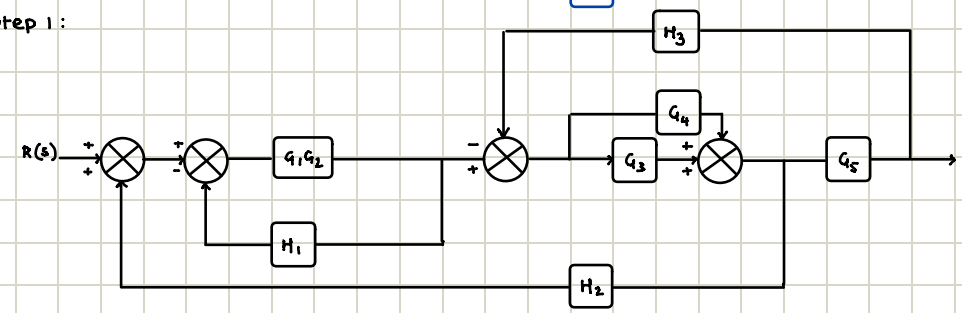



A. Step 1 :

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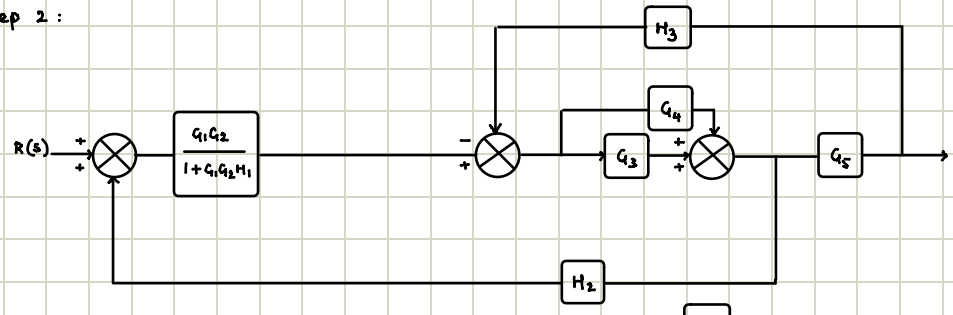


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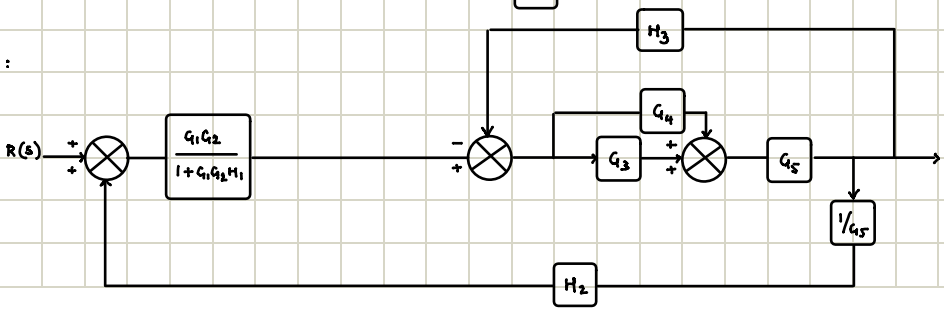

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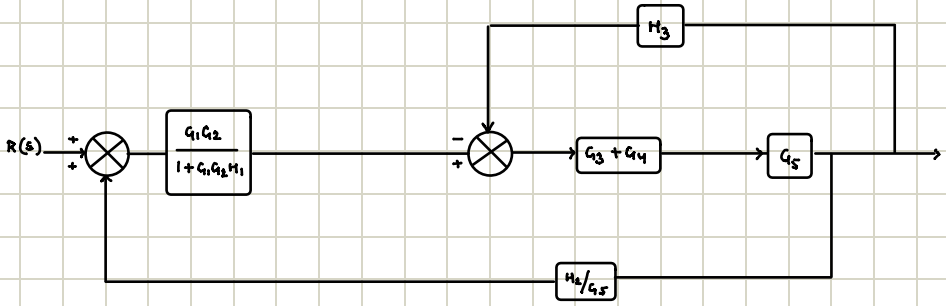


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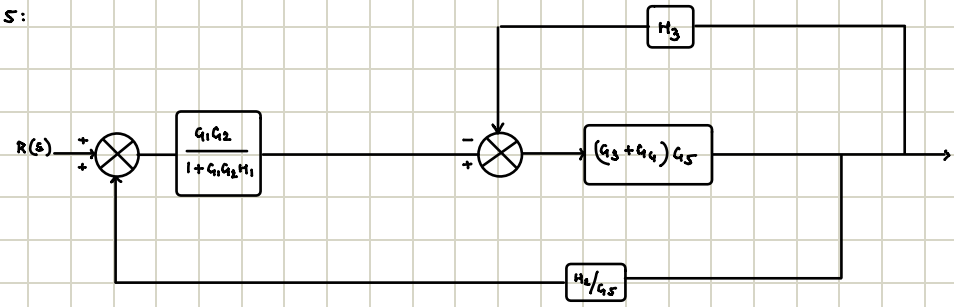


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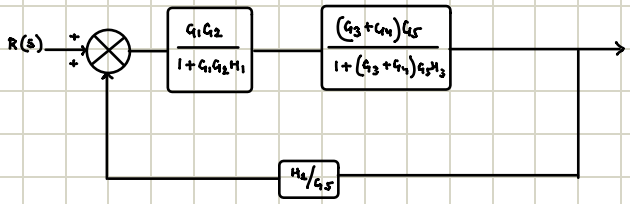
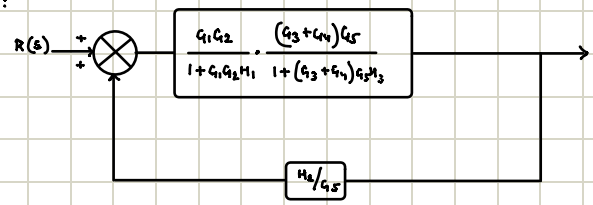


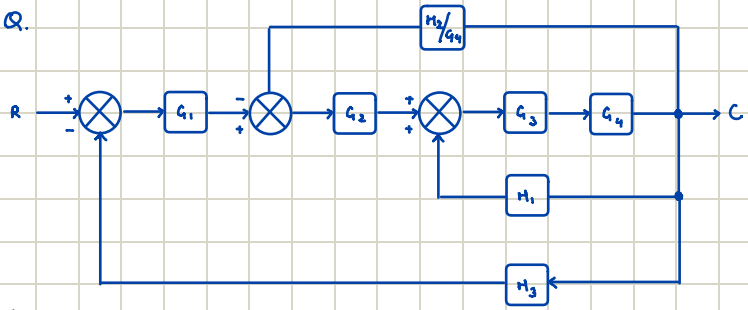

Step 7:



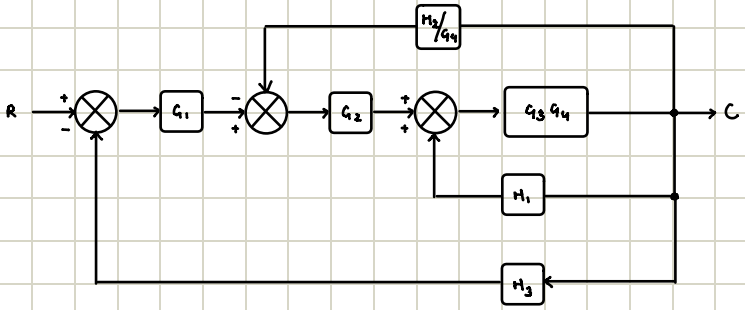

Step 8:

$$R(s) \longrightarrow \frac{\dfrac{G_1G_2}{1+G_1G_2H_1}\cdot\dfrac{(G_3+G_4)G_5}{1+(G_3+G_4)G_5H_3}}{1-\dfrac{G_1G_2}{1+G_1G_2H_1}\cdot\dfrac{(G_3+G_4)H_2}{1+(G_3+G_4)G_5H_3}}$$

Q.




A. Step 1:

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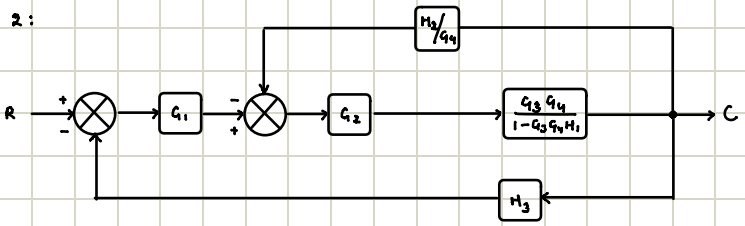


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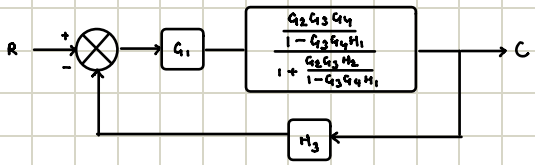


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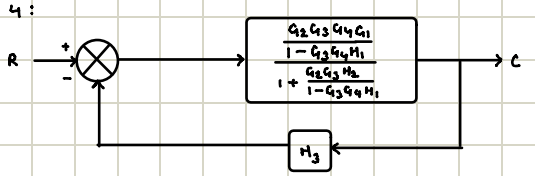
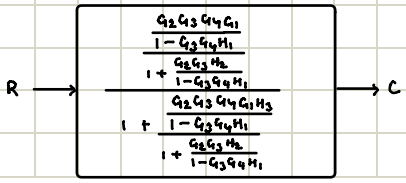



Step 5:



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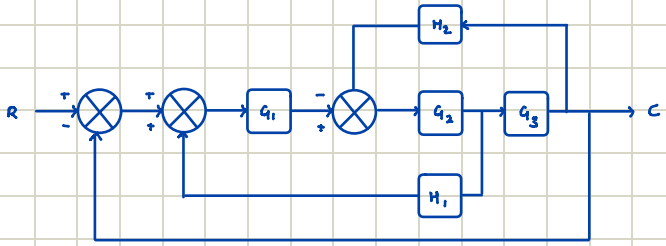
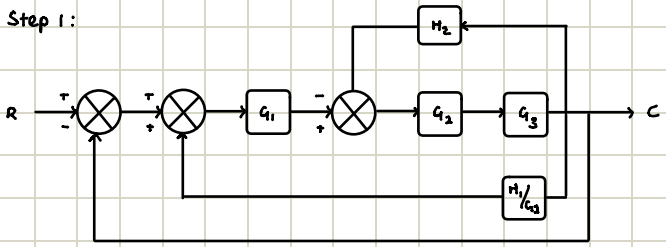



A. Step 1:

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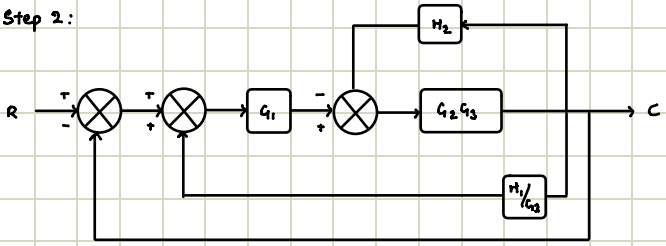


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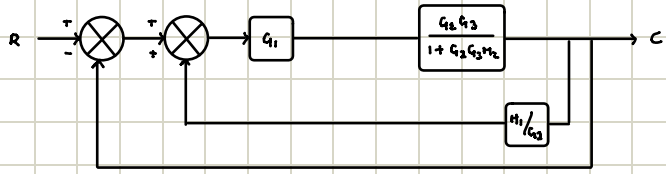


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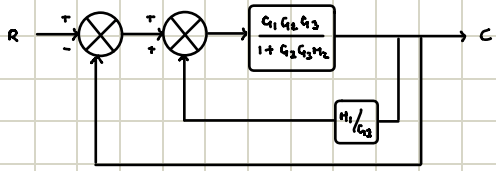
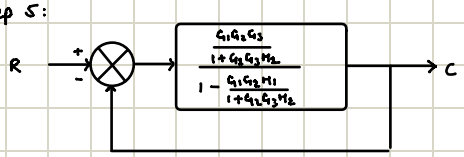
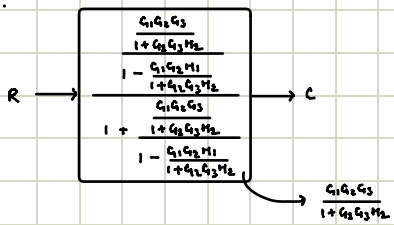



Step 5:




Step 6:



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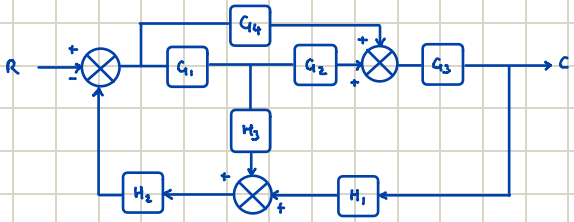
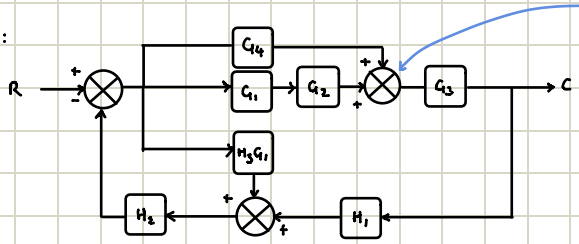
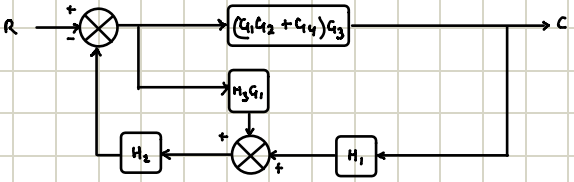



A. Step 1:

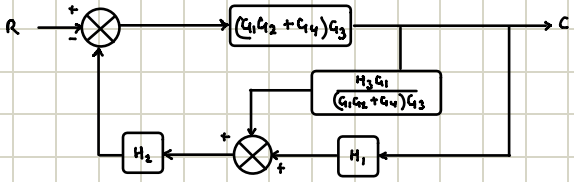



Both are adding cuz they are parallel

Step 2:




Step 3:

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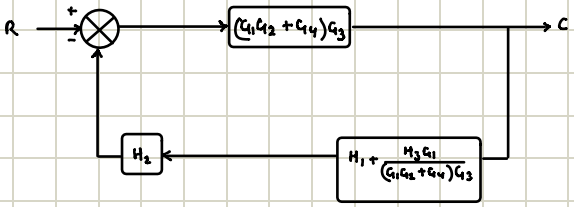
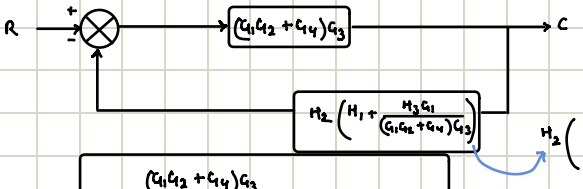



Step 5:

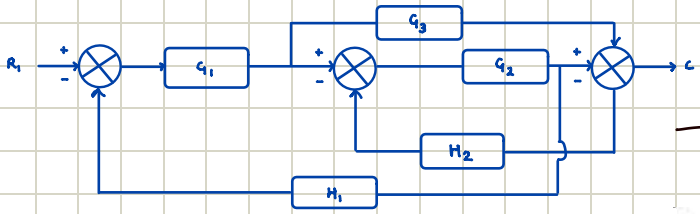


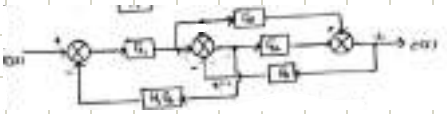

$H_2\left(\frac{H_1G_3(G_1G_2+G_4)+H_3G_1}{(G_1G_2+G_4)G_3}\right)$

Step 6:

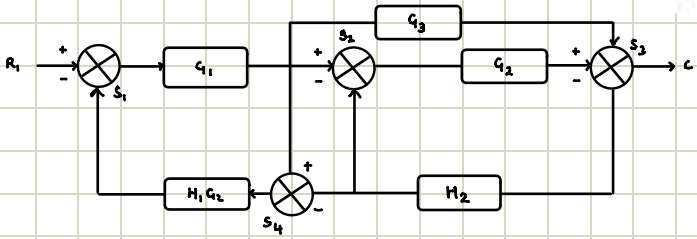
$$R \rightarrow \frac{(G_1G_2+G_4)G_3}{1+\frac{(G_1G_2+G_4)G_3H_2\left(H_1G_3(G_1G_2+G_4)+H_3G_1\right)}{(G_1G_2+G_4)G_3}} \rightarrow$$

Q.

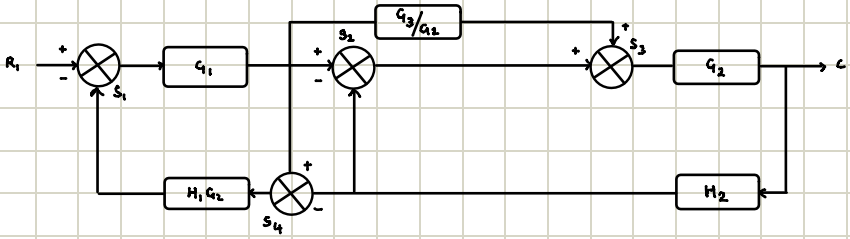


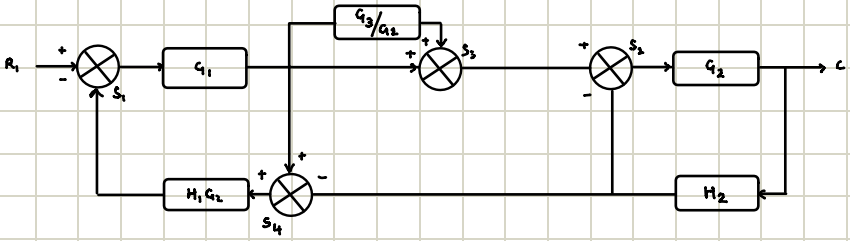
A. Step 1



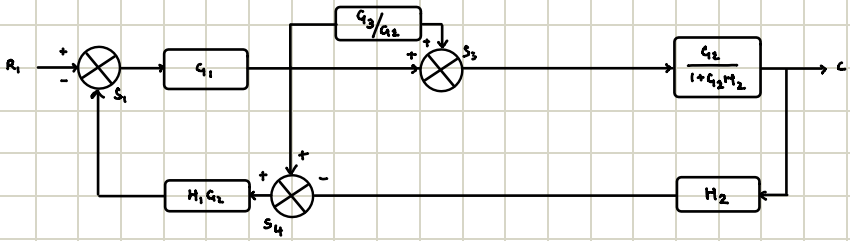

Step 2



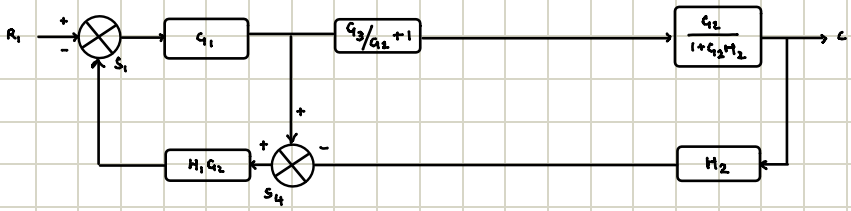

Step 3 (swap $S_2$ & $S_3$)



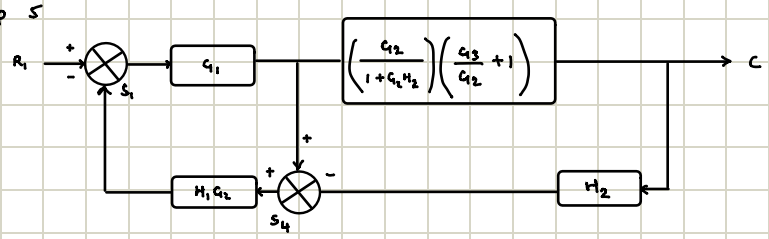

Step 4



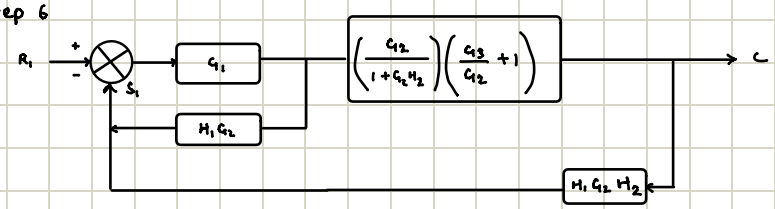
Step 4



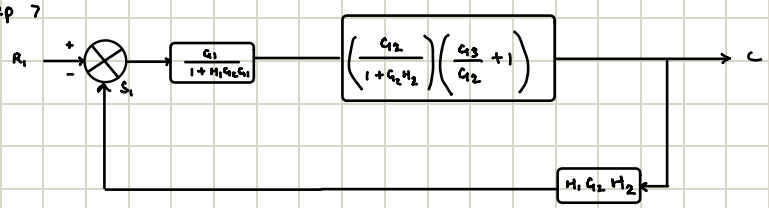

Step 5



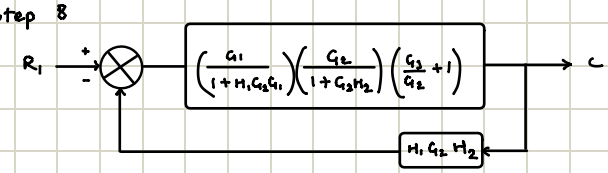

Step 6



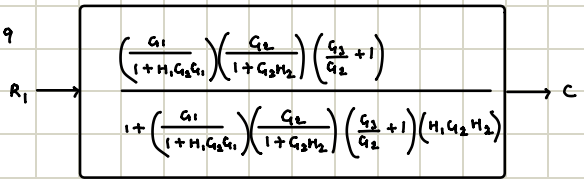

Step 7




Step 8



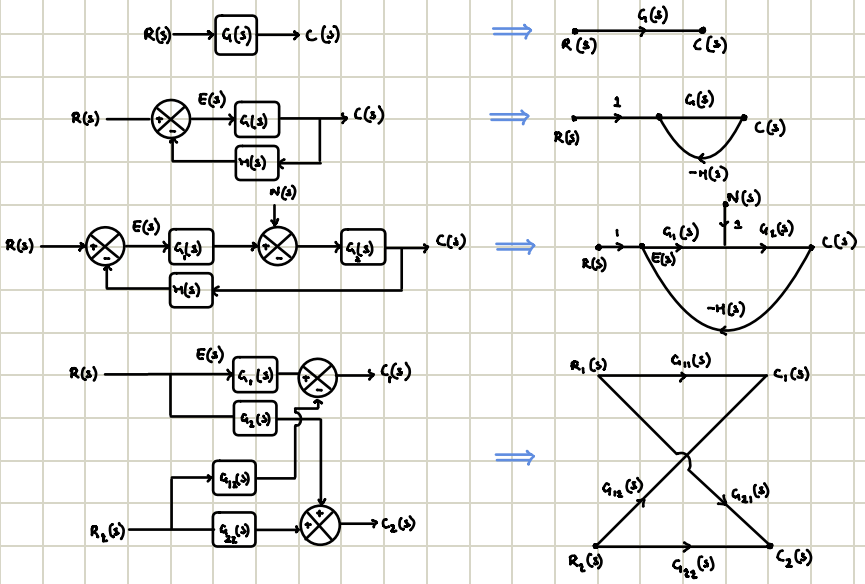

Step 9



## Signal Flow Graph

→ It is a diagram consisting of nodes that are connected by several directed branches and is a graphical representation of a set of linear relations

ex :




**Input node / Source node** – The node which has only outgoing signal

**Output node / sink** – The node which has only incoming signal

**Forward path** – It is a path traced from the input node to the output node in the direction of signal w/o passing through any node more than once

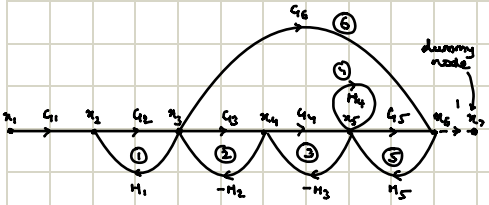
**Forward path gain** – The product of branch gains

**Loop** – It is a closed path that originates & terminates on the same node, with no node being met twice along the path

**Non-touching Loops** – Loops that don't share a common node

NTL : Non-Touching Loops

LG : Loop Gains




| | Loops | Loop Gains |
|---|---|---|
| 1) | $x_2 x_3 x_4$ | $G_2 H_1$ |
| 2) | $x_3 x_4 x_5$ | $-G_3 H_2$ |
| 3) | $x_4 x_5 x_4$ | $-G_4 H_3$ |
| 4) | $x_5 x_5$ | $H_4$ |
| 5) | $x_5 x_6 x_5$ | $G_5 H_5$ |
| 6) | $x_3 x_6 x_5 x_4 x_3$ | $G_6 H_5 H_3 H_2$ |

| NTL's | Product of LG's |
|---|---|
| 1 & 3 | $-G_2 H_1 G_4 H_3$ |
| 1 & 4 | $G_2 H_1 H_4$ |
| 1 & 5 | $G_2 H_1 G_5 H_5$ |
| 2 & 4 | $-G_3 H_2 H_4$ |
| 2 & 5 | $-G_3 H_2 G_5 H_5$ |

Forward paths : $x_1 x_2 x_3 x_4 x_5 x_6 x_7$

$x_1 x_2 x_3 x_6 x_7$

N : No. of forward paths

$P_k$ : $k^{th}$ Forward path

$\Delta_k$ : path factor of $k^{th}$ path

$\Delta$ : Determinant of the graph

→ This is a systematic approach to find T.F of LTI system

$$TF = \frac{C(s)}{R(s)} = \frac{\sum_{k=1}^{N} P_k \Delta_k}{\Delta}$$

$\Delta_k$ = 1 − (loop gains which doesn't touch the $k^{th}$ forward path)

$\Delta$ = 1 − (Sum of loop gains of individual loops) + (sum of loop gain products of all possible 2 non-touching loops) − (sum of loop gain products of all possible 3 non-touching) + ....

For the previous example,

$$TF = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

$\Delta_1 = 1 - 0 = 1$

$\Delta_2 = 1 - (H_4 - G_4H_3) = 1 - H_4 + G_4H_3$

$P_1 = G_1G_2G_3G_4G_5$

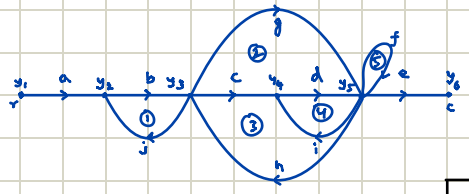
$P_2 = G_1G_2G_6$

$\Delta = 1 - (G_2H_1 - G_3H_2 - G_4H_3 + H_4 + G_5H_5 + G_6H_5H_3H_2) + (G_2H_1H_4 - G_2H_1G_4H_3 + G_2H_1G_5H_5 - G_3H_2H_4 - G_3H_2G_5H_5)$

$= 1 - G_2H_1 + G_3H_2 + G_4H_3 - H_4 - G_5H_5 - G_6H_5H_3H_2 + G_2H_1H_4 - G_2H_1G_4H_3 + G_2H_1G_5H_5 - G_3H_2H_4 - G_3H_2G_5H_5$

$$TF = \frac{G_1G_2G_3G_4G_5 + G_1G_2G_6(1 - H_4 + G_4H_3)}{1 - G_2H_1 + G_3H_2 + G_4H_3 - H_4 - G_5H_5 - G_6H_5H_3H_2 + G_2H_1H_4 - G_2H_1G_4H_3 + G_2H_1G_5H_5 - G_3H_2H_4 - G_3H_2G_5H_5}$$

Q.




A. Forward Paths ⇒ $y_1y_2y_3y_4y_5y_6$ → ①

$y_1y_2y_3y_5y_6$ → ②

Forward Path Gain ⇒ ① → abcde = $P_1$

② → abge = $P_2$

$\Delta_1 = 1 - 0 = 1$

$\Delta_2 = 1 - 0 = 1$

$$TF = \frac{\sum P_k\Delta_k}{\Delta}$$

$$= \frac{abcde + abge}{1 - (bj + gh + cdh + di + f) + bjdi + bjf} = \frac{abe(cd+g)}{1 - bj - gh - cdh - di - f + bjdi + bjf} = \frac{abe(cd+g)}{(bj-1)(f+di-1) - gh - cdh}$$

| Loops | | Loop Gains |
|---|---|---|
| ① | $y_1y_3y_2$ | bj |
| ② | $y_3y_5y_3$ | gh |
| ③ | $y_3y_4y_5y_3$ | cdh |
| ④ | $y_4y_5y_4$ | di |
| ⑤ | $y_5y_5$ | f |

| NTL's | Product of LG's |
|---|---|
| ① & ④ | bjdi |
| ① & ⑤ | bjf |

Q. Construct the SFG

$x_0$ is input $x_4$ is output

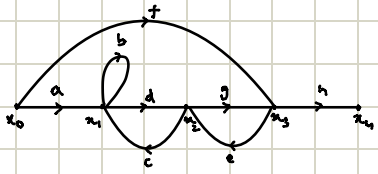
$x_1 = ax_0 + bx_1 + cx_2$

$x_2 = dx_1 + ex_3$

$x_3 = fx_0 + gx_2$

$x_4 = hx_3$

A.




Q. Construct the SFG

$x_2 = x_1 - x_6$

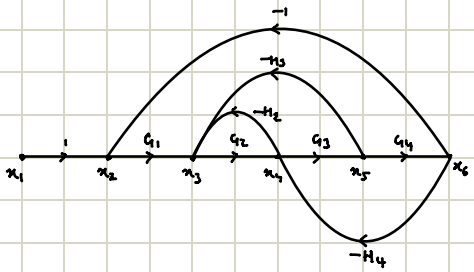
$x_3 = G_1x_2 - H_2x_4 - H_3x_5$

$x_4 = G_2x_3 - H_4x_6$

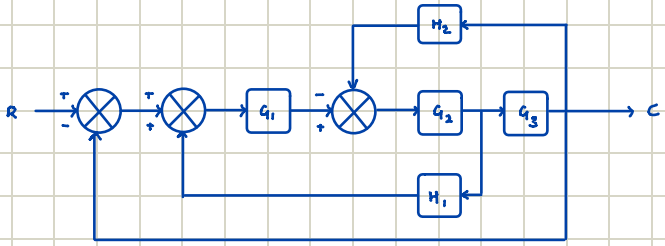
$x_5 = G_3x_4$

$x_6 = G_4x_5$

A.

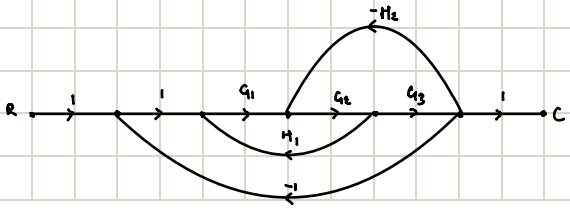


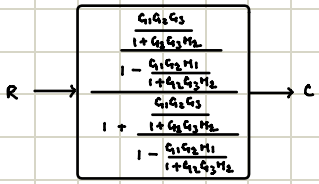

Q. Find the SFG for given block diagram




A. Refer pg 18 for Block diagram method ⇒

T.F ⇒






$$\frac{\frac{G_1G_2G_3}{1+G_2G_3H_2}}{1-\frac{G_1G_2H_1}{1+G_2G_3H_2}} \qquad 1+\frac{G_1G_2G_3}{1+G_2G_3H_2}$$

$P_1 \Rightarrow G_1G_2G_3$

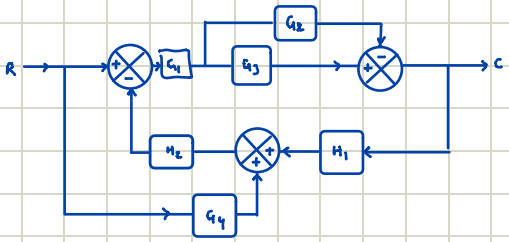
$\Delta_1 = 1-0 = 1$

$\Delta = 1-(-G_1G_2G_3 + G_1G_2H_1 - G_2G_3H_2)$

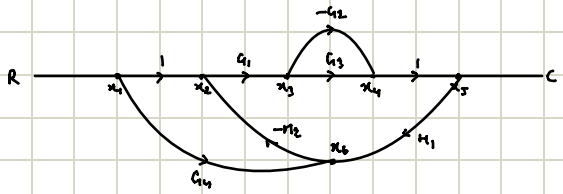
$= 1 + G_1G_2G_3 - G_1G_2H_1 + G_2G_3H_2$

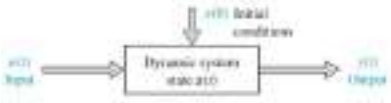
$$TF = \frac{G_1G_2G_3}{1+G_1G_2G_3 - G_1G_2H_1 + G_2G_3H_2}$$

Q. Convert given block diagram to SFG




A.





## Dynamic System

→ A time-varying control system in which one or more parameters of the system may vary as a function of time

## State of System

→ The state of system is a set of variables whose values, together with the input signals & the equations describing the dynamics, will provide the future state & output of the system

## State Variables

→ Variables that determine the future behavior of a system when the present state of the system and the inputs are known

→ $\vec{x}(t) = (x_1(t), x_2(t) \ldots, x_n(t))$

→ No. of state variables required = no. of storage elements present in system

## State Vector

→ Contains state variables as elements and will be represented as a matrix

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

## State Space Modelling

→ The dynamic system can be represented as

$$\dot{X} = AX + Bu$$

$$Y = CX + Du$$

They are called state differential equations

X: State vector

$\dot{X}$: Derivative of State vector

U: Input / Controlled Vector

Y: Output

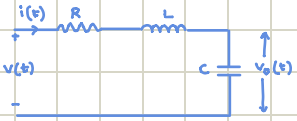
A: System Matrix

B: Input Matrix

C: Output Matrix

D: Feedforward Matrix

Q. Electrical System




A. Step 1 : Identify State Variables

2 State Variables $\Rightarrow$ $x_1 = V_c(t)$, $x_2 = i_L(t)$ $\Rightarrow X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Step 2: Formulate State Equations

From KVL, $v(t) = i(t)R + L\frac{di_L(t)}{dt} + V_o(t) \longrightarrow (1)$

$i_C(t) = C\frac{dV_c(t)}{dt}$

From (1), $L\frac{di_L(t)}{dt} = v(t) - i(t)R - V_o(t)$

$\frac{di_L(t)}{dt} = \frac{1}{L}\left(v(t) - i(t)R - V_o(t)\right)$

$\vec{x} = \begin{bmatrix} v_c(t) \\ i_L(t) \end{bmatrix}$

$\frac{d\vec{x}}{dt} = Ax + Bu(t)$

$A = \begin{bmatrix} 0 & 1/C \\ 1/L & -R/L \end{bmatrix}$ $B = \begin{bmatrix} 0 \\ -1/L \end{bmatrix}$

$u(t) = v(t)$

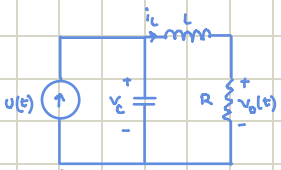
$y(t) = V_o(t)$

$y(t) = Cx + Du(t)$

$C = \begin{bmatrix} 1 & 0 \end{bmatrix}$

$D = \begin{bmatrix} 0 \end{bmatrix}$

Q.




A. $x = \begin{bmatrix} v_c(t) \\ i_L(t) \end{bmatrix}$

$v_L = L\frac{di_L(t)}{dt}$

$v_c(t) = L\frac{di_L(t)}{dt} + i_L R \Rightarrow \frac{di_L(t)}{dt} = \frac{1}{L}(v_c(t) - i_L R) \longrightarrow (1)$

$i_c(t) = C\frac{dv_c(t)}{dt}$

$u(t) = i_c(t) + i_L(t)$

$= C\frac{dv_c(t)}{dt} + i_L(t)$

$\frac{dv_c(t)}{dt} = \frac{1}{C}\left(u(t) - i_L(t)\right) \longrightarrow (2)$

$\frac{dx}{dt} = Ax + Bu(t)$

$\begin{bmatrix} \frac{dv_c}{dt} \\ \frac{di_L}{dt} \end{bmatrix} = \begin{bmatrix} 0 & -1/C \\ 1/L & -R/L \end{bmatrix}\begin{bmatrix} v_c(t) \\ i_L(t) \end{bmatrix} + \begin{bmatrix} 1/C \\ 0 \end{bmatrix}[u]$

$\dot{X} = A \quad X + B \quad U$

$Y = CX + D$

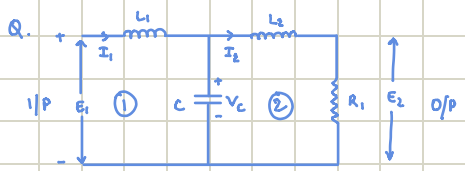
$Y = \begin{bmatrix} 0 & R \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0]$

$x_1 = v_c(t) \qquad \dot{x}_1 = \frac{dv_c(t)}{dt}$

$x_2 = i_L(t)$

$\dot{x}_2 = \frac{di_L(t)}{dt}$

Q.




A. 3 State Variables $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ V_c \end{bmatrix}$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{dI_1}{dt} \\ \frac{dI_2}{dt} \\ \frac{dV_c}{dt} \end{bmatrix}$$

$u = E_1$

$y = E_2$

KVL for ①, $E_1 = L_1 \frac{dI_1}{dt} + V_c$

$$\frac{dI_1}{dt} = \frac{1}{L_1}(E_1 - V_c) \longrightarrow ①$$

KVL for ②, $V_c = L_2 \frac{dI_2}{dt} + R_1 I_2$

$$\frac{dI_2}{dt} = \frac{1}{L_2}(V_c - R_1 I_2) \longrightarrow ②$$

KCL at node, $I_1 = I_c + I_2$

$$I_1 = C\frac{dV_c}{dt} + I_2 \Rightarrow \frac{dV_c}{dt} = \frac{1}{C}(I_1 - I_2) \longrightarrow ③$$

$E_2 = R_1 I_2$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1/L_1 \\ 0 & -R_1/L_2 & 1/L_2 \\ \frac{1}{C} & \frac{-1}{C} & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ V_c \end{bmatrix} + \begin{bmatrix} 1/L_1 \\ 0 \\ 0 \end{bmatrix} U$$

$$\dot{X} = AX + BU$$

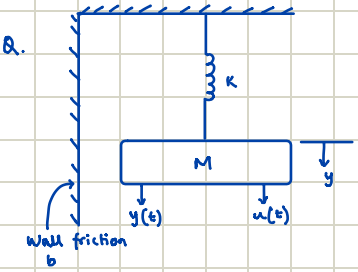
$$y = \begin{bmatrix} 0 & R & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ V_c \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} U$$

$$y = CX + DU$$

## Mechanical Systems

→ Displacement & Angular Velocity are state variables

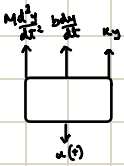
Q.




A. state variables, $x_1 = y$ (displacement)

$x_2 = \frac{dy}{dt} = \dot{y}$ (velocity)

FBD,




$\Rightarrow u(t) = M\frac{d^2y}{dt^2} + b\frac{dy}{dt} + Ky$

$u(t) = M\frac{dx_2}{dt} + bx_2 + Kx_1 \longrightarrow (1)$

From (1), $\frac{dx_2}{dt} = \frac{1}{M}\left(u(t) - bx_2 - Kx_1\right)$

and, $\frac{dx_1}{dt} = x_2$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{-K}{M} & \frac{-b}{M} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{M} \end{bmatrix} u$$

$\dot{x} = Ax + Bu$

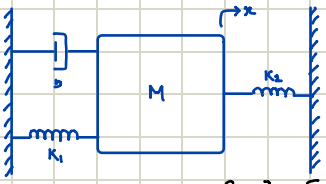
Output = $y(t)$

So,

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

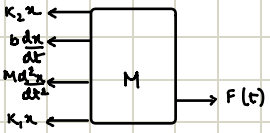
$y = Cx + Du$

Q.




A. State Variables $\Rightarrow \begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

FBD,




$$\Rightarrow F(t) = M\frac{d^2x}{dt^2} + b\frac{dx}{dt} + (K_1+K_2)x$$

$$F(t) = M\frac{dx_2}{dt} + bx_2 + (K_1+K_2)x_1 \rightarrow (i)$$

From (i), $$\frac{dx_2}{dt} = \frac{1}{M}\left(F(t) - bx_2 - (K_1+K_2)x_1\right)$$

$$\frac{dx_1}{dt} = x_2$$

So, $$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{-K_1-K_2}{M} & \frac{-b}{M} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{M} \end{bmatrix} F$$

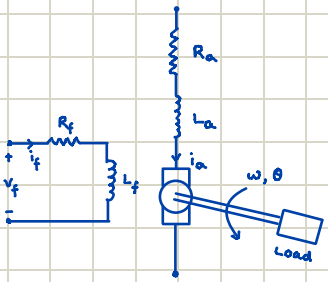
$$\dot{X} = A\,x + B\,u$$

and $$x(t) = x_1$$

So, $$y = \begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

$$y = C\,x + D\,u$$

Q. Field Controlled DC Motor




A.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} i_f \\ \theta \\ \frac{d\theta}{dt} \end{bmatrix} \qquad u = e_f \qquad y = \theta$$

$$e_f = i_f R_f + L_f \frac{di_f}{dt}$$

$$u = x_1 R_f + L_f \frac{dx_1}{dt} \Rightarrow \frac{dx_1}{dt} = \frac{u}{L_f} - \frac{R_f x_1}{L_f}$$

$$\frac{dx_2}{dt} = \frac{d\theta}{dt} = x_3$$

$$T = J\frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt}$$

$$K_T i_f = J\frac{dx_3}{dt} + Bx_3$$

$$\frac{dx_3}{dt} = \frac{K_T}{J}x_1 - \frac{B}{J}x_3$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -\frac{R}{L_f} & 0 & 0 \\ 0 & 0 & 1 \\ \frac{K_T}{J} & 0 & -\frac{B}{J} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1/L_f \\ 0 \\ 0 \end{bmatrix} v$$

$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix}$$

**Q. Armature Controlled DC Motor**

**A.** $x_1 = i_a$ $\quad u = V_a$ or $e_a$

$x_2 = \theta$ $\quad y = \theta$

$x_3 = \frac{d\theta}{dt}$

$$V_b = K_b \frac{d\theta}{dt}$$

$$V_a = i_a R_a + L_a \frac{di_a}{dt} + V_b$$

$$u = x_1 R_a + L_a \frac{dx_1}{dt} + K_b \frac{d\theta}{dt}$$

$$u = x_1 R_a + L_a \frac{dx_1}{dt} + K_b x_3 \Rightarrow \frac{dx_1}{dt} = \frac{1}{L_a}\left(u - x_1 R_a - K_b x_3\right) \longrightarrow (1)$$

$$\frac{dx_2}{dt} = \frac{d\theta}{dt} = x_3 \longrightarrow (2)$$

$$T = J\frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt}$$

$$K_T i_a = J\frac{dx_3}{dt} + B x_3$$

$$K_T x_1 = J\frac{dx_3}{dt} + B x_3 \Rightarrow \frac{dx_3}{dt} = \frac{K_T x_1}{J} - \frac{B}{J} x_3 \longrightarrow (3)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -\frac{R_a}{L_a} & 0 & -\frac{K_b}{L_a} \\ 0 & 0 & 1 \\ \frac{K_T}{J} & 0 & -\frac{B}{J} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} \frac{1}{L_a} \\ 0 \\ 0 \end{bmatrix} u$$

$$\dot{x} = A x + B u$$

$$y = \theta$$

$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

## Transfer Function from State equation

→ Consider SISO system

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

Take Laplace,

$$sX(s) = AX(s) + BU(s)$$

$$Y(s) = CX(s) + DU(s)$$

⇒ Initial conditions are zero, rearranging we get

$$(sI - A)X(s) = BU(s)$$

$$X(s) = \phi(s)\,BU(s) \qquad \left(\phi(s) = \frac{1}{sI - A}\right)$$

Then, $Y(s) = C\phi(s)BU(s) + DU(s)$

$$T(s) = \frac{Y(s)}{U(s)} = C\phi(s)B + D$$

Q. $$\dot{x} = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ -1 \end{bmatrix} u$$

$$y = \begin{bmatrix} -4 & 0 \end{bmatrix} x$$

A. $$T(s) = C(sI - A)^{-1}B + D$$

$$= C(sI - A)^{-1}B$$

$$sI = s\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 2 \\ -1 & s+3 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s(s+3)+2}\begin{bmatrix} s+3 & -2 \\ 1 & s \end{bmatrix} = \frac{1}{(s+1)(s+2)}\begin{bmatrix} s+3 & -2 \\ 1 & s \end{bmatrix}$$

$$C(sI - A)^{-1} = \begin{bmatrix} -4 & 0 \end{bmatrix}\frac{1}{(s+1)(s+2)}\begin{bmatrix} s+3 & -2 \\ 1 & s \end{bmatrix}$$

$$= \frac{1}{(s+1)(s+2)}\begin{bmatrix} -4(s+3) & 8 \end{bmatrix}$$

$$C(sI - A)^{-1}B = \frac{1}{(s+1)(s+2)}\begin{bmatrix} -4(s+3) & 8 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \frac{-4s - 20}{(s+1)(s+2)} = \frac{-4(s+5)}{(s+1)(s+2)}$$

→ Let $x_1 = Z(s)$

and $x_2 = sZ(s) = \dot{x}_1$

$x_3 = s^2 Z(s) = \dot{x}_2$

$x_4 = s^3 Z(s) = \dot{x}_3$

$Y(s) = b_3 s^3 Z(s) + b_2 s^2 Z(s) + b_1 s Z(s) + b_0 Z(s)$

$y(t) = b_3 x_4 + b_2 x_3 + b_1 x_2 + b_0 x_1$

$U(s) = s^4 Z(s) + a_3 s^3 Z(s) + a_2 s^2 Z(s) + a_1 s Z(s) + a_0 Z(s)$

$u(t) = \dot{x}_4 + a_3 x_4 + a_2 x_3 + a_1 x_2 + a_0 x_1$

$\dot{x}_4 = -a_3 x_4 - a_2 x_3 - a_1 x_2 - a_0 x_1 + u(t)$

$\dot{x} = Ax + Bu$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 & -a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u$$

$Y = CX + DU$

$y(t) = b_0 x_1 + b_1 x_2 + b_2 x_3 + b_3 x_4$

$$y = \begin{bmatrix} b_0 & b_1 & b_2 & b_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Q. $T(s) = \dfrac{12s^3 + 36s - 21}{3s^5 + 9s^3 - 15s^2 + 24s + 18}$

A. $T(s) = \dfrac{4s^3 + 12s - 7}{s^5 + 3s^3 - 5s^2 + 8s + 6}$

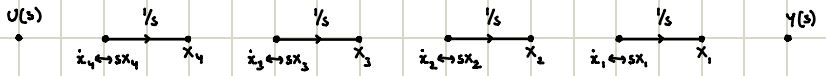
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -6 & -8 & 5 & -3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u$$

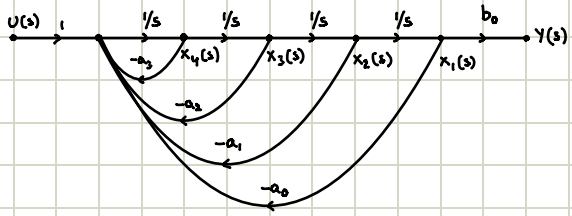
$$y = \begin{bmatrix} -7 & 12 & 0 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

# Signal Flow Graph State Models

→ $G(s) = \frac{Y(s)}{U(s)} = \frac{b_0}{s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0}$

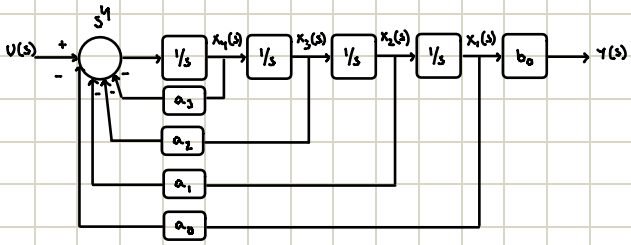
$= \frac{b_0 s^{-4}}{1 + a_3 s^{-1} + a_2 s^{-2} + a_1 s^{-3} + a_0 s^{-4}}$



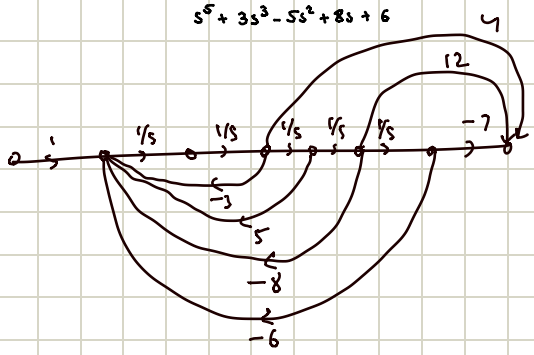




$x_4(t) = -a_0 x_1(t) - a_1 x_2(t) - a_2 x_3(t) - a_3 x_4(t) + u(t)$

$y(t) = b_0 x_1(t)$




$$\frac{4s^3 + 12s - 7}{s^5 + 3s^3 - 5s^2 + 8s + 6}$$



## Solution of State Equation

→ $\dot{X} = AX + BU$

when $U = 0$, $\dot{X} = AX$

$sX(s) - X(0) = AX(s)$

$X(s)(sI - A) = X(0)$

$X(s) = (sI - A)^{-1} x(0)$

$x(t) = \phi(t)\, x(0)$

Q. Find the solution of given state equation

$$\dot{x} = \begin{bmatrix} 0 & 2 \\ -1 & -3 \end{bmatrix} x \qquad x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

A. $$[sI - A]^{-1} = \frac{\begin{bmatrix} s+3 & -2 \\ 1 & s \end{bmatrix}}{(s+1)(s+2)} = \begin{bmatrix} \frac{s+3}{(s+1)(s+2)} & \frac{-2}{(s+1)(s+2)} \\ \frac{1}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix} = \begin{bmatrix} \frac{2}{s+1} - \frac{1}{s+2} & \frac{-2}{s+1} + \frac{2}{s+2} \\ \frac{1}{s+1} - \frac{1}{s+2} & \frac{-1}{s+1} + \frac{2}{s+2} \end{bmatrix}$$

$$\phi(t) = \mathcal{L}^{-1}[sI - A]^{-1} = \begin{bmatrix} 2e^{-t} - e^{-2t} & -2e^{-t} + 2e^{-2t} \\ e^{-t} - e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

$$x(t) = \phi(t)\, x(0) = \begin{bmatrix} 2e^{-t} - e^{-2t} & -2e^{-t} + 2e^{-2t} \\ e^{-t} - e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

# Transfer function from State Space Model.

We know the state space model of a LTI system is.

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

Apply Laplace transform on both sides of the State equation.

$$S \cdot X(s) = A\,X(s) + B\,U(s)$$

$$\Rightarrow \boxed{(sI - A)\,X(s) = B\,U(s)} \rightarrow \text{Multiply both sides by } (sI-A)^{-1}$$

$$\Rightarrow X(s) = (sI - A)^{-1} B\,U(s)$$

Apply Laplace transform on both sides of output equation

$$Y(s) = C \cdot X(s) + D \cdot U(s).$$

Substitute, $X(s)$ value in the above equation.

$$\Rightarrow Y(s) = C(sI - A)^{-1} B\,U(s) + D \cdot U(s).$$

$$\Rightarrow Y(s) = [C(sI - A)^{-1}B + D]\,U(s)$$

$$\Rightarrow \boxed{\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D}$$

The above equation represents the transfer function of the System. So, we can calculate the transfer function of the system by using this formula for the System represented in the State Space model.

Note: when $D = [0]$, the transfer function will be

$$\boxed{\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B}$$

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**Example:**

Let us calculate the transfer function of the system, represented in the state space model as.

$$\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} [u]$$

$$Y = [0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Sol$^n$ Here $A = \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $C = [0 \quad 1]$ & $D = [0]$.

The formula for the transfer function when $D = [0]$ is

$$\frac{Y(s)}{U(s)} = C(SI - A)^{-1} B.$$

Substitute A, B & C matrices in the above equation.

$$\Rightarrow \frac{Y(s)}{U(s)} = [0 \quad 1] \left[ \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix} \right]^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Note: Matrix inv = $\frac{\text{adj(matrix)}}{\text{determinant}}$

$$\Rightarrow \frac{Y(s)}{U(s)} = [0 \quad 1] \begin{bmatrix} s+1 & 1 \\ -1 & s \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \frac{Y(s)}{U(s)} = [0 \quad 1] \frac{\begin{bmatrix} s & -1 \\ 1 & s+1 \end{bmatrix}}{(s+1)s - (1)(-1)} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \frac{Y(s)}{U(s)} = [0 \quad 1] \frac{\begin{bmatrix} s-0 \\ 1+0(s+1) \end{bmatrix}}{s^2+s+1} = \frac{0 \times s + 1 \times 1}{s^2+s+1}$$

$$\Rightarrow \boxed{\frac{Y(s)}{U(s)} = \frac{1}{s^2+s+1}}$$ → transfer function of system for the given state space model

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## Solutions of Differential Equation.

How to obtain the solution of Equation $\dot{x} = A \cdot x$ ?

$$\dot{X} = A \cdot x.$$

Premultiply by $e^{-At}$

$$e^{-At}\dot{X} = A \cdot x \cdot e^{-At}$$

$$\Rightarrow e^{-At}\dot{x} - e^{-At}A \cdot x = 0 \qquad \left[\text{Note } \frac{d(u.v)}{dt}\right] = u \cdot \frac{dv}{dt} - v \cdot \frac{du}{dt}$$

$$\Rightarrow \frac{d}{dt}(e^{-At} \cdot x) = 0$$

$\Rightarrow$ integrating from $t = t_0$ to $t$

$$\int_{t_0}^{t} \frac{d}{dt}(e^{-A\tau} \cdot x(\tau)) \cdot d\tau = e^{-A\tau} \cdot x(\tau) \Big]_{t_0}^{t}$$

$$= e^{-At} \cdot x(t) - e^{-At_0} x(t_0) = 0$$

Multiplying through out by $e^{At}$

$$\therefore x(t) = e^{-At_0} \cdot e^{At} \cdot x(t_0)$$

$$\& \quad \boxed{x(t) = e^{A(t-t_0)} \cdot x(t_0)}$$

if $t_0 = 0$ then

$$\boxed{x(t) = e^{At} \cdot x(0)}$$

---

What is the solution of $\dot{X} = Ax + Bu$ ?

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$$\dot{x} = Ax + Bu$$

Multiply be $e^{-At}$.

$$e^{-At}\dot{x} = A \cdot e^{-At} \cdot x + e^{-At} \cdot B \cdot u$$

$$\left(e^{-At}\dot{x} - e^{-At} A \cdot x\right) = e^{-At} \cdot B \cdot u$$

$$\frac{d}{dt}\left(e^{-At} \cdot x\right) = e^{-At} \cdot B \cdot u$$

integrating from $t = t_0$ to $t$.

$$\int_{t_0}^{t} \frac{d}{dt}\left(e^{-A\tau} \cdot x(\tau)\right) = \int_{t_0}^{t} e^{-A\tau} B u(\tau) \cdot d\tau$$

$$e^{-At} \cdot x(t) - e^{-At_0} x(t_0) = B \int_{t_0}^{t} e^{-A\tau} \cdot u(\tau) \cdot d\tau$$

$$e^{-At} \cdot x(t) = e^{-At_0} x(t_0) + B \cdot \int_{t_0}^{t} e^{-A\tau} \cdot u(\tau) \cdot d\tau$$

Multiplying by $e^{At}$.

$$x(t) = e^{A \cdot t} \cdot e^{-At_0} \cdot x(t_0) + B \cdot e^{A \cdot t} \int_{t_0}^{t} e^{-A\tau} u(\tau)\, d\tau$$

$$\boxed{x(t) = e^{A(t-t_0)} x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)} \cdot B \cdot u(\tau) \cdot d\tau}$$

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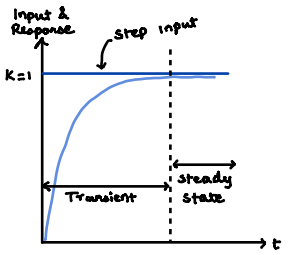
# Unit-2 Performance of Feedback Control System




## Time Response of Control System

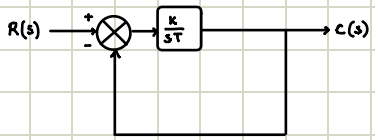
→ We have 2 types :

**Transient Response** - When the response of the system is changed from equilibrium, it takes some time to settle down.

**Steady-state Response** - The response of the system after transient response

## Time Response of First Order Control System

→




$$\frac{C(s)}{R(s)} = \frac{\frac{K}{ST}}{1+\frac{K}{ST}} = \frac{K}{ST+K}$$

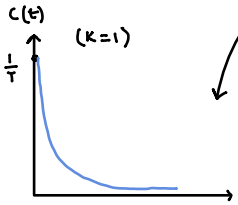
## Impulse Response of first order system

$$\frac{C(s)}{R(s)} = \frac{K}{ST+1}$$

If $r(t) = \delta(t) \Rightarrow R(s) = 1$, $C(s) = \frac{K}{ST+1} = \frac{K/T}{s+\frac{1}{T}}$

$$c(t) = \mathcal{L}^{-1}\left\{\frac{K/T}{s+\frac{1}{T}}\right\} = \frac{K}{T}e^{-t/T}$$

If $K=1$, it is called unit impulse response

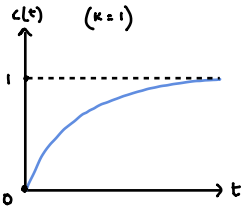



exponentially decreasing signal

If $r(t) = u(t)$, $R(s) = 1/s$, $C(s) = \frac{K}{1+ST}\cdot\frac{1}{S} = \frac{A}{s}+\frac{B}{1+ST}$

$$= \frac{K}{S} - \frac{KT}{1+ST} = K\left[\frac{1}{S} - \frac{1}{s+\frac{1}{T}}\right]$$

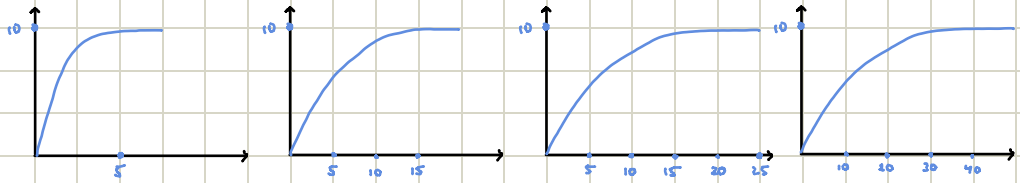
$$c(t) = K\left[u(t) - e^{-t/T}\right] = K\left[1-e^{-t/T}\right]$$




exponentially increasing signal

Q. Plot for $r(t) = u(t)$ & $c(t) = K(1-e^{-t/T})$ when $K = 10$, $T = 1, 3, 5, 7$

A.




Q. Plot the step response of first order system with $K = 1, 3, 5, 7$ & $T = 1$

## Relation b/w Impulse & Step Response

→ $\delta(t) = \begin{cases} 1, & t = 0 \\ 0, & \text{otherwise} \end{cases}$

$$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$$

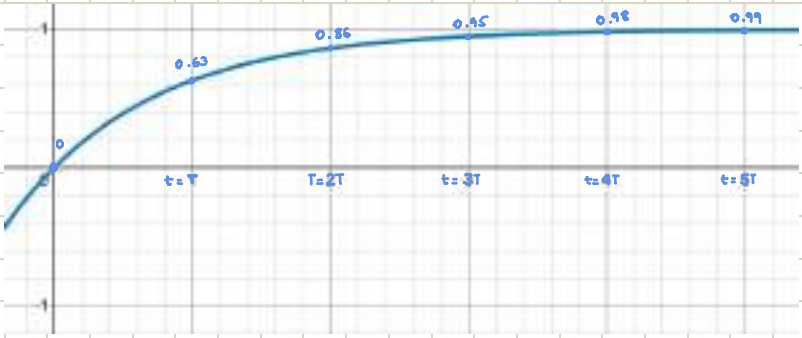
→ $$\int_{-\infty}^{t} \delta(t)\,dt = u(t)$$

$$\frac{du(t)}{dt} = \delta(t)$$

→ $$c(t) = K\left[1 - e^{-t/T}\right]$$

$$\frac{dc(t)}{dt} = \frac{Ke^{-t/T}}{T}$$

## Performance specifications of first order system (Step Response)




→ **Rise Time** $T_r$ – time taken by response to rise from 10% to 90% of its final value

ex – $c(t) = 1 - e^{-t/T}$

$0.1 = 1 - e^{-t_{r_1}/T} \Rightarrow t_{r_1} = 0.105T$

$0.9 = 1 - e^{-t_{r_2}/T} \Rightarrow t_{r_2} = 2.302T$

$T_r = t_{r_2} - t_{r_1} = (2.302 - 0.105)T = 2.197T \simeq 2.2T$

→ **Settling Time** $T_s$ – time required for system response to remain within a certain percent (2-5%) of its final value

ex – $c(t) = 1 - e^{-t/T}$

$0.98 = 1 - e^{-t/T} \Rightarrow T = 4T$

Q. Given impulse response of first order system $c(t) = 3e^{-0.5t}$. Find the time constant, DC Gain, Transfer function, Step Response

A. $c(t) = 3e^{-0.5t}$

$$C(s) = \frac{3}{s+0.5} = \frac{6}{2s+1} = \frac{K}{Ts+1}$$

i) $T = 2$

ii) $K = 6$

iii) $C(s) = \frac{6}{2s+1}$

iv) $\int 3e^{-0.5t} = \frac{3e^{-0.5t}}{-0.5} + C = -6e^{-0.5t} + C$

## Ramp Response of first order response

→ $r(t) = At$

$$R(s) = \frac{A}{s^2} \qquad C(s) = \frac{K}{ts+1} \cdot \frac{A}{s^2}$$

Let $A = 1$, $C(s) = \frac{K}{s^2(Ts+1)} = \frac{a}{s} + \frac{b}{s^2} + \frac{c}{Ts+1}$

$$\Rightarrow a = \frac{d}{ds}\left[\frac{K}{Ts+1}\right]_{s=0} = -K(Ts+1)^{-2}(T)\Big|_{s=0} = -KT$$

$$b = \left(\frac{K}{Ts+1}\right)_{s=0} = K$$

$$c = \frac{K}{s^2}\Big|_{s=-\frac{1}{T}} = KT^2$$

$$C(s) = \frac{K}{s^2} - \frac{KT}{s} + \frac{KT^2}{Ts+1}$$

$$c(t) = K\left[t - T + Te^{-t/T}\right]$$

Q. $r(t) = t^2/2$

A. $R(s) = \frac{1}{s^3}$

$$C(s) = \frac{1}{Ts+1} \cdot \frac{1}{s^3} \quad (\text{Take } K = 1)$$

$$= \frac{A}{Ts+1} + \frac{B}{s} + \frac{C}{s^2} + \frac{D}{s^3}$$

$$= \frac{-T^3}{Ts+1} + \frac{T^2}{s} - \frac{T}{s^2} + \frac{1}{s^3}$$

$\frac{-T^2}{s+1/T}$

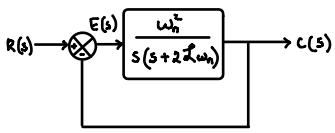
$$c(t) = -T^2e^{-t/T} + T^2 - Tt + \frac{t^2}{2}$$

$$A = \left(\frac{1}{s^3}\right)_{s=-\frac{1}{T}} = -T^3$$

$$B = \frac{d^2}{ds^2}\left(\frac{1}{Ts+1}\right)\Big|_{s=0} = \frac{1}{2!}\frac{d}{ds}(Ts+1)^{-2}(T)\Big|_{s=0} = \frac{2}{2}(Ts+1)^{-3}(T^2)\Big|_{s=0} = T^2$$

$$C = \frac{d}{ds}\left(\frac{1}{Ts+1}\right)\Big|_{s=0} = -(Ts+1)^{-2}(T)\Big|_{s=0} = -T$$

$$D = \frac{1}{Ts+1}\Big|_{s=0} = 1$$




## Second Order Control System

→ A second order system is characterized by the T.F

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} \qquad \frac{\omega_n^2}{s^2+2s\zeta\omega_n+\omega_n^2}$$

$\omega_n$ → undamped natural frequency : frequency of oscillation of system w/o damping

$\zeta$ → damping ratio : Measure of degree of resistance to change in system output

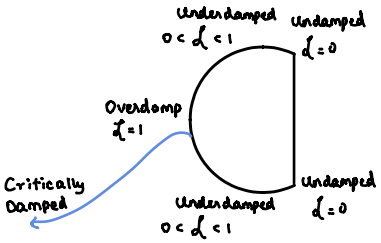
ex : RLC Circuit

→ Characteristics eqn : $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$

$as^2 + bs + c = 0 \Rightarrow \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$$\frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2}$$

$$= \frac{-2\zeta\omega_n \pm 2\omega_n\sqrt{\zeta^2-1}}{2} = \omega_n\left(-\zeta \pm \sqrt{\zeta^2-1}\right) \longrightarrow \text{Poles of T.F}$$




Case(i) $(\zeta > 1) \Rightarrow$ Overdamped (System has 2 real distinct poles)

Case (ii) $(0<\zeta<1) \Rightarrow$ Underdamped (System has 2 complex conjugate poles)

Case (iii) $(\zeta = 0) \Rightarrow$ Undamped (System has 2 imaginary poles)

Case (iv) $(\zeta = 1) \Rightarrow$ Critically Damped (System has 2 real but equal poles)

2 poles are : $\omega_n\left(-\zeta + \sqrt{\zeta^2-1}\right)$

$\omega_n\left(-\zeta - \sqrt{\zeta^2-1}\right)$

→ Step Response of 2nd order system

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} \qquad R(s) = \frac{1}{s}$$

$$C(s) = \frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)} = \frac{A}{s} + \frac{Bs+C}{s^2+2\zeta\omega_n s+\omega_n^2} = \frac{A(s^2+2\zeta\omega_n s+\omega_n^2) + (Bs+C)(s)}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$$

$$C(s) = \frac{1}{s} + \frac{-s-2\zeta\omega_n}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$A = 1 \Rightarrow A = 1$

$A + B = 0 \Rightarrow B = -1$

$2\zeta\omega_n A + C = 0 \Rightarrow C = -2\zeta\omega_n$

$$s^2 + 2\zeta\omega_n + \omega_n^2 = s^2 + 2\zeta\omega_n + \omega_n^2 + \zeta^2\omega_n^2 - \zeta^2\omega_n^2$$

$$= (s+\zeta\omega_n)^2 + \omega_n^2 - \zeta^2\omega_n^2$$

$$= (s+\zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)$$

$$= (s+\zeta\omega_n)^2 + \omega_d^2$$

$$C(s) = \frac{1}{s} - \left[\frac{s+2\zeta\omega_n}{(s+\zeta\omega)^2+\omega_d^2}\right] = \frac{1}{s} - \left[\frac{s+\zeta\omega_n}{(s+\zeta\omega)^2+\omega_d^2} + \frac{\zeta\omega_n}{(s+\zeta\omega)^2+\omega_d^2}\right]$$

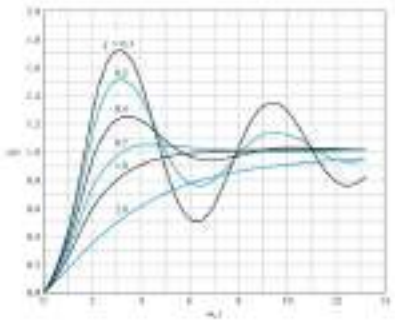
$$L^{-1}[C(s)] = c(t) = 1 - e^{-\zeta\omega_n t}\cos\omega_d t - \frac{\zeta\omega_n}{\omega_n\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\sin\omega_d t = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left(\sqrt{1-\zeta^2}\cos\omega_d t + \sin\omega_d t\right)$$

$\omega_d = \sqrt{\omega_n(1-\zeta^2)}$

$\omega_d$ : damping frequency

$L[e^{at}\cos Kt] = \frac{s-a}{(s-a)^2+K^2}$

$L[e^{at}\sin Kt] = \frac{K}{(s-a)^2+K^2}$



→ Let $\sin\theta = \sqrt{1-\zeta^2}$ & $\zeta = \cos\theta$

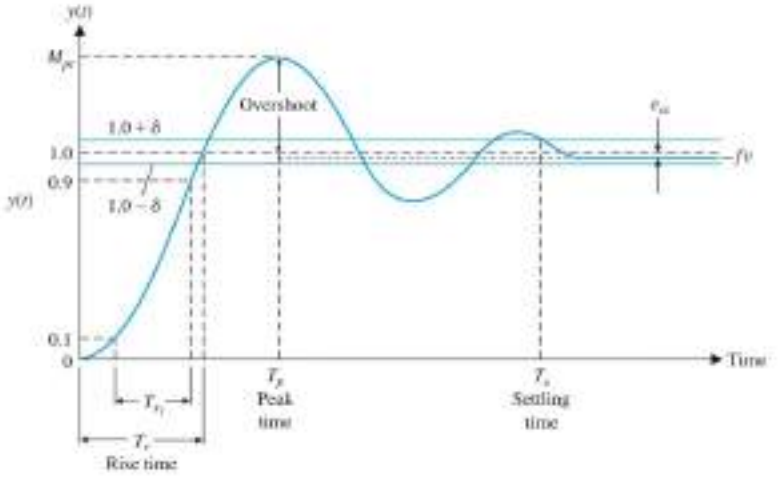
$\sin^2\theta + \cos^2\theta = 1$

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left[\sqrt{1-\zeta^2}\cos\omega_n t + \zeta\sin\omega_n t\right]$$

$$= 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left[\sin\theta\cos\omega_n t + \cos\theta\sin\omega_n t\right]$$

$$= 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_n t + \theta)$$

## Standard Performance parameters of second order system (underdamped)




**Delay Time ($t_d$) :** Time required to reach 50% of final value

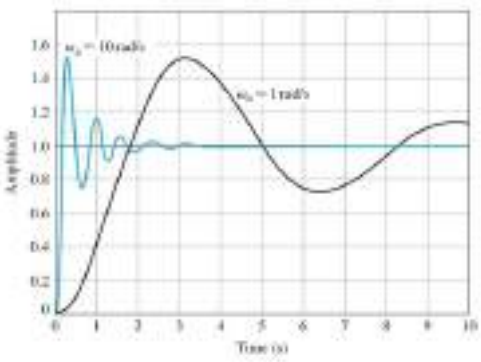
**Rise Time ($t_r$) :** Time required for response to rise from 10% to 90% of final value for overdamped systems & 0 to 100% for underdamped systems

**Peak Time ($t_p$) :** Time required for response to reach max peak for first time

**Max Overshoot ($M_p$):** Max peak value of response curve measured from unity

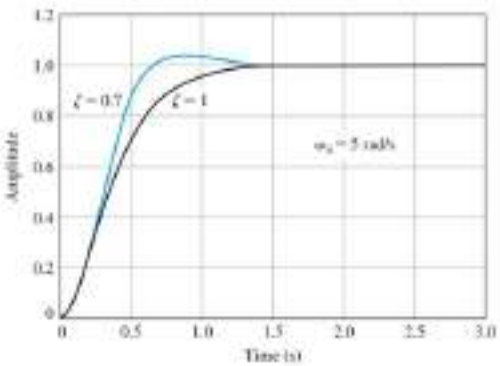
$$\%\ M_p = \frac{C(t_p) - C(\infty)}{C(\infty)} \times 100\%$$

**Settling Time ($t_s$):** Time required for response curve to reach & stay within a range about final value size specified by absolute % of the final value




$\omega_n$ : varying

$\zeta$ : constant




$\omega_n$ : constant

$\zeta$ : varying

**Q. Derive an equation for rise time**

A. $c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta)$

$c(t_r) = 1 - \frac{e^{-\zeta\omega_n t_r}}{\sqrt{1-\zeta^2}} \sin(\omega_d t_r + \theta) = 1$

$\Rightarrow \frac{e^{-\zeta\omega_n t_r}}{\sqrt{1-\zeta^2}} \sin(\omega_d t_r + \theta) = 0$

when underdamped, $0 < \zeta < 1$ .

so $\sqrt{1-\zeta^2} \neq 0$

then $\sin(\omega_d t_r + \theta) = 0$

$\omega_d t_r + \theta = n\pi \Rightarrow t_r = \frac{n\pi - \theta}{\omega_d}$

**Q. Derive an equation for peak time**

A. $c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta)$

at $t = t_p$, $\frac{d\,c(t)}{dt} = 0$

$\frac{d\,c(t)}{dt} = \frac{\zeta\omega_n e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta) - \frac{\omega_d e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \cos(\omega_d t + \theta)$

$= \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \left( \zeta\omega_n \sin(\omega_d t + \theta) - \omega_d \cos(\omega_d t + \theta) \right)$

Since under damped, $0 < \zeta < 1$

at $\frac{d\,c(t)}{dt} = 0$, $\left( \zeta\omega_n \sin(\omega_d t + \theta) - \omega_d \cos(\omega_d t + \theta) \right)$ must be 0

as $\sqrt{1-\zeta^2}$ can never be 0

$\Rightarrow \zeta\omega_n \sin(\omega_d t_p + \theta) = \omega_d \cos(\omega_d t + \theta)$

$\zeta\omega_n \sin(\omega_d t_p + \theta) = \omega_n \sqrt{1-\zeta^2} \cos(\omega_d t + \theta)$

$\sin(\theta - (\omega_d t_p + \theta)) = 0$

$\sin(-\omega_d t_p) = 0$

$\omega_d t_p = n\pi$

$t_p = \frac{n\pi}{\omega_d}$

**Q. Derive an equation for max overshoot**

A. $M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100$

$= c(t_p) - 1$

$= 1 - \frac{e^{-\zeta\omega_n t_p}}{\sqrt{1-\zeta^2}} \left( \sin(\omega_d t_p + \theta) \right) - 1$

$= \frac{-e^{-\zeta\omega_n \frac{\pi}{\omega_d}}}{\sqrt{1-\zeta^2}} \left( \sin\left(\omega_d \times \frac{\pi}{\omega_d}\right) + \theta \right)$

$= \frac{-e^{-\zeta\omega_n \frac{\pi}{\omega_d}}}{\sin\theta} (-\sin\theta)$

$= -e^{\frac{-\zeta\pi}{\sqrt{1+\zeta^2}}}$

**Q. Derive an equation for settling time ($t_s$)**

A. Let $c(t_s) = 0.98$ of the final value

$0.98 = 1 - e^{-\zeta\omega_n t_s}$

$e^{-\zeta\omega_n t_s} = 0.02$

$\zeta\omega_n t_s = 3.91$

$t_s = \frac{3.91}{\zeta\omega_n} = 3.91\tau \approx 4\tau \quad \left(\tau = \frac{1}{\zeta\omega_n} : \text{Time Constant}\right)$

$t_r = \frac{\pi - \theta}{\omega_d}$

$t_p = \frac{\pi}{\omega_d}$

$M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$

$t_s = \frac{4}{\zeta\omega_n} = 4\tau$

$\omega_d = \omega_n \sqrt{1-\zeta^2}$

Q. For a unity feedback system, having open loop Transfer func$^n$ $G(s) = \frac{1}{s(s+1)}$

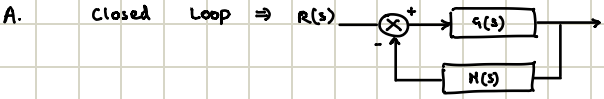
↳ $H(s) = 1$

Determine a) damping ratio

b) damp frequency oscillation

c) peak time

d) peak overshoot

A. Closed Loop ⇒ 

$\Rightarrow \frac{C(s)}{R(s)} = \frac{G(s)}{1 + H(s)G(s)}$

Then T.F of the function is $= \frac{\frac{1}{s(s+1)}}{1 + \frac{1}{s(s+1)}} = \frac{\frac{1}{s(s+1)}}{\frac{s(s+1)+1}{s(s+1)}} = \frac{1}{s(s+1)+1} = \frac{1}{s^2+s+1}$

Comparing with $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$\omega_n = 1$ & $\zeta = \frac{1}{2}$

a) $\zeta = 0.5$

b) $\omega_d = \omega_n\sqrt{1-\zeta_n^2} = \sqrt{1-0.25} = \frac{\sqrt{3}}{2} = 0.866$ rad/s

c) $t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\sqrt{3}/2} = 3.627$ s

d) $M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} = 0.163$

Q. $G(s) = \frac{4}{s+2}$ & $H(s) = \frac{1}{s}$ . Find $\zeta$, $\omega_d$, $t_p$

A. TF $= \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{4}{s+2}}{1 + \frac{4}{(s+2)s}} = \frac{\frac{4}{s+2}}{\frac{s^2+2s+4}{(s+2)s}} = \frac{4s}{s^2+2s+4}$

Comparing with $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

Neglect the zeros, $\omega_n = 2$ $\zeta = 1/2$

$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2\sqrt{1-0.25}$

$= 2\sqrt{0.75} = 1.732$ rad/s

$t_p = \frac{\pi}{\omega_d} = 1.187$ sec

$t_s = \frac{4}{\zeta\omega_n} = \frac{4}{\frac{1}{2} \times 2} = 4$ s

Q. The closed loop poles of a system $-2 \pm j3$. Calculate $\zeta$, $\omega_d$, $M_p$

A. $TF = \dfrac{G(s)}{(s+2-j3)(s+2+j3)} = \dfrac{G(s)}{s^2+4s+13}$

Comparing with $\dfrac{\omega_n^2}{s^2+2\zeta\omega_n+\omega_n^2}$

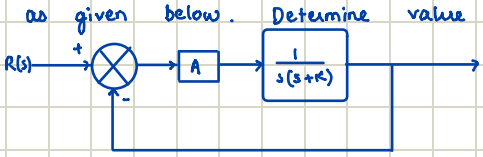
$\omega_n = \sqrt{13} = 3.605$ rad/s & $\zeta = \dfrac{2}{\sqrt{13}} = 0.554$

$\omega_d = \omega_n\sqrt{1-\zeta^2}$

$= \sqrt{13} \times \sqrt{1-\dfrac{4}{13}} = 3$ rad/s

$M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} = e^{\frac{-2/\sqrt{13}\cdot\pi}{3/\sqrt{13}}} = e^{-2\pi/3} = 0.123$

Q. A Step input of magnitude 2 is applied to a unity feedback system as given below. Determine value of A & K such that $\zeta = 0.6$ and $\omega_d = 8$ rad/s




A. $\dfrac{R(s)}{C(s)} = \dfrac{\frac{A}{s(s+K)}}{1+\frac{A}{s(s+K)}} = \dfrac{A}{s(s+K)+A} = \dfrac{A}{s^2+Ks+A}$

$A = \omega_n^2$ and $K = 2\zeta\omega_n$

$\omega_d = \omega_n\sqrt{1-\zeta^2}$

$8 = \omega_n\sqrt{1-0.6^2} \Rightarrow \omega_n = 10$ rad/s

$A = (10)^2 = 100$

$K = 2\times0.6\times10 = 12$

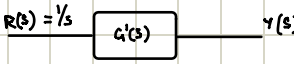
$M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} = e^{\frac{-0.6\pi}{\sqrt{1-0.6^2}}} = 0.0948$

## Effect of adding poles & zeros in second order system

→ $G(s) = \dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

adding a pole,

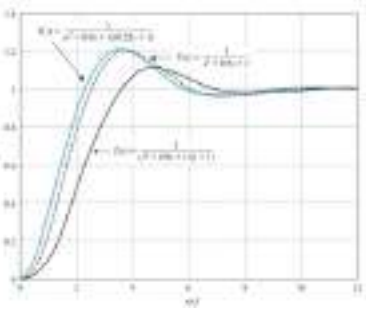
$G'(s) = \dfrac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)(s+a)}$




$Y(s) = R(s)\, G'(s)$

$= \frac{1}{s} \cdot \dfrac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)(s+a)}$ → Inverse Laplace

$y(t) = Ae^{-at} + B.1 + \underbrace{C.e^{-\zeta\omega_n t}\cos\omega_d t + De^{-\zeta\omega_n t}\sin\omega_d t}_{y(t)}$



→ When poles are added to existing second order system, the added poles completely determine the response of the system when it is closer to origin

→ But, when poles are away from origin, lesser the effect of additional poles.

Let s be system response of second order system & $s_1, s_2, s_3$ etc., be different systems with additional pole system

→ Additional poles make the system slower when they move towards the origin

→ Overshoot increases as they move away from the origin

→ Upon adding zeros

$G'(s) = \dfrac{\omega_n^2 (s+a)}{(s^2 + 2\zeta\omega_n s + \omega_n^2)}$ → 2 poles + 1 additional zero

applying inverse Laplace,

$g'(t) = \dot{g}(t) + g(t)$

↓

Response of new system

If a is very small (zeros closer to origin), $\dot{y}(t)$ term dominates (affects 2nd order response)

If a is very big, $\dot{y}(t)$ decreases & $\dot{y}(t) \simeq y(t)$

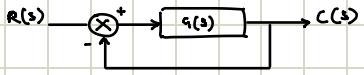
As the zero moves far from origin to $-\infty$ on splane, effect of adding zeros is negligible

When zeros lie on right hand side of s-plane, response starts from -ve value (undershoot)

# Steady State Errors and Error Constants

→ SSE is a measure of system accuracy

→ Consider unity feed back system,




$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} \longrightarrow (1)$$

$$C(s) = E(s)\,G(s) \longrightarrow (2)$$

↑ Error

Comparing (1) & (2), $E(s) = \frac{R(s)}{1+G(s)}$

→ Steady state error $(e_{ss}) = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s)$

$$\boxed{e_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}}$$

→ Steady state error for standard signals

i) Unit Step input

$r(t) = 1$

$R(s) = \frac{1}{s}$

$$e_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)} = \lim_{s\to 0} \frac{s\cdot 1/s}{1+G(s)} = \lim_{s\to 0} \frac{1}{1+G(s)}$$

Let $K_p = \lim_{s\to 0} G(s)$ (Possesion Error Constant)

Then $e_{ss} = \frac{1}{1+K_p}$

ii) Ramp input

$r(t) = s$

$R(s) = \frac{1}{s^2}$

$$e_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)} = \lim_{s\to 0} \frac{s\cdot 1/s^2}{1+G(s)} = \lim_{s\to 0} \frac{1}{s(1+G(s))} = \lim_{s\to 0} \frac{1}{s+sG(s)}$$

Let $\lim_{s\to 0} sG(s) = K_v$ (Velocity Error Constant)

$e_{ss} = \frac{1}{K_v}$

iii) Parabolic input

$r(t) = t^2/2$

$R(s) = \frac{1}{s^3}$

$e_{ss} = \lim_{s\to 0} \frac{s\,R(s)}{1+G(s)} = \frac{s \times 1/s^3}{1+G(s)} = \frac{1}{s^2(1+G(s))} = \frac{1}{s^2+s^2G(s)}$

Let $\lim_{s\to 0} s^2G(s) = K_a$ (Acceleration Error Constant)

$e_{ss} = \frac{1}{K_a}$

| | Input Signal | $e_{ss}$ | Error Constant |
|---|---|---|---|
| i) | Unit Step | $\frac{1}{1+K_p}$ | $K_p = \lim_{s\to 0} G(s)$ |
| ii) | Ramp Step | $\frac{1}{K_v}$ | $K_v = \lim_{s\to 0} sG(s)$ |
| iii) | Parabolic Step | $\frac{1}{K_a}$ | $K_a = \lim_{s\to 0} s^2G(s)$ |

Q. Given $r(t) = 5 + 2t + \frac{t^2}{2}$ & $G(s) = \frac{5(s+4)}{s^2(s+1)(s+20)}$. Find $K_p$, $K_v$ & $e_{ss}$

A.

| Input Signal | Error Constant | $e_{ss}$ |
|---|---|---|
| $5.u(t)$ | $K_p = \lim_{s\to 0} \frac{5(s+4)}{s^2(s+1)(s+20)} = \infty$ | $\frac{1}{1+K_p} = 0$ |
| $2t$ | $K_v = \lim_{s\to 0} \frac{5(s+4)}{s(s+1)(s+20)} = \infty$ | $\frac{1}{K_v} = 0$ |
| $t^2/2$ | $K_a = \lim_{s\to 0} \frac{5(s+4)}{(s+1)(s+20)} = 1$ | $\frac{1}{K_a} = 1$ |

## Types of Feedback C.S

→ Consider unity feedback system, with following open loop transfer function

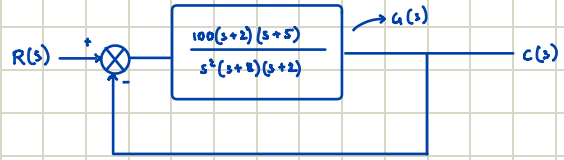
$$G(s) = \frac{K(T_as+1)(T_bs+1)\dots(T_ns+1)}{s^N(T_1s+1)(T_2s+1)\dots(T_ps+1)}$$

The term $s^N$ represents N poles at origin, depending on value of N, we have

- type 0: $N=0 \Rightarrow K_p = \lim_{s\to 0} G(s) = K$ — $e_{ss} = \frac{1}{1+K}$
- type 1: $N=1 \Rightarrow K_p = \lim_{s\to 0} G(s) = \frac{K}{5} = \infty$ — $e_{ss} = 0$
- type 2: $N=2 \Rightarrow K_p = \lim_{s\to 0} G(s) = \frac{K}{s^2} = \infty$ — $e_{ss} = 0$
- type 0: $N=0 \Rightarrow K_v = \lim_{s\to 0} sG(s) = 0$ — $e_{ss} = \infty$
- type 1: $N=1 \Rightarrow K_v = \lim_{s\to 0} sG(s) = K$ — $e_{ss} = \frac{1}{K}$
- type 2: $N=2 \Rightarrow K_v = \lim_{s\to 0} sG(s) = \frac{K}{s} = \infty$ — $e_{ss} = 0$
- type 0: $N=0 \Rightarrow K_a = \lim_{s\to 0} s^2G(s) = 0$ — $e_{ss} = \infty$
- type 1: $N=1 \Rightarrow K_a = \lim_{s\to 0} s^2G(s) = 0$ — $e_{ss} = \infty$
- type 2: $N=2 \Rightarrow K_a = \lim_{s\to 0} s^2G(s) = K$ — $e_{ss} = \frac{1}{K}$

| | Step Input | Ramp Input | Parabola |
|---|---|---|---|
| Type 0 | $\frac{1}{1+K}$ | $\infty$ | $\infty$ |
| Type 1 | 0 | $\frac{1}{K}$ | $\infty$ |
| Type 2 | 0 | 0 | $\frac{1}{K}$ |

Q. For the system shown below evaluate the static error constants & expected steady state errors for standard step, ramp & parabolic inputs




A.

$G(s)$

| | Step input | Ramp input | Parabola |
|---|---|---|---|
| Type 0 | $\frac{1}{0+3.5}$ | $\infty$ | $\infty$ |
| Type 1 | 0 | $\frac{1}{0+2.5}$ | $\infty$ |
| Type 2 | 0 | 0 | $\frac{1}{0+2.5}$ |

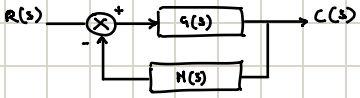
Q. $G(s) = \frac{K(s+2)}{s(s^3+7s^2+12s)}$ . Find $K_p$, $K_v$, $K_a$, $e_{ss}$

A. $G(s) = \frac{K(s+2)}{s^2(s^2+7s+12)} \Rightarrow$ Type 2

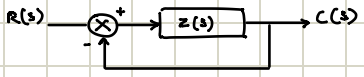
| Step input | Ramp input | Parabola |
|---|---|---|
| 0 | 0 | $\frac{6}{K}$ |

Steady State Error for non-unity feedback system

→




$$\frac{C(s)}{R(s)} = \frac{G}{1+GH}$$




$$\frac{C(s)}{R(s)} = \frac{Z}{1+Z}$$

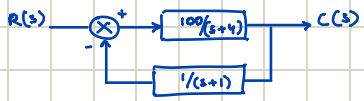
The error constants & steady state errors are defined for unity feedback system To extend the same for non-unity feedback systems, we compare with UFS

$$\frac{Z}{1+Z} \equiv \frac{G}{1+GH} \Rightarrow Z + ZGH = G + ZG \Rightarrow Z(1+GH-G) = G \Rightarrow Z = \frac{G}{1+GH-G}$$

The error constants will be

| | Step input | Ramp input | Parabola |
|---|---|---|---|
| Type 0 | $\frac{1}{1+K_p}$ | $\infty$ | $\infty$ |
| Type 1 | 0 | $\frac{1}{K}$ | $\infty$ |
| Type 2 | 0 | 0 | $\frac{1}{K}$ |

Q.




A. $H(s) = \frac{1}{s+1}$ $\quad G(s) = \frac{100}{s+4}$

$$Z = \frac{G}{1+GH-G} = \frac{\frac{100}{s+4}}{1+\frac{100}{(s+1)(s+4)}-\frac{100}{s+4}} = \frac{\frac{100}{s+4}}{\frac{(s+1)(s+4)+100-100(s+1)}{(s+1)(s+4)}} = \frac{100(s+1)}{(s+1)(s+4)-100s}$$

$K_p = \lim_{s\to 0} Z(s) = \frac{100}{4-0} = 25$ $\quad e_{ss} = \frac{1}{1+25} = \frac{1}{26}$

$K_v = \lim_{s\to 0} sZ(s) = 0$ $\quad e_{ss} = \infty$

$K_a = \lim_{s\to 0} s^2Z(s) = 0$ $\quad e_{ss} = \infty$

## Feedback Control System Characteristics

→ Feedback as a means of automatic regulation & control is inherent in nature

2 Types of system:

i) Open Loop systems

ii) Closed Loop (feedback) systems

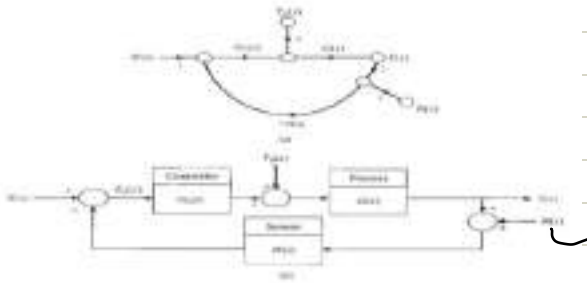
→ Open Loop Systems

→ Non-feedback control systems

→ Sensing of actual output & comparing of this output (through feedback) with desired input doesn't take place

→ Closed Loop System

→ Systems with feedback



↳ Unwanted noise signal

## Advantages of closed loop

→ Decreased sensitivity of the system to parameter variations

→ Improved rejection of the disturbances

→ Improved measurement of noise attenuation

→ Improved reduction of steady state error of system

→ Easy control & adjustment of the transient response of the system

→ In the closed loop system, Let $H(s) = 1$ & neglecting $T_d(s)$ & $N(s)$,

$E(s) = R(s) - Y(s)$

By principle of superposition, Output = $\Sigma$ each input

$$Y(s) = \frac{G(s)\,G_c(s)}{1+G(s)\,G_c(s)}\,R(s) + \frac{G(s)}{1+G_c(s)\,G(s)}\,T_d(s) - \frac{G(s)\,G_c(s)}{1+G(s)\,G_c(s)}\,N(s)$$

$$E(s) = Y(s) - R(s)$$

$$= \frac{1}{1+G(s)\,G_c(s)}\,R(s) - \frac{G(s)}{1+G_c(s)\,G(s)}\,T_d(s) + \frac{G(s)\,G_c(s)}{1+G(s)\,G_c(s)}\,N(s)$$

Let $L(s) = G_c(s)\,G(s)$ as Loop Gain

$$= \frac{1}{1+L(s)}\,R(s) - \frac{G(s)}{1+L(s)}\,T_d(s) + \frac{L(s)}{1+L(s)}\,N(s)$$

→ Define sensitivity function $S(s) = \frac{1}{1+L(s)}$

and complementary sensitivity function $C(s) = \frac{L(s)}{1+L(s)}$

Then, $E(s) = S(s)\,R(s) - S(s)\,G(s)\,T_d(s) + C(s)\,N(s)$

↳ Tracking error

→ To minimize tracking error, $S(s)$ & $C(s)$ should be small

and $S(s) + C(s) = 1$

Both $S(s)$ & $C(s)$ are both functions of controller, $G_c(s)$ which design engineer selects. and both can't be small, design compromises are made

## Design Compromises

→ To reduce $T_d(s)$ effect on $E(s)$, $L(s)$ should be large, so that $\frac{G(s)}{1+L(s)}$ should be reduced

→ To reduce $N(s)$, $L(s)$ should be decreased

→ Hence, $S(s)$ & $C(s)$ can't be small simultaneously. Hence design compromises

## Sensitivity of Control System to Parameter Variations

→ Suppose U(s) undergoes a change such that true plant model is $G(s) + \Delta G(s)$, then $E(s) \longrightarrow E(s) + \Delta E(s)$

Neglecting $T_d(s) + N(s)$,

$$E(s) + \Delta E(s) = \frac{1}{1 + G_c(s)\,(G(s) + \Delta G(s))}\, R(s)$$

$$\Delta E(s) = \frac{R(s)}{1 + G_c(s)\,(G(s) + \Delta G(s))} - E(s)$$

$$= \frac{R(s)}{1 + G_c(s)\,(G(s) + \Delta G(s))} - \frac{R(s)}{1 + G_c(s) G(s)}$$

$$= \frac{-G_c(s)\,\Delta G(s)\,R(s)}{\left(1 + G_c(s)G(s) + G_c(s)\Delta G(s)\right)\left(1 + G_c(s)G(s)\right)}$$

and $G_c(s)\,G(s) \gg G_c(s)\,\Delta G(s)$

$$\Delta E(s) \simeq \frac{-G_c(s)\,\Delta G(s)\,R(s)}{(1 + L(s))^2}$$

→ For large $L(s)$, $1 + L(s) \simeq L(s)$

$$\Delta E(s) \simeq \frac{1}{L(s)} \cdot \frac{\Delta G(s)}{G(s)}\, R(s)$$

→ When $L(s)$ increases, $\Delta E(s)$ decreases & sensitivity decreases

So, we need to define **Sensitivity**

## System Sensitivity

→ The ratio of % change in system transfer function to % change in process T.F

$$T(s) = \frac{Y(s)}{R(s)}$$

$$S_G^T = \left(\Delta T(s)/T(s)\right) \Big/ \left(\Delta G(s)/G(s)\right)$$

$$= \frac{\partial T/T}{\partial G/G} = \frac{\partial T}{\partial G} \times \frac{G}{T} = \frac{\partial}{\partial G}\left(\frac{G}{1+GH}\right) \times \frac{G}{\frac{G}{1+GH}} = \frac{(1+GH) - G(H)}{(1+GH)^2} \times (1+GH)$$

$$= \frac{1}{1+GH}$$

Q. A Closed loop system is used to track the sun to obtain maximum power from a photovoltic array. Tracking system represented as $G(s) = \frac{100}{\tau s + 1}$ where $\tau = 3$ sec

a) Calculate sensitivity for small change in $\tau$

b) Calculate time response of closed loop system response

A.

a) $S_{\tau}^{T} = S_{G}^{T} \cdot S_{\tau}^{G}$

$$S_{G}^{T} = \frac{1}{1+GH} = \frac{1}{1+\frac{100}{\tau s+1}} = \frac{\tau s+1}{\tau s+101}$$

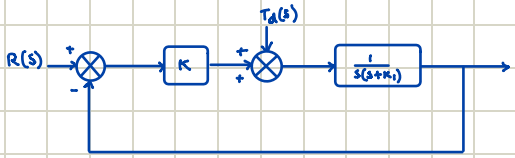
$$S_{\tau}^{G} = \frac{\partial G}{\partial \tau} \times \frac{\tau}{G} = \frac{\partial}{\partial \tau}\left(\frac{100}{\tau s+1}\right) \cdot \frac{\tau}{\frac{100}{\tau s+1}} = \frac{-100(s)}{(\tau s+1)^2} \cdot \frac{\tau(\tau s+1)}{100} = \frac{-\tau s}{\tau s+1}$$

$$S_{\tau}^{T} = \frac{\tau s+1}{\tau s+101} \times \frac{-\tau s}{\tau s+1} = \frac{-\tau s}{\tau s+101} = \frac{-3s}{3s+101}$$

b) $$\frac{Y(s)}{R(s)} = \frac{\frac{100}{\tau s+1}}{1+\frac{100}{\tau s+1}} = \frac{100}{\tau s+101} = \frac{100/101}{\frac{\tau s}{101}+1} = \frac{100/101}{\frac{3s}{101}+1}$$

Time Response $= \frac{3}{101} = 0.029\,s$

Q. Consider UFS shown below, The system has 2 parameters, Controller gain K and constant $K_1$ in the process. Calculate the insensitivity of closed loop transfer function to change in $K_1$. Assume $\tau_d = 0$




A. $$S_{K_1}^{T} = \frac{\partial T}{\partial K_1} \cdot \frac{K_1}{T}$$

$$T = \frac{\frac{K}{s(s+K_1)}}{1+\frac{K}{s(s+K_1)}} = \frac{K}{s^2+sK_1+K}$$

$$S_{K_1}^{T} = \frac{\partial}{\partial K_1}\left(\frac{K}{s^2+sK_1+K}\right) \times \frac{\frac{K_1}{K}}{\frac{\frac{s(s+K_1)}{1+\frac{K}{s(s+K_1)}}}}$$

## Cost of Feedback

→ Cost of Feedback increases as there are more no. of components & feedback systems need a sensor, which is most expensive component

→ Complexity of System also increases as no. of components & sensors

→ Open loop gain $G_c(s)\,G(s)$ is reduced to $\frac{G_c(s)\,G(s)}{1+G_c(s)G(s)}$ in unity -ve feedback system causing loss of gain

→ Open loop system is stable, closed-loop system may not be stable causing possibility of instability

Q. $c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$

Obtain the Closed Loop Transfer Function. Determine $\omega_n$, $\zeta$

A. $c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$

$\frac{dc(t)}{dt} = h(t) = -12e^{-60t} + 12e^{-10t}$

↳ Laplace, $H(s) = \frac{-12}{s+60} + \frac{12}{s+10} = \frac{12(-s-10+s+60)}{s^2+70s+600} = \frac{600}{s^2+70s+600} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

$\omega_n = \sqrt{600}$

$\zeta = \frac{70}{2\sqrt{600}} = 1.43$

Q. Find K & T for following:

settling time $t_s \le 4$ for 2% Tolerance

$G(s) = \frac{K}{s(Ts+1)}$. Minimum damping ratio for overshoot of 4.3, $\zeta = 0.707$

A. $t_s \le 4 \Rightarrow \frac{4}{\zeta\omega_n} \le 4 \Rightarrow \zeta\omega_n \ge 1 \Rightarrow \omega_n \ge \frac{1}{0.707} \Rightarrow \omega_n \ge \sqrt{2}$

T.F $\Rightarrow \frac{G(s)}{1+G(s)C(s)} = \frac{\frac{K}{s(Ts+1)}}{1+\frac{K}{s(Ts+1)}} = \frac{K}{Ts^2+s+K} = \frac{K/T}{s^2+\frac{s}{T}+\frac{K}{T}}$

comparing with $\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

$\omega_n^2 = \frac{K}{T} \Rightarrow \omega_n = \sqrt{\frac{K}{T}}$

$2\zeta\omega_n = \frac{1}{T} \Rightarrow \omega_n = \frac{1}{\frac{2}{\sqrt{2}}T} = \frac{1}{\sqrt{2}T} = \sqrt{\frac{K}{T}} \Rightarrow K = \frac{1}{2T} \Rightarrow \omega_n = \sqrt{\frac{K}{T}} = \sqrt{2} \Rightarrow \frac{K}{T} = 2$

$2 \times \frac{1}{\sqrt{2}} \times \sqrt{2} = \frac{1}{T} \Rightarrow T = \frac{1}{2}$

$K = 1$

Q. $G(s) = \frac{25}{s(s+1)}$ . Find $t_r$ , $t_s$ , $t_p$ , $\%\ M_p$

A. $TF = \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{25}{s(s+1)}}{1 + \frac{25}{s(s+1)}} = \frac{25}{s^2+s+25}$

Comparing with $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$\omega_n = 5$

$\zeta = \frac{1}{10} = 0.1$

$\omega_d = \omega_n\sqrt{1-\zeta^2} = 5\sqrt{0.99} = 4.98$

$\theta = \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right) = 84.26^\circ$ degrees
$= 1.47$ rad

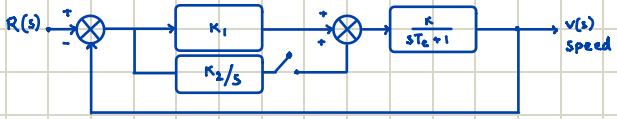
$t_r = \frac{\pi - \theta}{\omega_d} = \frac{3.14 - 1.47}{4.98} = 0.335$ s

$t_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.1 \times 5} = 8$s

$t_p = \frac{\pi}{\omega_d} = 0.63$s

$\%\ M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100 = 72.93\ \%$

Q. An automobile speed control system is as shown below




Determine static error constant & steady state error for unit step input for 2 cases

a) Switch is open

b) Switch is closed

Also identify the type of system in both cases

A. a) $G(s) = \frac{K_1 K}{sT_e + 1} \Rightarrow$ Type 0

$K_p = \lim_{s\to 0} G(s) = \frac{K_1 K}{0+1} = K_1 K$ ; $e_{ss} = \frac{1}{1 + K_1 K}$

b) $G(s) = \frac{(K_1 + \frac{K_2}{s})K}{sT_e + 1} = \frac{(K_1 s + K_2)K}{s^2 T_e + s} \Rightarrow$ Type 1

$K_p = \lim_{s\to 0} G(s) = \frac{K_2 K}{0} = \infty$ ; $e_{ss} = 0$

(Starting of ISA 2 portions)

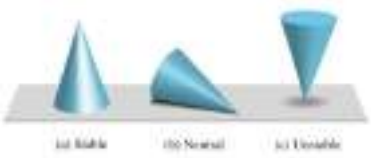


Illustration of stability

## Stability

→ Important in design & control of feedback control systems

→ A Stable system is defined as a system with a bounded system response (if system is subjected to bounded input / disturbance & response is bounded in magnitude, system is stable)

→ A Stable system is dynamic system with bounded response to bounded input

## Determining Stability

→ $g(t)$ is stable iff $\int_0^\infty |g(t)| < \infty$ (BIBO)

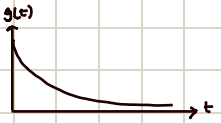
→ The location of poles in s-plane of system indicates resulting transient response

There are 4 types based on pole location & transient response

i) Poles in left hand portion of s-plane ⇒ decreasing response for disturbance inputs

ex: $G(s) = \frac{1}{s+a}$ ⇒ poles at $s = -a$

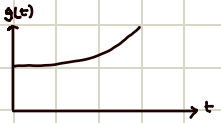
$g(t) = e^{-at}$, $t \geq 0$




ii) Poles in right hand portion of s-plane ⇒ increasing response for disturbance inputs

ex: $G(s) = \frac{1}{s-a}$ ⇒ poles at $s = a$

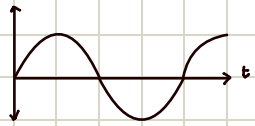
$g(t) = e^{at}$, $t \geq 0$




iii) Simple poles on jω-axis ⇒ neutral response for disturbance inputs

ex: $G(s) = \frac{a}{s^2+a^2}$ ⇒ poles at $s = \pm ja$

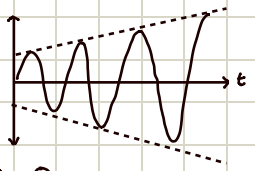
$g(t) = \sin at$, $t \geq 0$




iv) Multiple poles on jω-axis ⇒ increasing response for disturbance inputs

ex: $G(s) = \frac{2as}{(s^2+a^2)^2}$ ⇒ poles at $s = \pm ja, \pm ja$

$g(t) = t \sin at$, $t \geq 0$




→ Therefore, most desirable case is ①, poles at left hand side of s-plane to make system stable

## Methods to determine Stability

i) Routh - Hurwitz → Time Domain
ii) Root - Locus
iii) Bode Plot → Frequency Domain
iv) Nyquist Criterion

## R-H Criterion

→ Consider a characteristic equation of SISO LTI System

$\Delta(s) = q(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0$

→ The necessary condition for stability is

i) the coefficient of positive real function is positive & non-zero
ii) all the roots of characteristic equation should have real & negative part

→ **Procedure :**

1) Construct R-H Table

| | | | |
|---|---|---|---|
| $s^n$ | $a_n$ | $a_{n-2}$ | $a_{n-4}$ |
| $s^{n-1}$ | $a_{n-1}$ | $a_{n-3}$ | $a_{n-5}$ |
| $s^{n-2}$ | $b_{n-1}$ | $b_{n-3}$ | $b_{n-5}$ |
| $s^{n-3}$ | $c_{n-1}$ | $c_{n-3}$ | $c_{n-5}$ |
| ⋮ | ⋮ | ⋮ | ⋮ |
| $s^0$ | $h_{n-1}$ | | |

where

$$b_{n-1} = \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}} = \frac{-1}{a_{n-1}}\begin{vmatrix} a_n & a_{n-2} \\ a_{n-1} & a_{n-3} \end{vmatrix}$$

$$b_{n-3} = \frac{-1}{a_{n-1}}\begin{vmatrix} a_n & a_{n-4} \\ a_{n-1} & a_{n-5} \end{vmatrix}$$

$$c_{n-1} = \frac{-1}{b_{n-1}}\begin{vmatrix} a_{n-1} & a_{n-3} \\ b_{n-1} & b_{n-3} \end{vmatrix}$$

2) No of sign changes in the first column will determine no. of roots of $q(s)$ consisting positive real parts

**Q.** $q(s) = s^3 + 2s^2 + 3s + 1 = 0$

**A.**

| | | | |
|---|---|---|---|
| $s^3$ | 1 | 3 | 0 |
| $s^2$ | 2 | 1 | 0 |
| $s^1$ | 5/2 | 0 | 0 |
| $s^0$ | 1 | 0 | |

No sign changes, so no poles on RHS of s-plane
System is stable

**Q.** $q(s) = 2s^4 + s^3 + 3s^2 + 5s + 10 = 0$

| | | | |
|---|---|---|---|
| $s^4$ | 2 | 3 | 10 |
| $s^3$ | 1 | 5 | 0 |
| $s^2$ | -7 | 10 | 0 |
| $s^1$ | 45/7 | 0 | |
| $s^0$ | 10 | | |

2 sign changes, so 2 poles on RHS of s-plane
System is unstable

Q. Consider $T(s) = \dfrac{1000}{s^3 + 10s^2 + 31s + 1030}$

A.

| | | | |
|---|---|---|---|
| $s^3$ | 1 | 31 | 0 |
| $s^2$ | 10 | 1030 | 0 |
| $s^1$ | -72 | 0 | |
| $s^0$ | 1030 | | |

2 sign changes, so 2 poles on RHS of s-plane

System is unstable

Q. $q(s) = (s-1+j\sqrt{7})(s-1-j\sqrt{7})(s+3) = 0$

A. $(s^2 - 2s + 8)(s+3) = 0$

$s^3 - 2s^2 + 8s + 3s^2 - 6s + 24 = 0$

$s^3 + s^2 + 2s + 24 = 0$

| | | | | |
|---|---|---|---|---|
| $s^3$ | 1 | 2 | 0 | 0 |
| $s^2$ | 1 | 24 | 0 | 0 |
| $s^1$ | -22 | 0 | | |
| $s^0$ | 24 | | | |

2 sign changes, so 2 poles on RHS of s-plane

System is unstable

RH Criterion if first column has zeros

→ Replace 0 with $\varepsilon$ where $\varepsilon$ is some small +ve constant greater than 0 then analyse by making $\varepsilon \to 0$ from left to right

Q. $s^4 + s^3 + 2s^2 + 2s + 3 = 0$

A. Method 1:

| | | | |
|---|---|---|---|
| $s^4$ | 1 | 2 | 3 |
| $s^3$ | 1 | 2 | 0 |
| $s^2$ | $0^{\varepsilon}$ | 3 | |
| $s^1$ | $\frac{2\varepsilon - 3}{\varepsilon}$ | 0 | |
| $s^0$ | 3 | 0 | |

$\lim_{\varepsilon \to 0} \frac{2\varepsilon - 3}{\varepsilon} = 2 - \lim_{\varepsilon \to 0} \frac{3}{\varepsilon} \Rightarrow$ -ve number

2 sign changes, so 2 poles on RHS of s-plane

System is unstable

Method 2: Substitute $s = 1/z \Rightarrow \frac{1}{z^4} + \frac{1}{z^3} + \frac{2}{z^2} + \frac{2}{z} + 3 = 0$

$\hookrightarrow 3z^4 + 2z^3 + 2z^2 + z + 1 = 0$

| | | | |
|---|---|---|---|
| $s^4$ | 3 | 2 | 1 |
| $s^3$ | 2 | 1 | 0 |
| $s^2$ | 1/2 | 1 | 0 |
| $s^1$ | -3 | 0 | |
| $s^0$ | | | |

$2 - 1/2 = \frac{-3/2}{1/2}$

Q. $q(s) = s^5 + 2s^4 + 2s^3 + 4s^2 - 11s + 10$

| | | | |
|---|---|---|---|
| $s^5$ | 1 | 2 | -11 |
| $s^4$ | 2 | 4 | 10 |
| $s^3$ | $\overset{\varepsilon}{0}$ | -16 | 0 |
| $s^2$ | $\frac{32+4\varepsilon}{\varepsilon}$ | 10 | |
| $s^1$ | $d_1$ | 0 | |
| $s^0$ | 10 | | |

$$d_1 = \frac{-\left(10\varepsilon + 16\left(\frac{32+4\varepsilon}{\varepsilon}\right)\right)}{\frac{32+4\varepsilon}{\varepsilon}}$$

$$\lim_{\varepsilon \to 0} \frac{32+4\varepsilon}{\varepsilon} = \lim_{\varepsilon \to 0} \frac{32}{\varepsilon} + 4 = +ve$$

$$\lim_{\varepsilon \to 0} d_1 = \lim_{\varepsilon \to 0} \frac{-\left(10\varepsilon + 16\left(\frac{32+4\varepsilon}{\varepsilon}\right)\right)}{\frac{32+4\varepsilon}{\varepsilon}} = \lim_{\varepsilon \to 0} \frac{-10\varepsilon}{\frac{32+4\varepsilon}{\varepsilon}} - 16 = -16$$

2 sign changes, so 2 poles on RHS of s-plane

System is unstable

RH Criterion if entire row is 0

→ If all the elements are 0 in a particular row, it is because:

i) equation has one pair of real roots with equal magnitude but opposite sign

ii) one or more pair of imaginary roots

iii) pair of complex conjugate roots forming symmetry about origin of s-plane

$s = -1 \pm j$ , $1 \pm j$

iv) non-repeated pairs of roots located on jw axis

$s^4 + 3s^2 + 2 = 0 \Rightarrow s = \pm j, \pm j\sqrt{2}$

v) repeated pairs of roots located on jw axis

$s^4 + 2s^2 + 1 = 0 \Rightarrow s = \pm j, \pm j$

Q. $q(s) = s^5 + 4s^4 + 8s^3 + 8s^2 + 7s + 4 = 0$

| | | | |
|---|---|---|---|
| $s^5$ | 1 | 8 | 7 |
| $s^4$ | 4 | 8 | 4 |
| $s^3$ | 6 | 6 | 0 |
| $s^2$ | 4 | 4 | 0 |
| $s^1$ | 0 | 0 | 0 |
| $s^0$ | | | |

From auxillary eqn from the row above zero row $\Rightarrow A(s) = 4s^2 + 4 = 0$ → $s = \pm j$

Differentiate $\frac{dA}{ds}$ and replace zero row by coefficients of $\frac{dA}{ds} = 8s + 0$

From $A(s)$, $s = \pm j \Rightarrow$ 2 roots on imaginary axis & system is marginally stable

To get further roots, divide $\frac{q(s)}{s^2+1}$

$$
\begin{array}{r}
s^3 + 4s^2 + 7s + 4 \\
\hline
s^2+1 \;\big)\; s^5 + 4s^4 + 8s^3 + 8s^2 + 7s + 4 \\
s^5 + s^3 \\
\hline
4s^4 + 7s^3 + 8s^2 + 7s + 4 \\
4s^4 + 4s^2 \\
\hline
7s^3 + 4s^2 + 7s + 4 \\
7s^3 + 7s \\
\hline
4s^2 + 4 \\
4s^2 + 4 \\
\hline
0
\end{array}
$$

| | | |
|---|---|---|
| $s^3$ | 1 | 7 |
| $s^2$ | 4 | 4 |
| $s^1$ | 6 | 0 |
| $s^0$ | 4 | |

$\Rightarrow$ No sign changes $\Rightarrow$ Then in final, system is marginally stable

Q. Given $G(s) = \frac{K(s+2)(s+1)}{(s+0.1)(s-1)}$. Assume $H(s) = 1$ Find the value of K for system to stable

A. $\frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{K(s+2)(s+1)}{(s+0.1)(s-1)}}{1 + \frac{K(s+2)(s+1)}{(s+0.1)(s-1)}} = \frac{K(s+2)(s+1)}{(s+0.1)(s-1) + K(s+2)(s+1)}$

$q(s) = (s+0.1)(s-1) + K(s+2)(s+1)$

$= s^2(1+K) + s(3K - 0.9) + (2K - 0.1)$

| | | |
|---|---|---|
| $s^2$ | $1+K$ | $2K-0.1$ |
| $s^1$ | $3K-0.9$ | 0 |
| $s^0$ | $2K-0.1$ | |

i) No poles on RHS

$K+1 > 0 \Rightarrow K > -1$

$3K - 0.9 > 0 \Rightarrow K > 0.3$

$2K - 0.1 > 0 \Rightarrow K > 0.05$

which means if $K > 0.3 \Rightarrow$ stable

ii) 1 pole on RHS

$K+1 > 0 \Rightarrow K > -1$

$2K - 0.1 < 0 \Rightarrow K < 0.05$

iii) 2 poles on RHS

RH Criterion if repeated roots on jω axis

Q. $q(s) = s^5 + s^4 + 2s^3 + 2s^2 + s + 1 = 0$

A.

| | | | |
|---|---|---|---|
| $s^5$ | 1 | 2 | 1 |
| $s^4$ | 1 | 2 | 1 |
| $s^3$ | ⓪ | ⓪ | 0 |
| $s^2$ | 1 | 1 | |
| $s^1$ | ⓪ | | |
| $s^0$ | 1 | | |

$A(s) = s^4 + 2s^2 + 1 = (s^2+1)^2$ ⇒ Multiple roots on jω axis ⇒ Unstable

$\frac{dA(s)}{ds} = 4s^3 + 4s$

$A_2(s) = s^2 + 1$

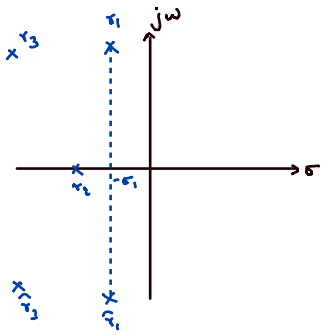
$\frac{dA_2}{ds} = 2s$

## Relative Stability of Feedback Control Systems

→ If the system satisfies the RH criterion and is stable, it is desirable to determine the relative stability

→ The relative stability of a system can be defined as the property that is measured by the relative real path of each root or pair of roots

→ Consider a set of roots as shown in the s-plane in the left ($r_2$ is more stable than $r_1$, $\hat{r}_1$)




Using RH Criterion if we need to comment on relative stability, it is done by utilising a change of variable which shifts the s-plane axis on a trial and error basis. A shift of the vertical axis to $-\sigma_1$ will result in the roots $r_1$ and $\hat{r}_1$ appearing on shifted axis

ex: $q(s) = s^3 + 7s^2 + 25s + 9 = 0$

Let $\sigma_1 = 1$ and define $s = z - 1$

$q(z-1) = (z-1)^3 + 7(z-1)^2 + 25(z-1) + 9$

$= z^3 - 3z^2 + 3z - 1 + 7(z^2 - 2z + 1) + 25z - 25 + 9$

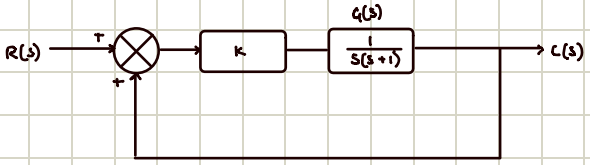
$= z^3 + 4z^2 + 14z - 10 = 0$

This kind of shift with different σ values & determining relative stability is a tedious process therefore RH method is not suggested to determine relative stability

# Unit - 3

## Root Locus

→




$$\frac{C(s)}{R(s)} = \frac{KG(s)}{1+KG(s)}$$

$K \rightarrow 0$ to $\infty$

Characteristic equation, $1 + G(s)K = 0$

$$G(s) = \frac{-1}{K}$$

→ Root locus is the path of the roots of the characteristic equation traced out in the s-plane as a system parameter varies from 0 to ∞

→ This method provides a measure of sensitivity of roots to a variation in parameter K being considered. It is used in conjungtion to RH Criterion

→ When $K=0$, roots are at the poles of $G(s)$

$K=\infty$, roots are at the zeros of $G(s)$

Locus of the roots of the characteristic equation $1+KG(s)=0$ begins at the poles of $G(s)$ and ends at the zeros of $G(s)$ as K varies from 0 to ∞

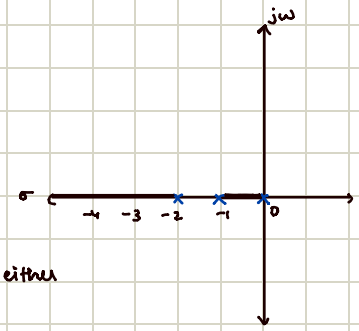
## Root Locus Procedure

Q. Given $G(s) = \frac{K}{s(s+1)(s+2)}$

A. 1) Identify poles and zeros

poles : $0, -1, -2 \Rightarrow n=3$ (n : no. of open loop poles)

zeros : Nil $\Rightarrow m=0$ (m: no. of open loop zeros)




2) Root locus is always symmetrical about real axis

3) Always start from the open loop pole & terminate on either finite open loop zero or ∞

4) A point on real axis lies on the root locus branch if total no. of poles and zeros on the right side of the point is odd

5) No. of branches = $\begin{cases} n & \text{, if } n>m \\ m & \text{, if } m>n \\ n \text{ or } m & \text{, if } n=m \end{cases}$

6) Asymptotes

$$\phi = \frac{(2q+1)\pi}{n-m}, \quad q = 0, 1, 2 \ldots (n-m-1)$$

$\phi_1 = \frac{180}{3} = 60°$ ; $\phi_2 = \frac{3\times180}{3} = 180°$ ; $\phi_3 = \frac{5\times180}{3} = 300°$

7) Finding centroid : The point at which asymptotes meet on the real axis

Centroid $(\sigma) = \left(\Sigma(\text{real part of poles}) - \Sigma(\text{real parts of zeros})\right) \div (n-m)$ $\left(\sigma = \frac{-3-0}{3} = -1\right)$

https://youtu.be/MIVkzOiFvhI?si=VSRAZ-1rd1T1V7aC

Personal Recommendation for Root Locus!

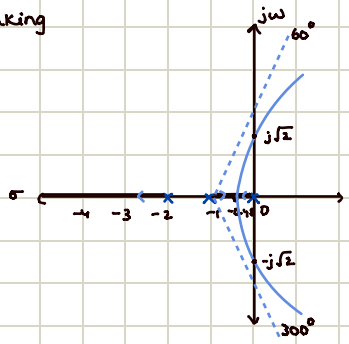
8) The breakaway point on real axis is calculated by making $\frac{dK}{ds} = 0$ and solving for s

$1 + G(s)H(s) = 0 \Rightarrow 1 + \frac{K}{s(s+1)(s+2)} = 0 \Rightarrow K = -(s+1)(s+2)s$

$K = -s^3 - 3s^2 - 2s$

$\frac{dK}{ds} = -3s^2 - 6s - 2 = 0$

$s = -0.42$ ✓ or $-1.57$ ✗




9) To find the point at which root locus intersects imaginary axis, use RH criterion and find if its marginally stable

$s^3 + 3s^2 + 2s + K = 0$

| | | |
|---|---|---|
| $s^3$ | 1 | 2 |
| $s^2$ | 3 | K |
| $s^1$ | $\frac{6-K}{3}$ | 0 |
| $s^0$ | K | 0 |

For it to be marginally stable, $K = 6$

$3s^2 + 6 = 0$

$3s^2 = -6 \Rightarrow s = \pm j\sqrt{2}$

10) If all elements of any row of Routh array are zero, then root locus branch intersects the imaginary axis and the value of K at that point is calculated using RH method, substitute this K value in the auxillary equation to get intersection of root locus branch with imaginary axis

Here, we have only breakaway point as we have only poles. If zeros are there, we need to find break in point

11) If there are complex conjugate of poles or zeros, we need to calculate angle of departure from pole & angle of arrival to zero in s-plane

angle of departure $\phi_p = 180 - \phi$

angle of arrival $\phi_z = 180 + \phi$

where $\phi = \sum \theta_1 - \sum \theta_2$

= Sum of angles of poles − Sum of angles of zeros

Q. The open loop T.F of unity -ve feedback system is given by $\frac{K(s+1)}{(s^2+4s+13)}$

Draw root locus as K varies from 0 to $\infty$

A. $\frac{K(s+1)}{s^2+4s+13}$

zeros $\Rightarrow -1$ ; poles $\Rightarrow -2 \pm 3j$

No. of branches $= n = 2$

$\sigma = \frac{-2-2-(-1)}{1} = -3$

$\theta = \frac{(2q+1)\pi}{n-m} \Rightarrow q = 0 \Rightarrow \theta = \pi$

Root locus lies b/w $-1$ & $-\infty$

RH array not required because root locus doesn't cut real axis

Now to find breakaway point, $1 + G(s)H(s) = 0$

$1 + \frac{K(s+1)}{s^2+4s+13} = 0 \Rightarrow s^2 + (4+K)s + (13+K) = 0$

$K = -\frac{(s^2+4s+13)}{s+1}$

$\frac{dK}{ds} = -\frac{(2s+4)(s+1) + (s^2+4s+13)}{(s+1)^2} = 0$

$0 = -(2s^2+6s+4) + (s^2+4s+13)$

$s^2 + 2s - 9 = 0 \Rightarrow s = -4.16$ ✓, $2.16$ ✗

$K_{s=-4.16} = 4.3$ ✓

$K_{s=-2.16} = -8.3$ ✗

Angle of departure at $-2+j3$ $\theta_d = 180 - (\theta_1 - \theta_2)$

$\theta_1 = 90$

$\theta_2 = 180 - \tan^{-1}\left(\frac{3}{1}\right) = 108.4$

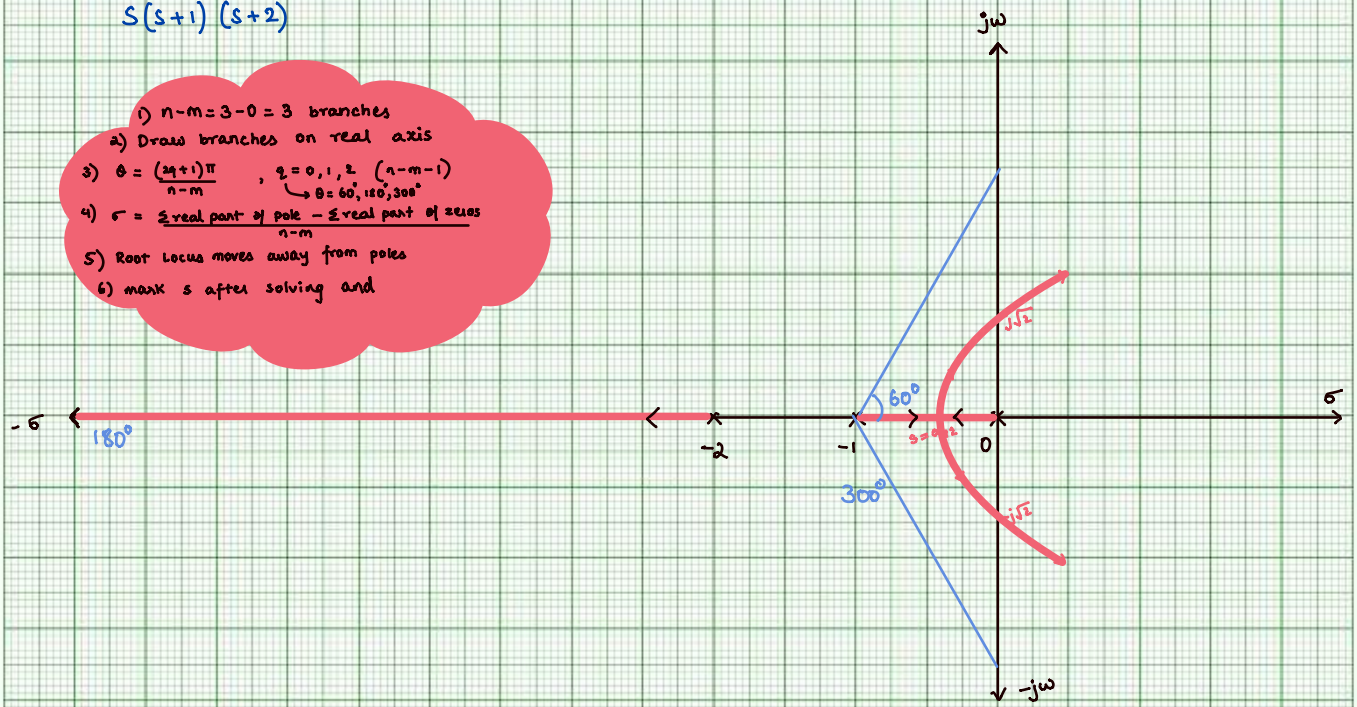
$\theta_d = 180 - (90 - 108.4) = 198.4$

at $-2-j3$ $\theta_d = -198.4$

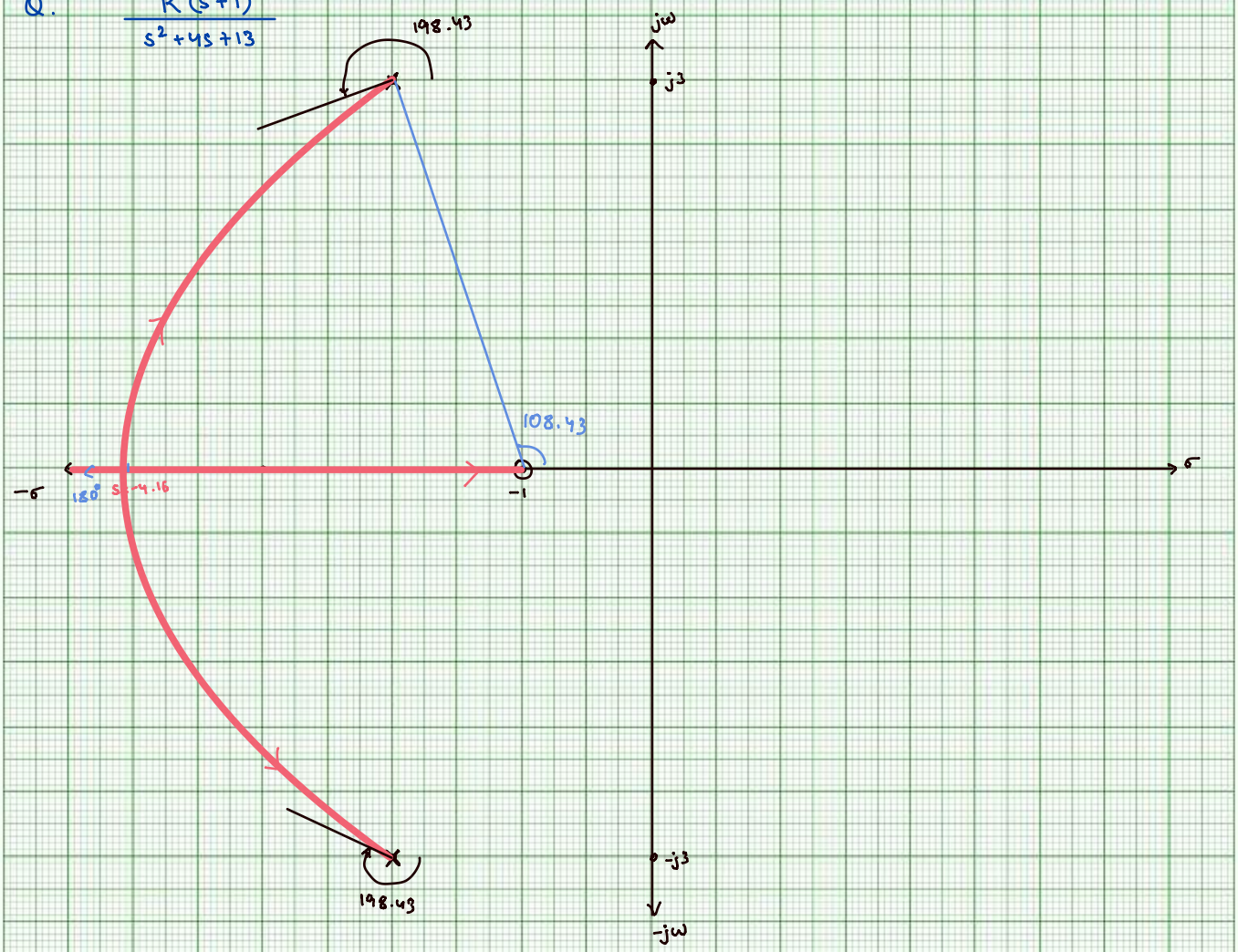
Angle of arrival $\Rightarrow$ No complex zeros

Q. $\frac{K}{S(S+1)(S+2)}$

1) $n-m = 3-0 = 3$ branches
2) Draw branches on real axis
3) $\theta = \frac{(2q+1)\pi}{n-m}$ , $q = 0, 1, 2 \ldots (n-m-1)$
   → $\theta = 60°, 180°, 300°$
4) $\sigma = \frac{\Sigma \text{real part of pole} - \Sigma \text{real part of zeros}}{n-m}$
5) Root Locus moves away from poles
6) mark s after solving and




Q. $\frac{K(S+1)}{S^2+4S+13}$



Q. The open loop TF of unity -ve feedback system is given by $\frac{K}{s(s+4)(s^2+4s+20)}$
Draw root locus as K varies from 0 to $\infty$

A. $\frac{K}{s(s+4)(s^2+4s+20)}$

Poles $\Rightarrow$ $0, -4, -2+j4, -2-j4$

no. of branches $= 4$

$\sigma = \frac{(0-4-2-2)}{4} = -2$

$\theta = \frac{(2q+1)\pi}{n-m}$ ; $q = 0, 1, 2, 3 \Rightarrow \theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

$1 + G(s)H(s) = 0 \Rightarrow 1 + \frac{K}{s(s+4)(s^2+4s+20)} = 0 \Rightarrow K + s^4 + 8s^3 + 36s^2 + 80s = 0$

$K = -(s^4 + 8s^3 + 36s^2 + 80s)$

$\frac{dK}{ds} = -(4s^3 + 24s^2 + 72s + 80)$

$s = -2, -2+j2.45, -2-j2.45$

→ breakaway points

| | | | |
|---|---|---|---|
| $s^4$ | 1 | 36 | K |
| $s^3$ | 8 | 80 | 0 |
| $s^2$ | 26 | K | |
| $s^1$ | $\frac{2080-8K}{26}$ | 0 | |
| $s^0$ | K | | |

$\frac{2080-8K}{26} = 0 \Rightarrow K = 260$

$26s^2 + 260 = 0 \Rightarrow s^2 + 10 = 0 \Rightarrow s = \pm j\sqrt{10}$

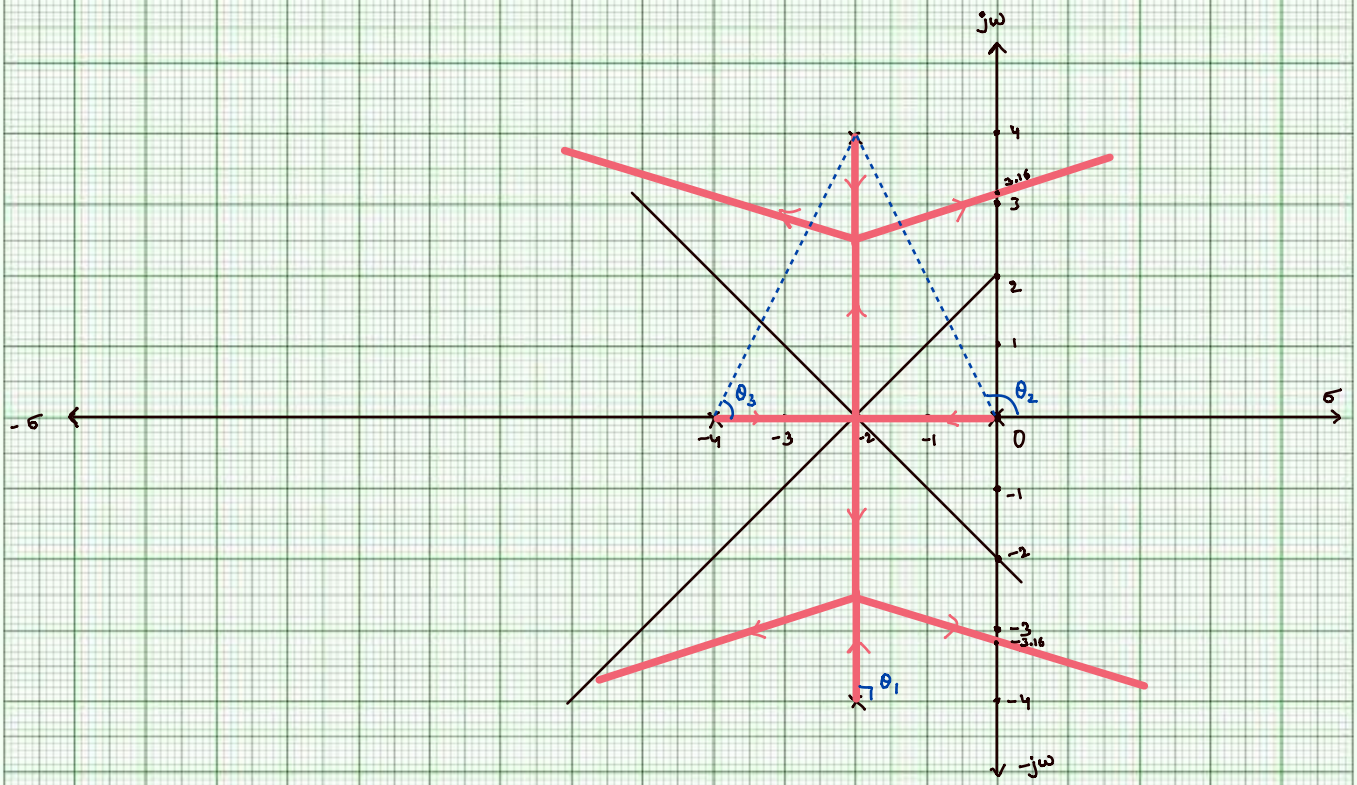
Angle of departure $\theta_d = 180 - (\theta_1 + \theta_2 + \theta_3)$

$= 180 - \left(90 + \left(180 - \tan^{-1}\left(\frac{4}{2}\right)\right) + \tan^{-1}\left(\frac{4}{2}\right)\right)$

$= 180 - (90 + 116.56 + 63.4)$

$= -90$

Q. $\frac{K}{s(s+4)(s^2+4s+20)}$



Q. Plot the root locus for $G(s) = \dfrac{s^2 - 2s + 2}{(s+2)(s+3)(s+4)}$

A. Poles $\Rightarrow -2, -3, -4$

Zeros $\Rightarrow 1 \pm j$

No. of branches $= 3$

$$\sigma = \frac{(-2-3-4)-(1+1)}{3-2} = -11$$

$$\theta = \frac{(2q+1)\pi}{n-m} = \pi$$

$$1 + G(s)H(s) = 0 \Rightarrow 1 + \frac{K(s^2-2s+2)}{(s+2)(s+3)(s+4)} = 0 \Rightarrow K = -\frac{(s^3+9s^2+26s+24)}{s^2-2s+2}$$

$$(s^2+5s+6)(s+4)$$

$$s^3+5s^2+6s+4s^2+20s+24$$

$$(s^3+9s^2+26s+24) + K(s^2-2s+2)$$

$$\frac{dK}{ds} = -\left(\frac{(3s^2+18s+26)(s^2-2s+2) - (2s-2)(s^3+9s^2+26s+24)}{(s^2-2s+2)^2}\right) = 0$$

$$= (3s^4 - 6s^3 + 6s^2 + 18s^3 - 36s^2 + 36s + 26s^2 - 52s + 52) - (2s^4 + 18s^3 + 52s^2 + 48s - 2s^3 - 18s^2 - 52s - 48) = 0$$

$$\frac{dK}{ds} = s^4 - 4s^3 - 38s^2 - 12s + 100$$

$s = 8.482,\ 1.412,\ -2.365,\ -3.529$

| | | |
|---|---|---|
| $s^3$ | 1 | $26-2K$ |
| $s^2$ | $9+K$ | $24+2K$ |
| $s^1$ | $\frac{(9+K)(26-2K)-(24+2K)}{9+K}$ | 0 |
| $s^0$ | $24+2K$ | |

$$\frac{(9+K)(26-2K)-(24+2K)}{9+K} = 0 \Rightarrow K = -8.8 \text{ (X)} \text{ or } 11.8 \text{ (✓)}$$

$$(9+K)s^2 + (24+2K) = 0$$

$$20.8s^2 + 47.6 = 0 \Rightarrow s = \pm j1.512$$

$\theta_4 = 90^\circ$

$\theta_1 = \tan^{-1}\left(\frac{1}{3}\right) = 18.43$

$\theta_2 = \tan^{-1}\left(\frac{1}{4}\right) = 14.03$

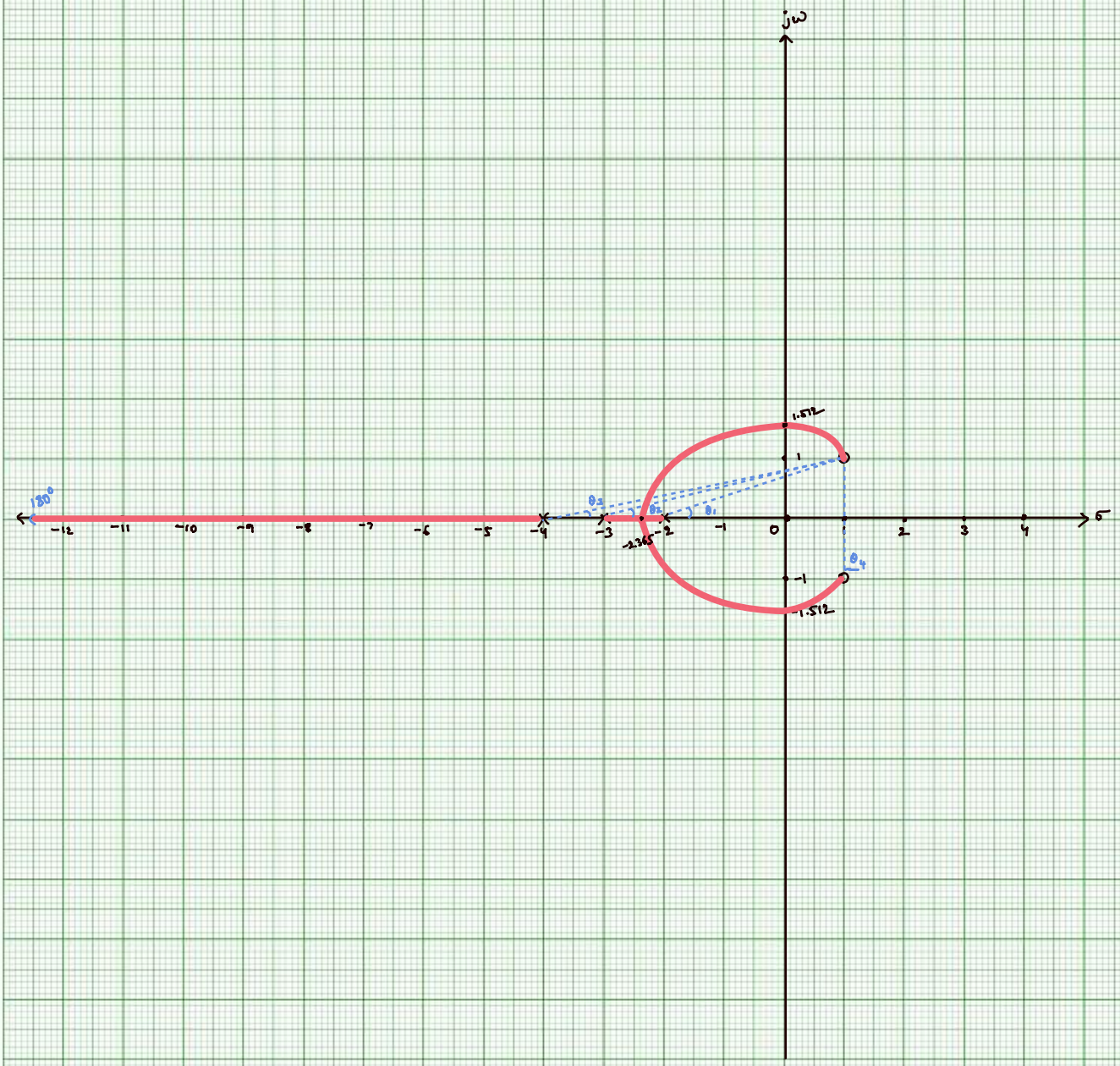
$\theta_3 = \tan^{-1}\left(\frac{1}{5}\right) = 11.309$

$\theta_a = 180 + \phi = 180 + (\text{angle of poles} - \text{angle of zeros})$

$= 180 + (18.43 + 14.03 + 11.309 - 90^\circ)$

$= 133.78$

Q. $\frac{s^2 - 2s + 2}{(s+2)(s+3)(s+4)}$



Q. Plot the root locus for $G(s) = \frac{K}{s(s^2+8s+32)}$

A. Poles $= 0, -4 \pm j4$

No. of branches $= 3$

$\sigma = \frac{0-4-4-(0)}{3} = -2.67$

$\theta = \frac{(2q+1)\pi}{n-m} \Rightarrow 0 \text{ to } 2 \Rightarrow \frac{\pi}{3}, \pi, \frac{5\pi}{3}$

$1 + G(s)H(s) = 0 \Rightarrow 1 + \frac{K}{s(s^2+8s+32)} = 0 \Rightarrow s^3 + 8s^2 + 32s + K = 0 \Rightarrow K = -(s^3 + 8s^2 + 32s)$

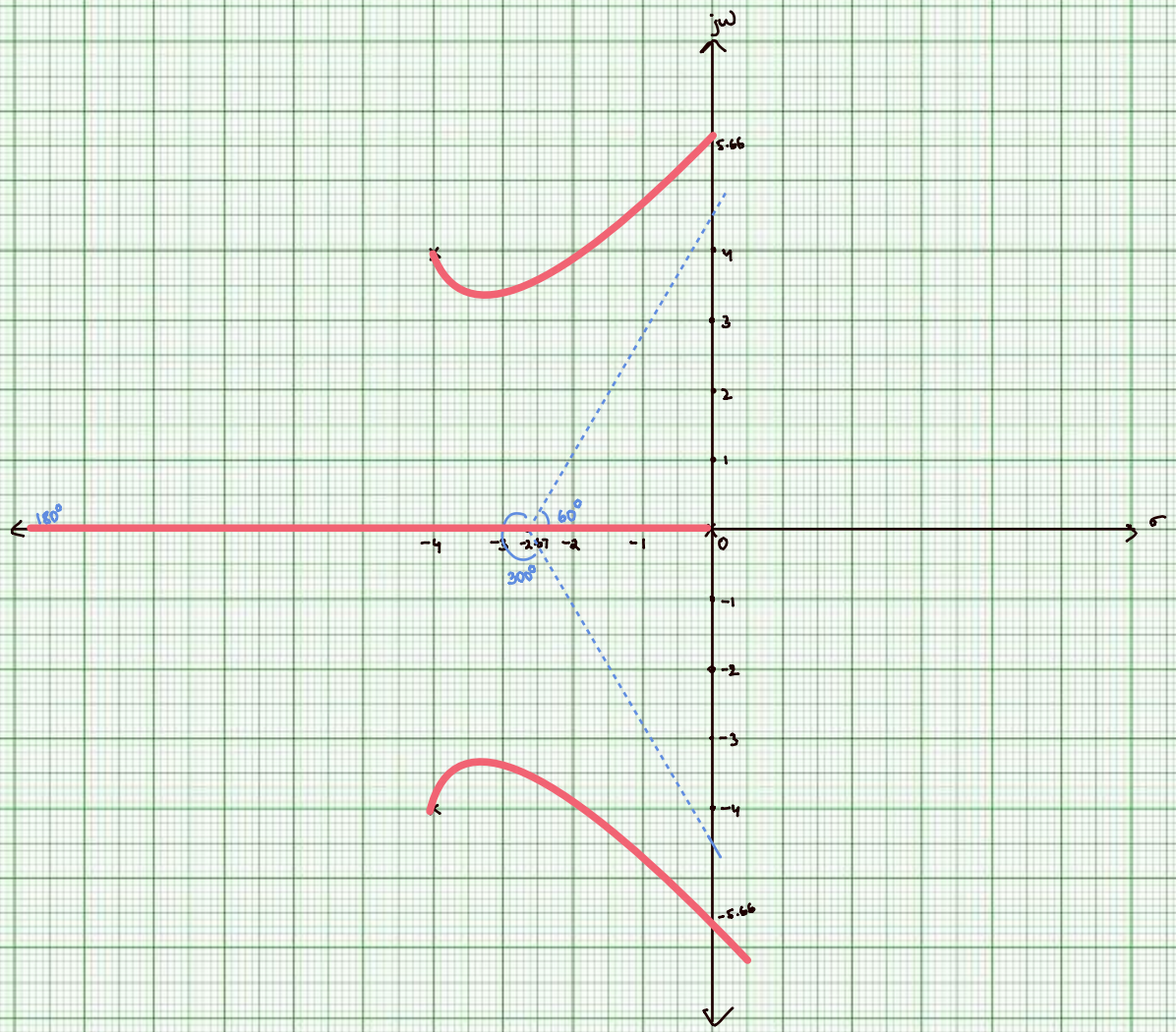
$\frac{dK}{ds} = -(3s^2 + 16s + 32) = 0 \Rightarrow s = -1.69 \text{ or } -6.31$

| | | |
|---|---|---|
| $s^3$ | 1 | 32 |
| $s^2$ | 8 | K |
| $s^1$ | $\frac{K-256}{8}$ | 0 |
| $s^0$ | K | 0 |

$\frac{K-256}{8} = 0 \Rightarrow K = 256$

Auxiliary eqn $\Rightarrow 8s^2 + 256 = 0 \Rightarrow s = \pm j4\sqrt{2} = \pm j5.66$

$$\frac{K}{s(s^2+8s+32)}$$



Q. Plot the root locus for $G(s) = \frac{K(s+2)}{s^2+2s+13}$

A. Poles at $-1 \pm j2\sqrt{3}$

Zeros at $-2$

$$\sigma = \frac{-1-1-(-2)}{1} = 0$$

$$\theta = \frac{(2q+1)\pi}{n-m} \qquad 0 \text{ to } 0 \Rightarrow \pi$$

$$1 + G(s)H(s) = 1 + \frac{K(s+2)}{s^2+2s+13} = 0 \Rightarrow \frac{-(s^2+2s+13)}{s+2} = K$$

won't cut imaginary axis $\Rightarrow$ no need to do RH (no poles on real axis)

$$\frac{dK}{ds} = -\left(\frac{(2s+2)(s+2) - (s^2+2s+13)}{(s+2)^2}\right) = 0$$

$$= 2s^2+6s+4-s^2-2s-13 = s^2+4s-9 = 0 \Rightarrow s = 1.6 \text{ or } -5.6 \checkmark$$

$$\tan\theta_1 = \frac{2\sqrt{3}}{1} \Rightarrow \theta_1 = 73.8^\circ$$

$$\theta_2 = 180 - (90^\circ - \theta_1)$$
$$= 180 - (90 - 73.8)$$
$$= 163.8^\circ$$

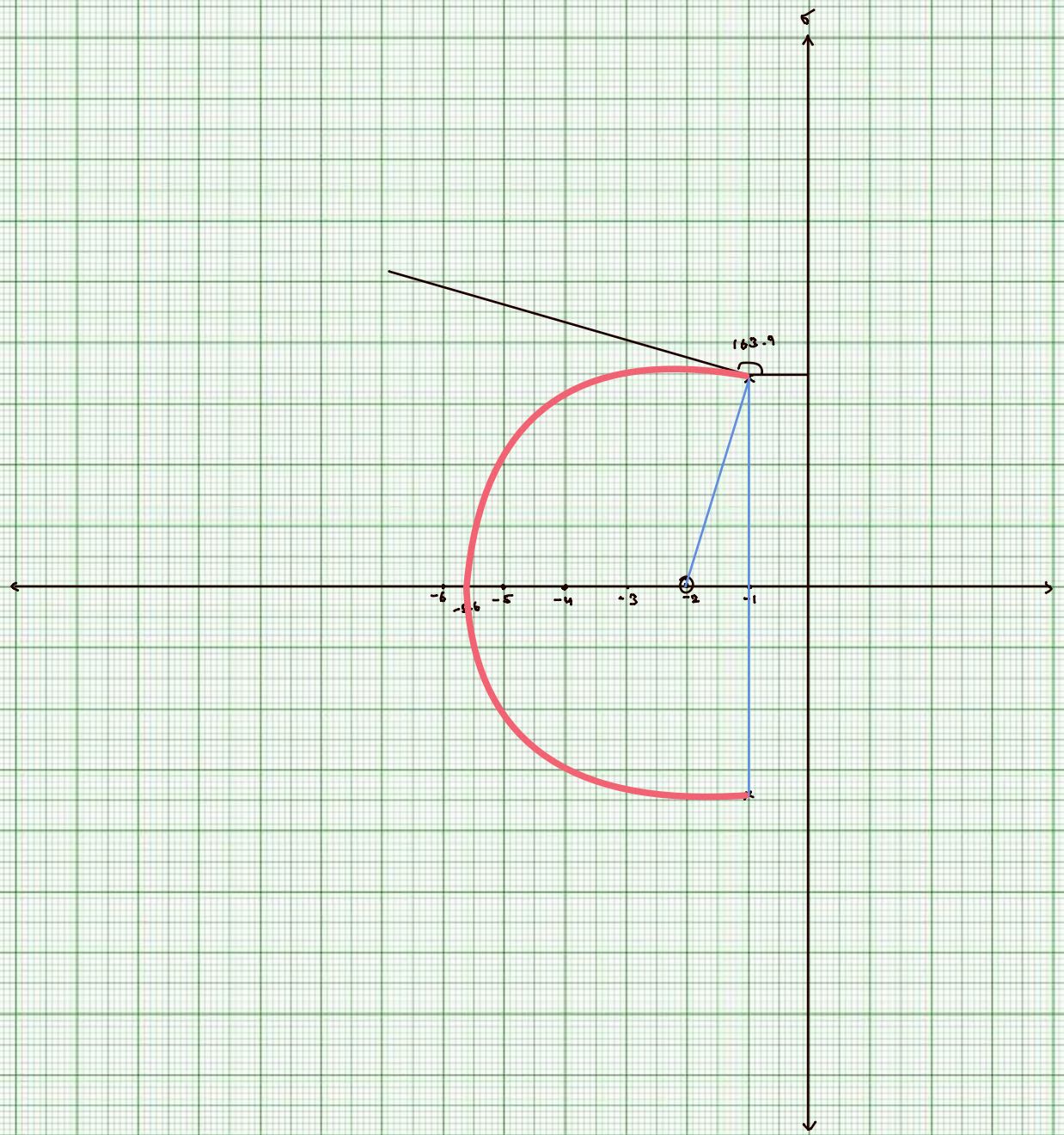
Effect of adding poles to open loop T.F

→ When more poles are added, root locus shifts towards the right tending to lower the system's relative stability and slow down settling time

Effect of adding zeros to open loop T.F

→ When more zeros are added, root locus shifts towards the left tending to increase the system's relative stability and speed up settling time

$$\frac{K(s+2)}{s^2+2s+13}$$



# Frequency Response Methods

→ Frequency response of a system is defined as steady-state response of the system to a sinusoidal input signal. The sinusoid is a unique input signal, and resulting output signal for a linear system is sinusoidal in the steady state; it differs from input only in amplitude & phase angle

→ $R(s) \longrightarrow \boxed{G(s)} \longrightarrow C(s)$ $\quad Y(s) = G(s)R(s)$

$r(t) = A\sin\omega t$

$R(s) = \frac{A\omega}{s^2+\omega^2}$

$$Y(s) = G(s)\frac{A\omega}{s^2+\omega^2} = \frac{A\,G(s)\,\omega}{(s+j\omega)(s-j\omega)} = \frac{A_1}{(s+j\omega)} + \frac{A_2}{(s-j\omega)} = \frac{G(-j\omega)A}{-2j(s+j\omega)} + \frac{A\,G(j\omega)}{2j(s-j\omega)}$$

$$= \frac{A}{2j}\left[\frac{-|G(j\omega)|e^{-j\phi}}{(s+j\omega)} + \frac{|G(j\omega)|e^{j\phi}}{s-j\omega}\right] = \frac{A|G(j\omega)|}{2j}\left[\frac{-e^{-j\phi}}{(s+j\omega)} + \frac{e^{j\phi}}{(s-j\omega)}\right]$$

$$y(t) = \frac{A|G(j\omega)|}{2j}\left[-e^{-j\phi}e^{-j\omega t} + e^{j\phi}e^{j\omega t}\right]$$

$$= A|G(j\omega)|\sin(\omega t+\phi)$$

Comparing $y(t)$ and $r(t)$, only amplitude & phase angle differ

→ Let general open loop TF be $H(s)G(s) = \frac{K(s+z_1)(s+z_2)\cdots(s+z_n)}{s^n(s+p_1)(s+p_2)\cdots(s+p_n)}$

Let $s = j\omega \Rightarrow G(j\omega)H(j\omega) = \frac{K(j\omega+z_1)(j\omega+z_2)\cdots(j\omega+z_n)}{(j\omega)^n(j\omega+p_1)(j\omega+p_2)\cdots(j\omega+p_n)} = \frac{Kz_1\left(\frac{j\omega}{z_1}+1\right)z_2\left(\frac{j\omega}{z_2}+1\right)\cdots z_n\left(\frac{j\omega}{z_n}+1\right)}{(j\omega)^n p_1\left(\frac{j\omega}{p_1}+1\right)\cdots p_n\left(\frac{j\omega}{p_n}+1\right)}$

$$G(j\omega)H(j\omega) = \frac{K\left(\frac{j\omega}{z_1}+1\right)\left(\frac{j\omega}{z_2}+1\right)\cdots\left(\frac{j\omega}{z_n}+1\right)}{(j\omega)^n\left(\frac{j\omega}{p_1}+1\right)\left(\frac{j\omega}{p_2}+1\right)\cdots\left(\frac{j\omega}{p_n}+1\right)} \quad \left(K = \frac{Kz_1z_2\cdots z_n}{p_1p_2\cdots p_n}\right)$$

$$= \frac{K(j\omega T_1+1)(j\omega T_2+1)\cdots(j\omega T_n+1)}{(j\omega)^n(j\omega T_a+1)(j\omega T_b+1)\cdots(j\omega T_n+1)} \quad \left(T_1 = \frac{1}{z_1},\ T_2 = \frac{1}{z_2}\cdots,\ T_a = \frac{1}{p_1},\ T_b = \frac{1}{p_2}\cdots\right)$$

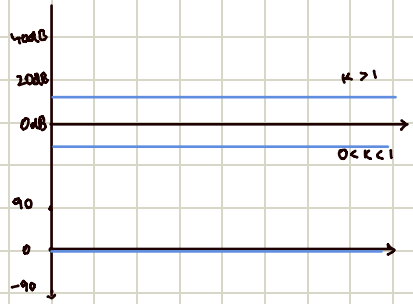
**Magnitude** $\Rightarrow$ $20\log|G(j\omega)H(j\omega)| = 20\log_{10}K + 20\log_{10}\left(\sqrt{1+(\omega T_1)^2}\right) + \cdots - 20n\log_{10}(\omega) - 20\log_{10}\left(\sqrt{1+(\omega T_a)^2}\right) + \cdots$

**Phase** $\Rightarrow$ $\tan^{-1}(\omega T_1) + \tan^{-1}(\omega T_2) + \cdots - n\tan^{-1}(\infty) - \tan^{-1}(\omega T_a) - \tan^{-1}(\omega T_b) \cdots - \tan^{-1}(\omega T_n)$

1) $|G(s)H(s)| = K$

Magnitude : $20\log_{10} K$

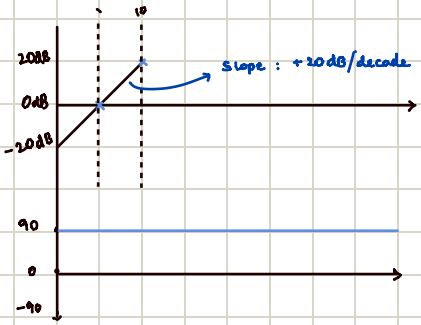
Phase : 0




2) $|G(s)H(s)| = s$ $\Rightarrow$ 1 zero at origin

Magnitude : $20\log_{10}|G(s)H(s)| = 20\log_{10}|j\omega| = 20\log_{10}\omega$

Phase : $\tan^{-1}\left(\frac{\omega}{0}\right) = 90^\circ$

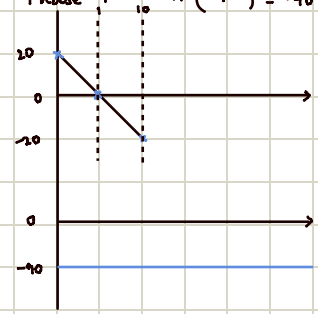



3) $G(s) = 1/s$ $\Rightarrow$ 1 pole at origin

Magnitude : $20\log_{10}|G(j\omega)| = 20\log_{10}\left|\frac{1}{j\omega}\right| = 20\log_{10}\left(\frac{1}{\omega}\right)$

Phase : $\tan^{-1}\left(\frac{-1}{\omega}\right) = -90^\circ$

Slope : - 20 dB/decade



Q. $G(s) = \frac{10}{s(1+0.4s)(1+0.1s)}$

A. 1) Try to make sure it is in standard time constant form $(1+sT)$

2) Put $s = j\omega$

and find corner frequency

$\omega = \frac{1}{T} \Rightarrow \omega_{c_1} = \frac{1}{0.4} = 2.5$ rad/sec

$\omega_{c_2} = \frac{1}{0.1} = 10$ rad/sec

https://youtu.be/qnsznGf70eU?si=n7VzZjALrGI2Tdce

Highly Recommended for Bode Plot

3) Fill this table ↓

| Term | Corner Freq | Slope | Cummulative Slope |
|---|---|---|---|
| $10/j\omega$ | – | -20 | – |
| $1/1+0.4j\omega$ | $\omega_{c_1} = 2.5$ | -20 | -40 |
| $1/1+0.1j\omega$ | $\omega_{c_2} = 10$ | -20 | -60 |

↳ Makes sure corner freq in ascending order

4) Select 2 frequencies $\omega_L$ & $\omega_H$ (Any value lower or higher than corner freq respectively)

↳ Lower ↳ Higher

Assume $\omega_L = 0.1$ (Can't be 1)

$\omega_H = 100$

$\omega = \omega_L \Rightarrow A = 20\log\left|\frac{10}{j\omega}\right| = 20\log\left|\frac{10}{0.1}\right| = 40$ dB

$\omega = \omega_{c_1} \Rightarrow A = 20\log\left|\frac{10}{j\omega}\right| = 20\log\left|\frac{10}{2.5}\right| = 12$ dB

$\omega = \omega_{c_2} \Rightarrow A = \left[\text{slope from } \omega_{c_1} \text{ to } \omega_{c_2} \times \log\frac{\omega_{c_2}}{\omega_{c_1}}\right] + A\omega_{c_1} = -40\log\left(\frac{10}{2.5}\right) + 12 = -12$ dB

$\omega = \omega_H \Rightarrow A = \left[\text{slope from } \omega_{c_2} \text{ to } \omega_H \times \log\frac{\omega_H}{\omega_{c_1}}\right] + A\omega_{c_2} = -60\log\left(\frac{100}{10}\right) - 12 = -72$ dB

For reference,

if $s \Rightarrow +90$

$1/s \Rightarrow -90$

$s^2 \Rightarrow +180$

$1/s^2 \Rightarrow -180$

5) Plot on semilog graph

6) Now we find phase

$\phi = -90 - \tan^{-1}(0.4\omega) - \tan^{-1}(0.1\omega)$

↳ s in den

| | $\omega$ | $\phi$ |
|---|---|---|
| ✓: take values from prev table | ✓ 0.1 | -93 |
| | 0.5 | -104 |
| | 1 | -118 |
| | ✓ 2.5 | -149 |
| extra values for precision | 3 | -157 |
| | ✓ 10 | -210 |
| | 20 | -236 |
| | ✓ 100 | -262 |

↳ Not plotted cuz too many already plotted

→ Keep angle unit as degree & find values using calc

→ Use free hand drawing for phase plot

GCF → Draw a line from 0 & find intersection point with magnitude plot

PCF → Draw a line from -180 & find intersection point with phase plot

Stable : GM & PM = +ve (GCF < PCF)

Marginally stable : GM & PM are zero or GM = PM (GCF = PCF)

Unstable : GM & PM = -ve (GCF > PCF)

Gain Crossover Frequency : Frequency at which $|G(j\omega)H(j\omega)| = 1$

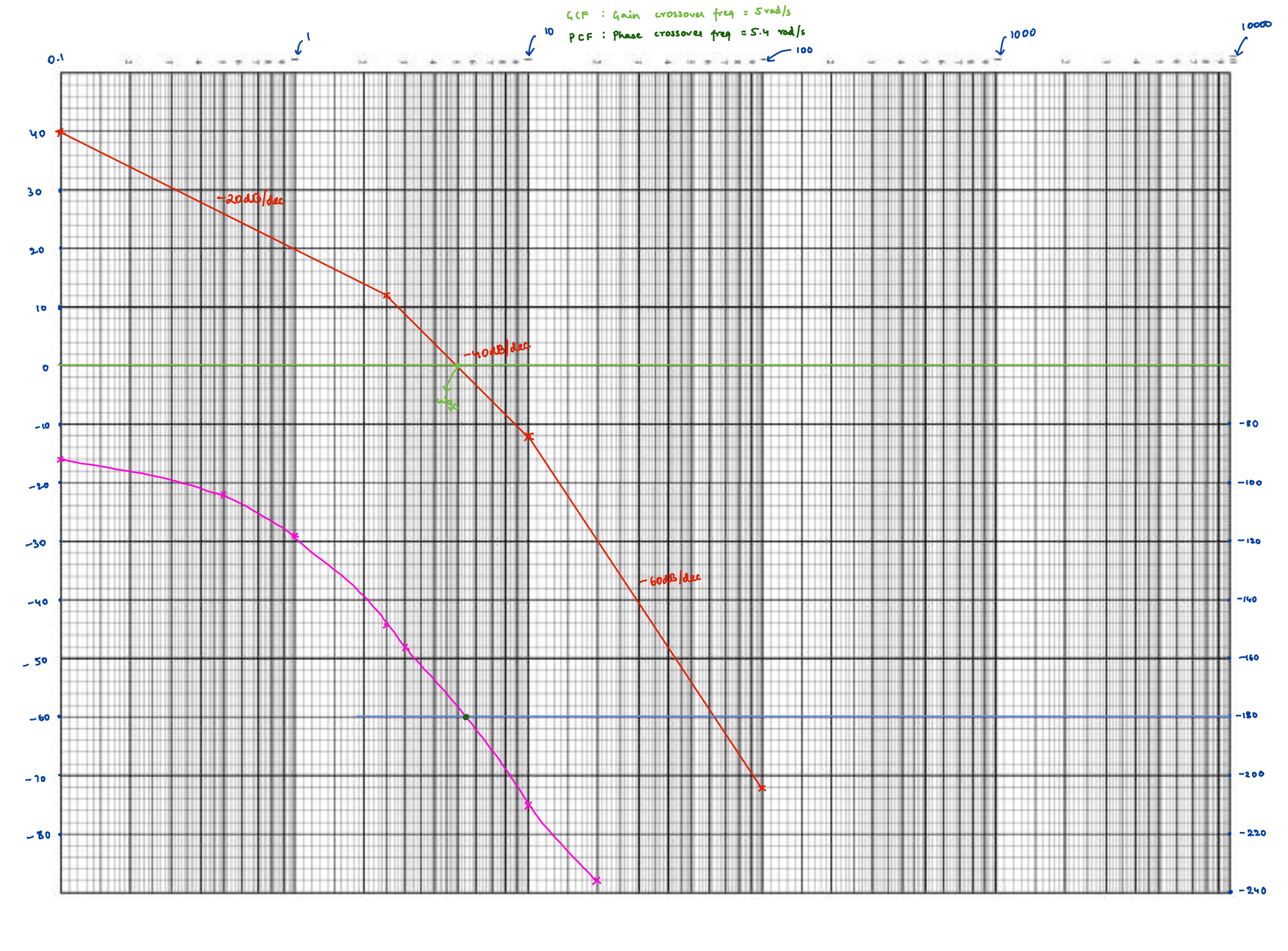
Frequency at which gain is 0dB (or) magnitude plot crosses 0dB line

Phase Crossover Frequency : Frequency at which $\angle G(j\omega)H(j\omega) = -180^\circ$

Frequency where angle is $-180^\circ$ or phase plot crosses $180^\circ$ line

GM = 0 dB − G (PCF)

PM = $\phi$ (GCF) + 180°


GCF : Gain crossover freq = 5 rad/s
PCF : Phase crossover freq = 5.4 rad/s
0.1
1
10
100
1000
10000
40
30
20
10
0
-10
-20
-30
-40
-50
-60
-70
-80
-20dB/dec
-40dB/dec
-60dB/dec
ωgc
-80
-100
-120
-140
-160
-180
-200
-220
-240

Q. $G(s) = \dfrac{800(s+2)}{s^2(s+10)(s+40)}$

A. $G(s) = \dfrac{800\left(1+\frac{s}{2}\right)\times 2}{s^2\left(1+\frac{s}{10}\right)\left(1+\frac{s}{40}\right)\times 400} = \dfrac{4\left(1+\frac{s}{2}\right)}{s^2\left(1+\frac{s}{10}\right)\left(1+\frac{s}{40}\right)}$ $\quad K = 4$

$s = j\omega$

$\omega = \frac{1}{T} \Rightarrow \omega_{c_1} = \frac{1}{\frac{1}{2}} = 2$ , $\omega_{c_2} = \frac{1}{\frac{1}{10}} = 10$ , $\omega_{c_3} = \frac{1}{\frac{1}{40}} = 40$

| Term | Corner Freq | Slope | Cummulative Slope |
|---|---|---|---|
| $\frac{4}{(j\omega)^2}$ | – | −40 | – |
| $1+0.5j\omega$ | $\omega_{c_1} = 2$ | +20 | −20 |
| $\frac{1}{1+0.1j\omega}$ | $\omega_{c_2} = 10$ | −20 | −40 |
| $\frac{1}{1+0.025j\omega}$ | $\omega_{c_3} = 40$ | −20 | −60 |

$\omega_L = 0.1$ & $\omega_h = 100$

$\omega = \omega_L \Rightarrow A = 20\log\left|\frac{4}{(j\omega)^2}\right| = 20\log\left(\frac{4}{0.01}\right) = 52\,dB$

$\omega = \omega_{c_1} \Rightarrow A = 20\log\left|\frac{4}{(j\omega)^2}\right| = 20\log\left(\frac{4}{(2.5)^2}\right) = -4\,dB$

$\omega = \omega_{c_2} \Rightarrow A = -20\log\left|\frac{10}{2}\right| - 4 = -18\,dB$

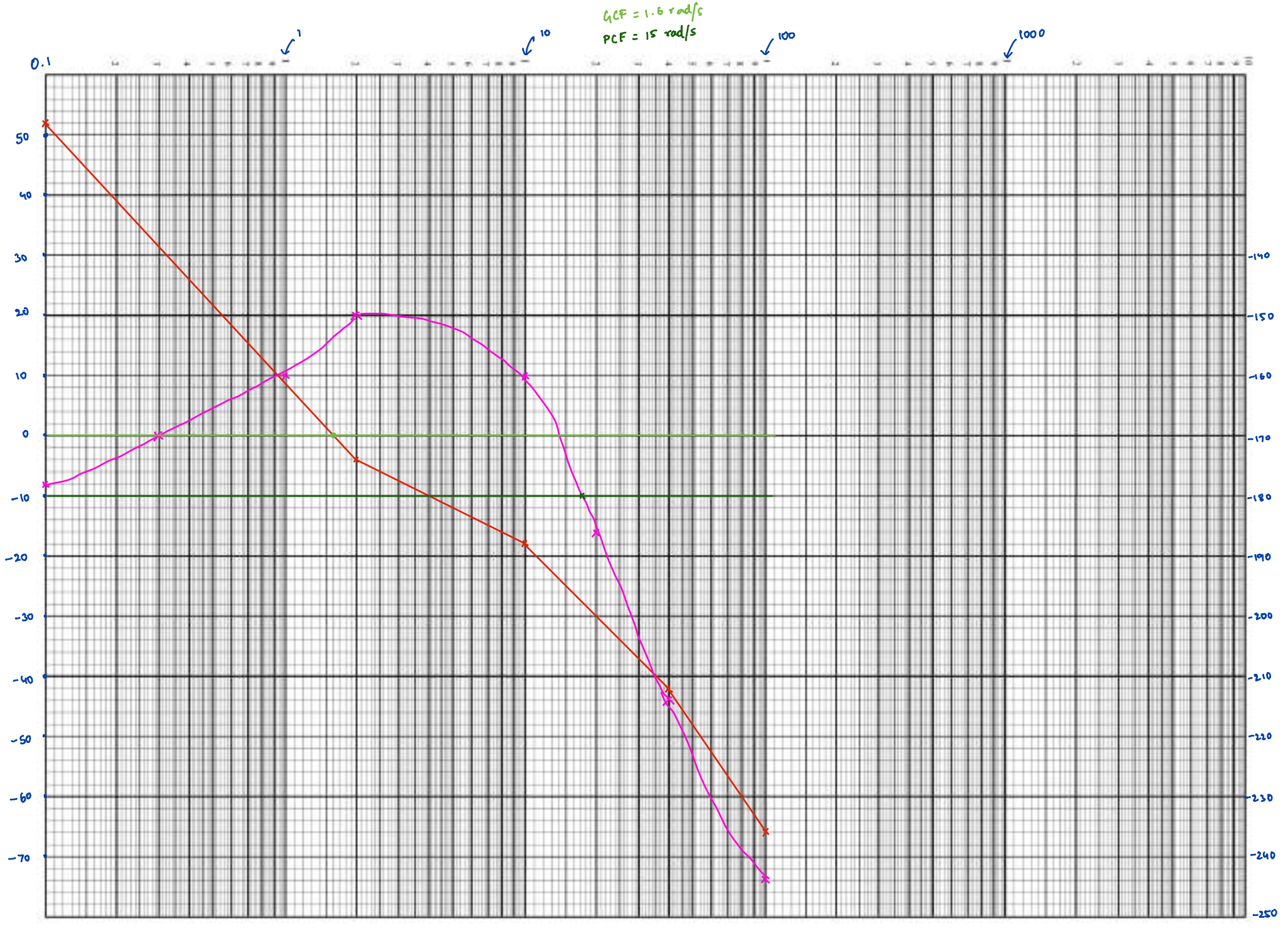
$\omega = \omega_{c_3} \Rightarrow A = -40\log\left|\frac{40}{10}\right| - 18 = -42\,dB$

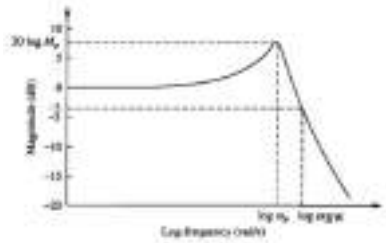
$\omega = \omega_h \Rightarrow A = -60\log\left|\frac{100}{40}\right| - 42 = -60\,dB$

$\phi = -180 + \tan^{-1}(0.5\omega) - \tan^{-1}(0.1\omega) - \tan^{-1}(0.025\omega)$

| $\omega$ | $\phi$ |
|---|---|
| 0.1 | −177.85 |
| 0.5 | −169.54 |
| 1 | −160.57 |
| 2 | −149.17 |
| 5 | −145.5 |
| 10 | −160.34 |
| 20 | −185.71 |
| 40 | −213.83 |
| 100 | −243.63 |

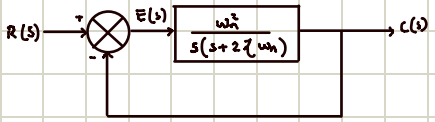
GCF = 1.6 rad/s
PCF = 15 rad/s
} → system is stable


GCF = 1.6 rad/s
PCF = 15 rad/s
0.1
1
10
100
1000
50
40
30
20
10
0
-10
-20
-30
-40
-50
-60
-70
-140
-150
-160
-170
-180
-190
-200
-210
-220
-230
-240
-250



## Second Order Closed loop System

→ Consider 2nd order closed loop system




$$\frac{C(s)}{R(s)} = T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Take $s = j\omega$ & find $T(j\omega)$

$$M = |T(j\omega)| = \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2}} = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]^2 + 4\zeta^2\frac{\omega^2}{\omega_n^2}}} \quad (\text{if } \zeta << 1)$$

differentiate wrt $\omega^2$ & equating to 0 gives Max value $M_p$

$$M_p = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

where, $\omega_p = \omega_n\sqrt{1-2\zeta^2}$

**Cutoff Frequency** – The frequency at which magnitude of closed loop frequency response is -3dB below its zero frequency is called cutoff frequency

**Bandwidth** – Bandwidth is the frequency at which the frequency has declined 3dB from its low frequency value

$$M = 1/\sqrt{2}$$

$$|G(j\omega)| = \left|\frac{1}{(1-u^2) + j2\zeta u}\right| = \frac{1}{\sqrt{2}}$$

Solving for $u$,

$$u = \frac{\omega_B}{\omega_n} \Rightarrow \omega_B = \omega_n\sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

$$= \frac{[1 + 2\alpha s]}{\alpha[1 + 2s]} \qquad (\alpha > 1, c)$$

$$\therefore \frac{V_0(s)}{V_2(s)} = \frac{1 + 2\alpha s}{\alpha[1 + 2s]}$$

To find the magnitude at the maximum phase, substitute $\omega = \frac{1}{2\sqrt{\alpha}}$ in magnitude expression.

$$|G_c(j\omega)| = \frac{k[\sqrt{1 + \omega^2\alpha^2 2^2}]}{\alpha[\sqrt{1 + \omega^2 2^2}]} \Bigg|_{\omega = \frac{1}{2\sqrt{\alpha}}}$$

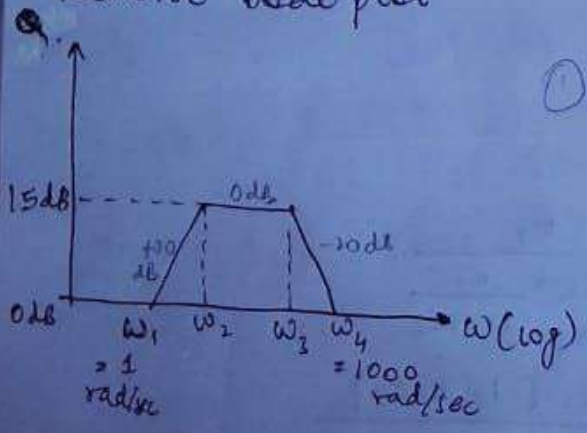
$$|G_c(j\omega)| = \frac{1}{\sqrt{\alpha}}$$

$$20\log_{10}|G_c(j\omega)| = 20\log_{10}\frac{1}{\sqrt{\alpha}}$$

$$= -10\log\alpha$$

7|A|25

Reverse bode plot




① initial slope: 0 dB

$20\log_{10}k = 0$ dB

$k = 1$

② C.F: 1 rad/sec, $\omega_2$, $\omega_3$, 1000 rad/sec

$$G(s)H(s) = \frac{k(1 + j\omega/\omega_1)(1 + j\omega/\omega_4)}{(1 + j\omega/\omega_2)(1 + j\omega/\omega_3)}$$

$$= \frac{k(1 + j\omega/1)(1 + j\omega/1000)}{(1 + j\omega/\omega_2)(1 + j\omega/\omega_3)}$$

① Finding $\omega_2$, $\omega_3$

$y_2 = 15$, $y_1 = 0$

$x_2 = \omega_2$, $x_1 = 1$

$m = 20$

$(y_2 - y_1) = m\log\left(\frac{x_2}{x_1}\right)$

$15 = 20 \log \omega_2$

$\log \omega_2 = 0.75$

$\omega_2 = 5.62$ rad/sec

$-15 = -20 \log \left(\frac{1000}{\omega_3}\right)$

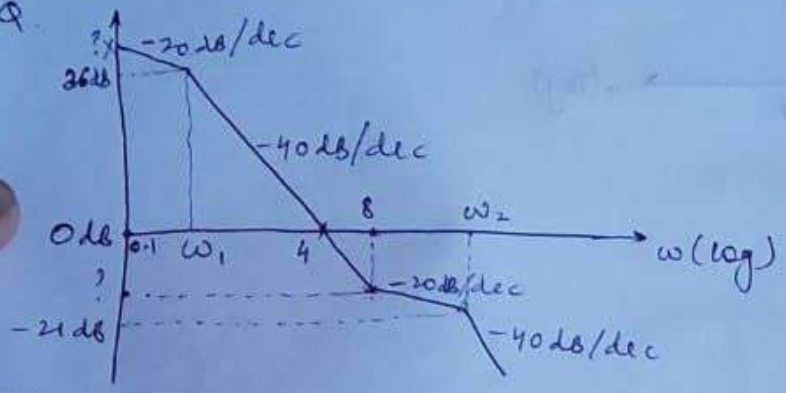
$\log \frac{1000}{\omega_3} = +0.75$

$\frac{1000}{\omega_3} = 5.62.$

$\omega_3 = 177.8$

$\therefore G(s)H(s) = \frac{k(1+j\omega)(1+j\omega/1000)}{(1+j\omega/5.62)(1+j\omega/177.8)}$

Q.




① Initial slope = −20 dB/dec

② CF:

$\omega_1$, 8 rad/s, $\omega_2$

$G(j\omega)H(j\omega) = \frac{k(1+j\omega/8)}{j\omega(1+j\omega/\omega_1)(1+j\omega/\omega_2)}$

$0 - 36 = -40 \log \frac{4}{\omega_1}$

$\frac{36}{40} = \log \frac{4}{\omega_1}$

$\omega_1 = 0.503$ rad/sec

$36 - y = -20 \log \left(\frac{0.501}{0.1}\right)$

$\therefore y = 50$

$20 \log k - 20 \log_{10}(0.1) = 50$

$\therefore k = 31.62$

$y_2 - 0 = -40 \log\left(\frac{8}{4}\right)$

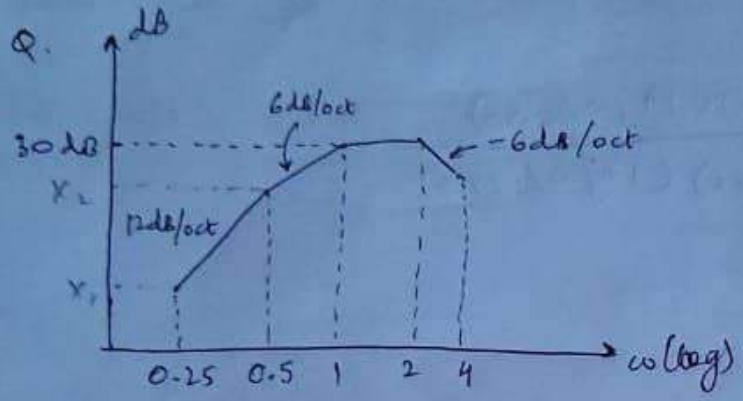
$y_2 = -12.04$

$\therefore -24 - (-12.04) = -20 \log\left(\frac{\omega_2}{8}\right)$

$\omega_2 = 22.4$ rad/sec

$G(j\omega)H(j\omega) = \frac{31.62(1+j\omega/8)}{j\omega(1+j\omega/0.503)(1+j\omega/22.4)}$

Q.




① initial slope = 12dB/oct

② C.F. : 0.25, 0.5, 1, 2, 4

$X_2 - X_1 = 12 \log_2\left(\frac{0.5}{0.25}\right)$

$X_2 - X_1 = 12$ —— ①

$30 - X_2 = 6 \log_2\left(\frac{1}{0.5}\right)$

$30 - 6 = X_2$

$X_2 = 24$ $\quad X_1 = 12$

$6 \log_2^k + 12 \log_2^{0.25} = 12$

$6 \log_2^k = 36$

$\log_2^k = 6$ $\quad \therefore k = 64$

$G(s) = \frac{64s^2}{(1+\frac{s}{2})(1+\frac{s}{1})(1+\frac{s}{0.5})}$

Q. $G(s) = \frac{k}{s(s+2)}$

↳ Given OL TF where $H(s) = 1$, it is ~~compensated~~ derived to design a compensator for the system such that the static velocity constant is given as $K_v = 20$ and phase margin must be atleast 50°.

Ans: $K_v = \lim_{s \to 0} s \cdot G(s) H(s)$

$= \lim_{s \to 0} s \cdot \frac{k}{s(s+2)} \cdot 1$

$= \lim_{s \to 0} \frac{k}{s+2} \cdot 1$

$20 = \frac{k}{2}$

$k = 40$

$G(s) H(s) = \frac{40}{s(s+2)}$

$G(j\omega) H(j\omega) = \frac{40}{j\omega(j\omega+2)} = \frac{40}{2j\omega\left(\frac{j\omega}{2}+1\right)} = \frac{20}{j\omega\left(\frac{j\omega}{2}+1\right)}$

initial slope → $\frac{20}{s}$

$20\log_{10} 20 - 20\log_{10} 0.1 = 46.02$ dB

CF: 2 rad/sec

| factors | C.F | slope | Net slope | freq. range |
|---|---|---|---|---|
| $20/j\omega$ | – | -20 dB/dec | -20 dB/dec | 0.1 to 2 rad/sec |
| $\frac{1}{1+\frac{j\omega}{2}}$ | 2 | -20 dB/dec | -40 dB/dec | 2 to ∞ |

$\angle G(j\omega) H(j\omega) = -90° - \tan^{-1}\frac{\omega}{2}$

Let's say PM = 20

$\phi_m = 50 - 20 + 9°$ (Lag comp)

$= 39°$

$\sin\phi_m = \frac{\alpha-1}{\alpha+1}$ → $0.629(\alpha+1) = \alpha - 1$

$1.629 = 0.371\alpha$ ∴ $\alpha = 4.39$

$|G(j\omega)| \cdot \frac{1}{\sqrt{\alpha}}$

$20\log_{10}|G(j\omega)| - 20\log\frac{1}{\sqrt{\alpha}} = -10\log_{10}(\alpha)$

$= -6.92$

$G(j\omega)H(j\omega) = \frac{k[1+2\tau\omega]}{\alpha[1+\omega 2]}$

$\omega_m = 9$ rad/sec

↳ new gain crossover

$\omega_m = \frac{1}{2\sqrt{\alpha}} \quad \therefore \quad 2 = 0.0527$

$P = \frac{1}{2} = 20$

$z = \frac{1}{\alpha 2} = 4.54$

$\therefore G(j\omega)H(j\omega) = \frac{1}{4.44}\frac{(s+4.55)}{(s+20)}$

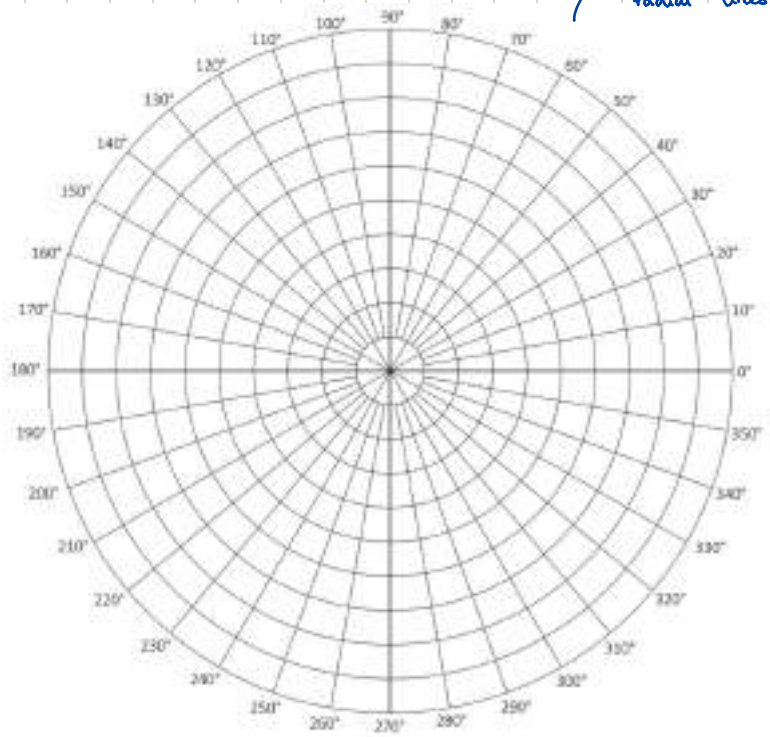
$\frac{1}{4.44}\frac{(s+4.55)}{(s+20)} \times \frac{20}{s(s+2)}$ → the phase margin you get from this must be $\geq 50$. otherwise repeat the process. (change 9 to 12) ↳ 9

# Unit-4

## Polar Plot

→ Plot which can be drawn b/w magnitude & phase angle of $G(j\omega)H(j\omega)$ by varying $\omega$ from 0 to $\infty$

concentric circles : magnitude
radial lines : phase angle




Polar Form : $G(j\omega)H(j\omega) = |G(j\omega)H(j\omega)| \angle G(j\omega)H(j\omega)$

Rectangular Form : $G(j\omega)H(j\omega) = Re[G(j\omega)H(j\omega)] + Im[G(j\omega)H(j\omega)]$

## Steps to Draw Polar Plots

1) Determine $G(s)H(s)$
2) Put $s = j\omega$
3) Write expression for magnitude & phase
4) Calculate magnitude & phase by changing value of freq $\omega$ from 0 to $\infty$ & construct table below.

(In the questions, only 2 values, but find like 6-7 to get correct plot)

https://youtu.be/xWw7z5QzYgE?si=X7r-uPYhVX6D76ZO

Check out for Polar Plots!

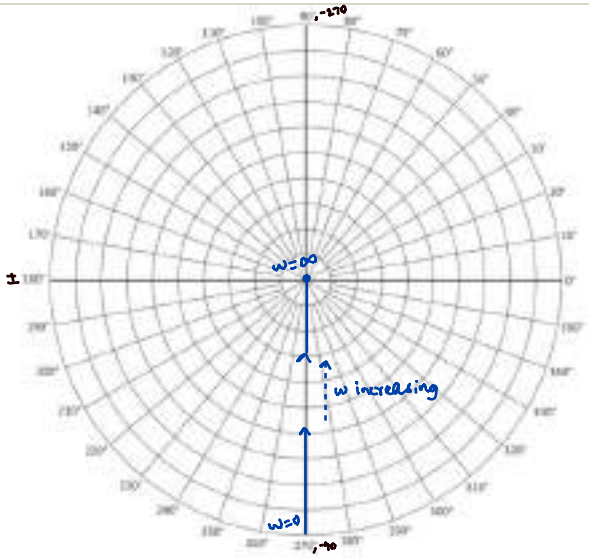
Q. $G(s)H(s) = 1/s$

A. 1) $G(s)H(s) = 1/s$

2) $G(j\omega)H(j\omega) = \frac{1}{j\omega}$

$|G(j\omega)H(j\omega)| = \frac{1}{\omega}$ $\quad \angle G(j\omega)H(j\omega) = -90$

| $\omega$ | M | $\phi$ |
|---|---|---|
| 0 | $\infty$ | -90 |
| $\infty$ | 0 | -90 |



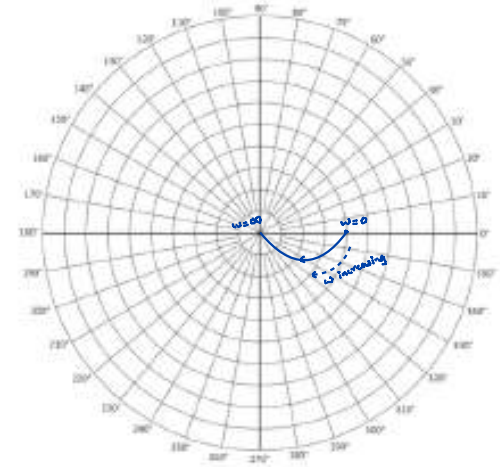

Q. $G(s)H(s) = \frac{1}{1+sT}$

A. 1) $G(s)H(s) = \frac{1}{1+sT}$

2) $G(j\omega)H(j\omega) = \frac{1}{1+j\omega T}$

$|G(j\omega)H(j\omega)| = \frac{1}{\sqrt{1+\omega^2T^2}}$ $\quad \angle G(j\omega)H(j\omega) = -\tan^{-1}(\omega T)$

| $\omega$ | M | $\phi$ |
|---|---|---|
| 0 | 1 | 0 |
| $\infty$ | 0 | -90 |



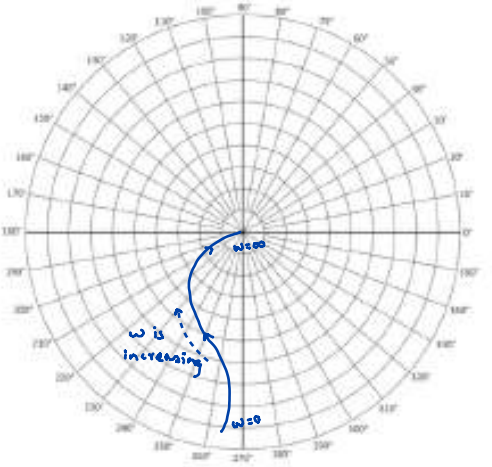

Q. $G(s)H(s) = \frac{1}{s(1+sT)}$

A. 1) $G(s)H(s) = \frac{1}{s(1+sT)}$

2) $G(j\omega)H(j\omega) = \frac{1}{j\omega(1+j\omega T)}$

$|G(j\omega)H(j\omega)| = \frac{1}{\omega\sqrt{1+\omega^2T^2}}$ $\quad \angle G(j\omega)H(j\omega) = -90 - \tan^{-1}(\omega T)$

| $\omega$ | M | $\phi$ |
|---|---|---|
| 0 | $\infty$ | -90 |
| $\infty$ | 0 | -180 |



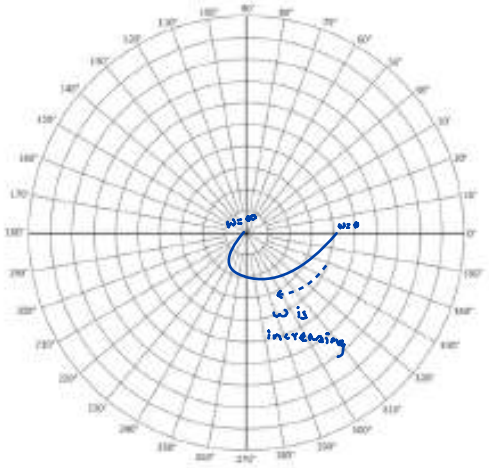
Q. $G(s)H(s) = \frac{1}{(1+sT_1)(1+sT_2)}$

A. 1) $G(s)H(s) = \frac{1}{(1+sT_1)(1+sT_2)}$

2) $G(j\omega)H(j\omega) = \frac{1}{(1+j\omega T_1)(1+j\omega T_2)}$

$|G(j\omega)H(j\omega)| = \frac{1}{\sqrt{1+\omega^2T_1^2}\sqrt{1+\omega^2T_2^2}}$ $\quad \angle G(j\omega)H(j\omega) = -\tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2)$

| $\omega$ | M | $\phi$ |
|---|---|---|
| 0 | 1 | 0 |
| $\infty$ | 0 | -180 |



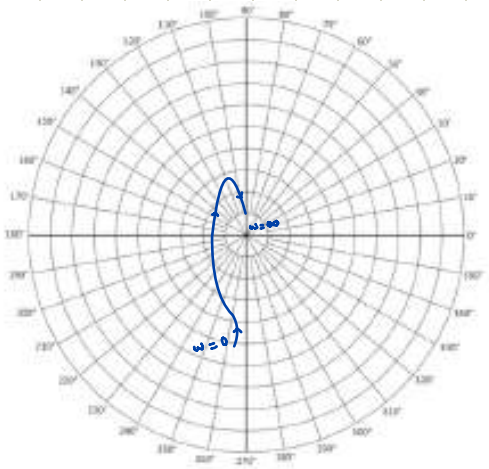

Q. $G(s)H(s) = \frac{1}{s(1+sT_1)(1+sT_2)}$

A. 1) $G(s)H(s) = \frac{1}{s(1+sT_1)(1+sT_2)}$

2) $G(j\omega)H(j\omega) = \frac{1}{j\omega(1+j\omega T_1)(1+j\omega T_2)}$

$|G(j\omega)H(j\omega)| = \frac{1}{\omega\sqrt{1+\omega^2T_1^2}\sqrt{1+\omega^2T_2^2}}$ $\quad \angle G(j\omega)H(j\omega) = -90 - \tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2)$

| $\omega$ | M | $\phi$ |
|---|---|---|
| 0 | $\infty$ | -90 |
| $\infty$ | 0 | -270 |



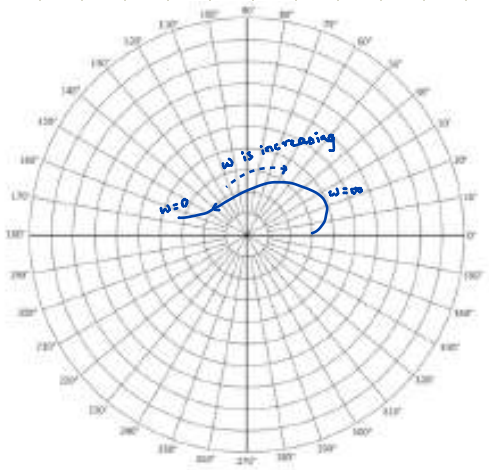

Q. $G(s)H(s) = \frac{1}{s^2(1+sT_1)(1+sT_2)}$

A. 1) $G(s)H(s) = \frac{1}{s^2(1+sT_1)(1+sT_2)}$

2) $G(j\omega)H(j\omega) = \frac{1}{(j\omega)^2(1+j\omega T_1)(1+j\omega T_2)}$

$|G(j\omega)H(j\omega)| = \frac{1}{\omega^2\sqrt{1+\omega^2T_1^2}\sqrt{1+\omega^2T_2^2}}$ $\quad \angle G(j\omega)H(j\omega) = -180 - \tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2)$

| $\omega$ | M | $\phi$ |
|---|---|---|
| 0 | $\infty$ | -180 |
| $\infty$ | 0 | -360 |



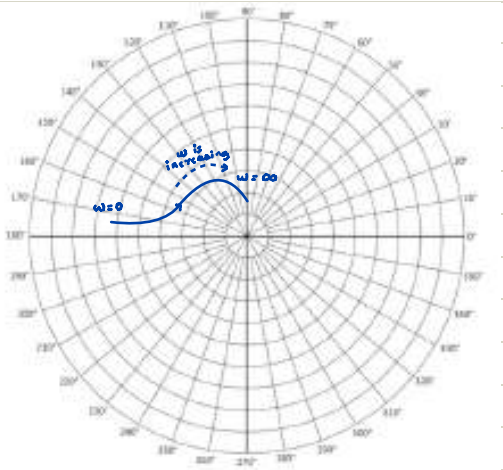
Q. $G(s)H(s) = \frac{1}{s^2(1+sT_1)(1+sT_2)}$

A. 1) $G(s)H(s) = \frac{1}{s^2(1+sT_1)}$

2) $G(j\omega)H(j\omega) = \frac{1}{(j\omega)^2(1+j\omega T_1)}$

$|G(j\omega)H(j\omega)| = \frac{1}{\omega^2\sqrt{1+\omega^2T_1^2}}$ $\quad \angle G(j\omega)H(j\omega) = -180 - \tan^{-1}(\omega T_1)$

| $\omega$ | M | $\phi$ |
|---|---|---|
| 0 | $\infty$ | -180 |
| $\infty$ | 0 | -270 |




Q. $G(s)H(s) = \frac{10}{s(s+1)(s+2)}$. Sketch the polar plot & calculate GM in dB & Comment stability

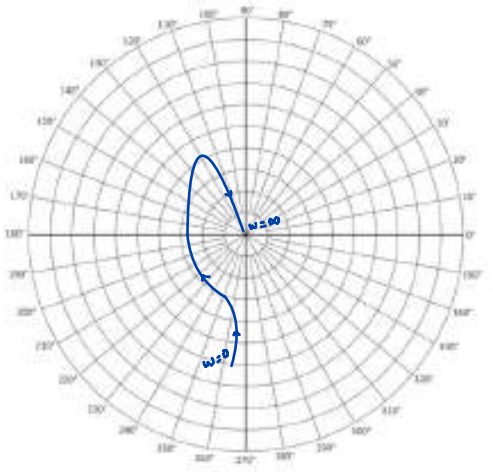
A. 1) $G(s)H(s) = \frac{10}{s(s+1)(s+2)}$

2) $G(j\omega)H(j\omega) = \frac{10}{j\omega(j\omega+1)(j\omega+2)}$

3) $|G(j\omega)H(j\omega)| = \frac{10}{\omega\sqrt{\omega^2+1}\sqrt{\omega^2+4}}$

$\angle G(j\omega)H(j\omega) = -90 - \tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{2}\right)$

| $\omega$ | M | $\phi$ |
|---|---|---|
| 0 | $\infty$ | -90 |
| $\infty$ | 0 | -270 |




$\underbrace{\frac{-j\,10\omega(2-\omega^2)}{\omega^2(1+\omega^2)(4+\omega^2)}}_{\text{imag}} - \underbrace{\frac{30\omega^2}{\omega^2(1+\omega^2)(4+\omega^2)}}_{\text{real}}$

## Mapping Contours in the s-plane

→ Contour Map is a trajectory in one plane mapped to another plane

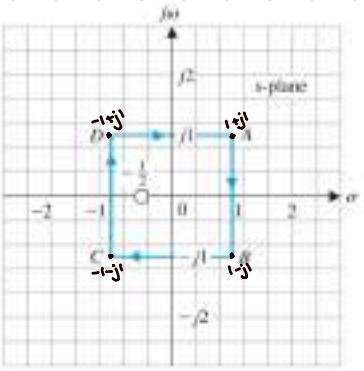
$s = \sigma + j\omega$

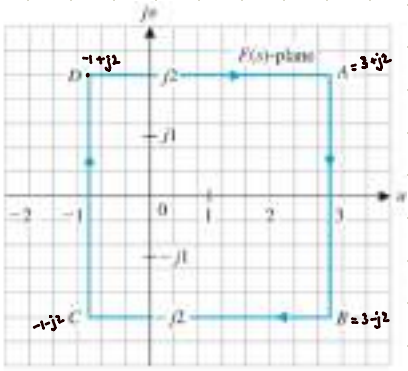
$F(s) = u + jv$

ex: $F(s) = 2s + 1$

$= 2(\sigma + j\omega) + 1$

$= (2\sigma + 1) + j(2\omega)$




| | s | F(s) |
|---|---|---|
| A | $1+j1$ | $3+j2$ |
| | $1$ | $3$ |
| B | $1-j1$ | $3-j2$ |
| | $-j1$ | $1-j2$ |
| C | $-1-j1$ | $-1-j2$ |
| | $-1$ | $1$ |
| D | $-1+j1$ | $-1+j2$ |
| | $j1$ | $1+j2$ |

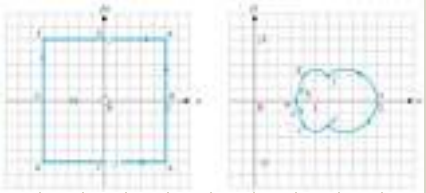
ex: $F(s) = \frac{s}{s+2} = \frac{\sigma+j\omega}{\sigma+j\omega+2} = \frac{(\sigma+j\omega)(\sigma+2-j\omega)}{(\sigma+2)^2+\omega^2} = \underbrace{\frac{\sigma^2+2\sigma+\omega^2}{(\sigma+2)^2+\omega^2}}_{u} + \underbrace{\frac{j2\omega}{(\sigma+2)^2+\omega^2}}_{v}$

## Cauchy's Theorem (Principle of Argument)

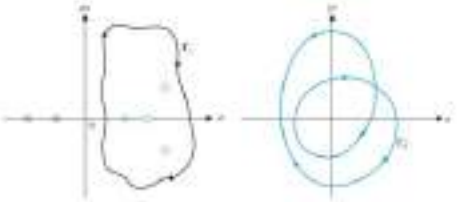
→ If a contour encircles z zeros & p poles of F(s) & doesn't pass through any pole (or) zero of F(s) & it travels in clockwise direction along contour } Then the contour encircles origin of F(s) $N = z - p$ times (ccw)

ccw: counter-clockwise / Anti-clockwise ↺

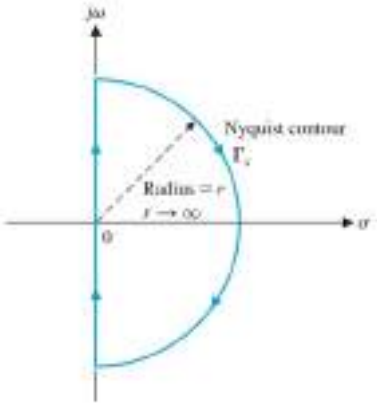
ex:



$F(s) = \frac{s}{s+\frac{1}{2}} \Rightarrow N = z - p = 1 - 1 = 0$



$\Rightarrow N = 3 - 1 = 2$




## Nyquist Criterion

→ For a system to be stable, all the zeros of F(s) must lie on left hand of s-plane

So, choose a contour that encircles entire right hand of s-plane and then

Use cauchy's theorem, $Z = N + P$

So, in general,

$$G(s)H(s) = \frac{K(s+z_1)(s+z_2)\dots(s+z_n)}{(s+p_1)(s+p_2)\dots(s+p_n)}$$

Characteristic eqⁿ,

$$F(s) = 1 + G(s)H(s) = 1 + \frac{K(s+z_1)(s+z_2)\dots(s+z_n)}{(s+p_1)(s+p_2)\dots(s+p_n)}$$

$$= \frac{(s+z_1')(s+z_2')\dots(s+z_n')}{(s+p_1)(s+p_2)\dots(s+p_n)}$$

→ Nyquist Stable Criterion :

A feedback system is stable iff contour in F(s) doesn't encircle the (-1, 0) point when no. of poles of F(s) in right hand side of s-plane = 0

(or)

A feedback system is stable iff for contour the no. of c.c.w encirclements of (-1, 0) point = no. of poles of F(s) with positive real parts

ex: $G(s)H(s) = \frac{K}{(s+1)(0.1s+1)}$

$s+1=0 \Rightarrow s=-1$

$0.1s+1=0 \Rightarrow s=-10$

$s = j\omega$, $\omega \to 0$ to $\infty$

$$L(j\omega) = \frac{K}{(j\omega+1)(0.1j\omega+1)}$$

Assume $K = 100$,

$$|L(j\omega)| = \frac{100}{\sqrt{\omega^2+1}\sqrt{(0.1\omega)^2+1}}$$

$$\angle L(j\omega) = 0 - \tan^{-1}(\omega) - \tan^{-1}(0.1\omega)$$

these can be any values

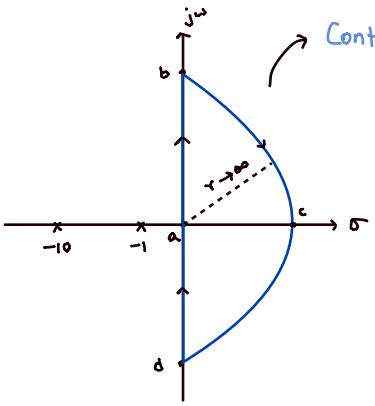
| ω | $\lvert L(j\omega)\rvert$ | $\angle L(j\omega)$ | $Re(L(j\omega))$ | $Im(L(j\omega))$ |
|---|---|---|---|---|
| 0 | 100 | 0 | 100 | 0 |
| 1 | 70.36 | -50.71 | 44.55 | 54.45 |
| 10 | 7.036 | -129.3 | -4.45 | 5.44 |
| ∞ | 0 | -180 | 0 | 0 |

$$L(j\omega) = \frac{100(1-j\omega)(1-0.1j\omega)}{(1+\omega^2)(1+0.01\omega^2)} = \frac{100(1-0.1\omega^2)+j110\omega}{(1+\omega^2)(1+0.01\omega^2)}$$

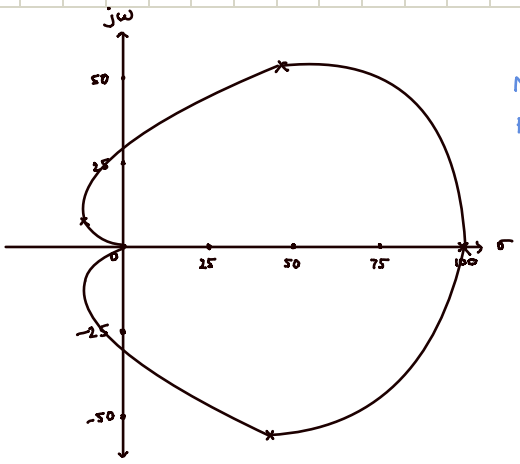
To know where it cuts jω axis, $Re(j\omega) = 0 \Rightarrow \frac{100(1-0.1\omega^2)}{(1+\omega^2)(1+0.01\omega^2)} = 0 \Rightarrow \omega^2 = 10$

$\omega = 3.16$ rad/s

$$Im(j\omega)\big|_{\omega=3.16} = \frac{j110(-3.16)}{(1+10)(1+0.1)} = -j28.76$$




Contour in s-plane




Nyquist Diagram From the table

Image is symmetric cuz

$Re[G(-j\omega)] = Re[G(j\omega)]$

$Im[G(-j\omega)] = -Im[G(j\omega)]$

Continuation of problem,

Let $s = Re^{j\theta}$

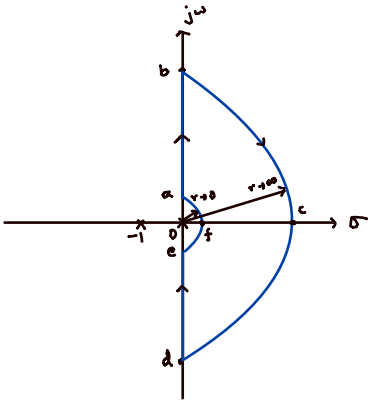
$$L(s) = \frac{100}{(1+Re^{j\theta})(0.1Re^{j\theta}+1)}$$

$$\lim_{R\to\infty} L(s) = \lim_{R\to\infty} \frac{100}{0.1R^2e^{j2\theta}\left(1+\frac{1}{Re^{j\theta}}\right)\left(1+\frac{1}{0.1Re^{j\theta}}\right)}$$

$$= 0 \times e^{-j2\theta}$$

$\theta$ varies from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ $\Rightarrow$ $L(s)$ varies from $\pi$ to $-\pi$

Q. $L(s) = \frac{5}{s(s+1)}$




A. Assume $s = j\omega$,

$$L(j\omega) = \frac{5}{j\omega(j\omega+1)} = \frac{5(-j\omega)(1-j\omega)}{\omega^2(1+\omega^2)} = \frac{5(-\omega^2-j\omega)}{\omega^2+\omega^4}$$

$$|L(j\omega)| = \frac{5}{\omega\sqrt{1+\omega^2}}$$

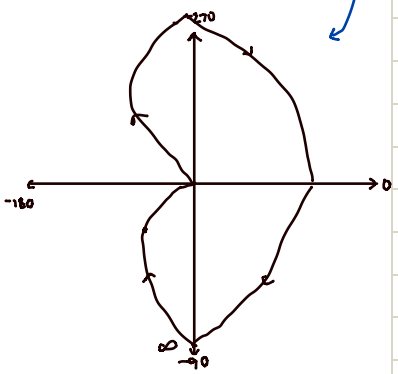
$$\angle L(j\omega) = -\tan^{-1}(\infty) - \tan^{-1}(\omega)$$

$$= -90 - \tan^{-1}(\omega)$$

$Re(j\omega) = 0 \Rightarrow \frac{-5\omega^2}{\omega^2+\omega^4} = 0 \Rightarrow \omega = 0$

$Im(j\omega)\big|_{\omega=0} = \infty \Rightarrow$ Point at which it cuts jω axis

| ω | $\lvert L(j\omega)\rvert$ | $\angle L(j\omega)$ | $Re(L(j\omega))$ | $Im(L(j\omega))$ |
|---|---|---|---|---|
| 0 | ∞ | -90 | ∞ | ∞ |
| 1 | 3.53 | -135 | -2.5 | -2.5 |
| 10 | 0.05 | -174.3 | -0.0495 | $-4.95\times10^{-3}$ |
| ∞ | 0 | -180 | 0 | 0 |




If $s = -j\omega$ $\Rightarrow$ It is mirror image of 'ab' is 'ed'

Now, $s = Re^{j\theta}$ & $\theta \to 90^\circ$ to $-90^\circ$ (Path bcd)

$$\lim_{R\to\infty} L(j\omega) = \lim_{R\to\infty} \frac{5}{Re^{j\theta}(Re^{j\theta}+1)} = \lim_{R\to\infty} \frac{5}{R^2e^{j2\theta}\left(1+\frac{1}{Re^{j\theta}}\right)} = 0\angle -2\theta \Rightarrow -180 \text{ to } 180$$

Now, $s = re^{j\theta}$ & $\theta \to -90$ to $90$ (Path efa) $\Rightarrow$

$$\lim_{r\to0} \frac{5}{re^{j\phi}(re^{j\phi}+1)} = \lim_{r\to0} \frac{5}{re^{j\phi}} = \infty e^{-j\phi} \Rightarrow \infty \text{ from } 90^\circ \text{ to } -90^\circ$$

→ This part is done to verify how we have to draw different paths in the plot

Stability Analysis

→ Looking at $L(s)$, no. of poles of $L(s)$ in RHP is 0 $(P = 0)$

→ Looking at the contour, it doesn't encircle $(-1, 0)$ $(N = 0)$

→ Since $Z = N+P = 0$, there isn't a closed loop hole in RHP of s-plane

→ Closed loop system is stable for all values of $K > 0$

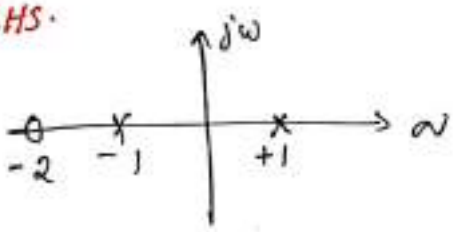
RHP: Right hand plane

So, from prev q, $Z = N+P = 0+0 = 0$, closed loop system is stable

∴ The closed loop system is stable

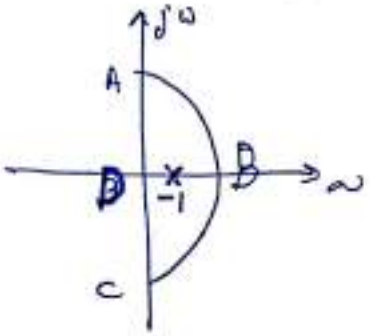
③ Eg: System with open loop TF poles on RHS.

$$G(s)H(s) = \frac{s+2}{(s+1)(s-1)}$$




Poles = +1, -1, zeros = -2, open loop TF unstable

Hence P = 1




① Section DA  Put $s = j\omega$.

$$G(j\omega)H(j\omega) = \frac{(j\omega+2)}{(j\omega+1)(j\omega-1)}$$

$$|G(j\omega)H(j\omega)| = \frac{\sqrt{\omega^2+4}}{\sqrt{\omega^2+1}\sqrt{\omega^2+1}}$$

$$\angle G(j\omega)H(j\omega) = \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}(\omega) + \tan^{-1}(-\omega)$$

$$\left(180^\circ - \tan^{-1}\omega.\right)$$

$$= \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}(\omega) - 180^\circ + \tan^{-1}(\omega)$$

$$= -180 + \tan^{-1}\left(\frac{\omega}{2}\right)$$

| When ω | M | ϕ |
|---|---|---|
| 0 | $\sqrt{4} = 2$. | $-180^\circ$. |
| ∞ | 0 | $-180^\circ + 90^\circ = -90^\circ$ |

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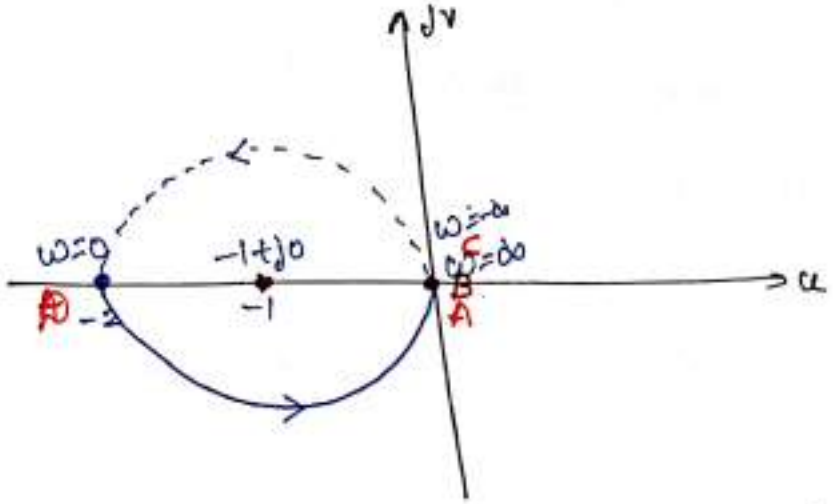


Section Portion / ABC :

$\omega \Rightarrow 90^\circ \to 0 \to -90^\circ$

$$\lim_{R\to\infty} L(j\omega) = \lim_{R\to\infty} \frac{Re^{j\omega}+2}{(Re^{j\omega}+1)(Re^{j\omega}-1)} = \lim_{R\to\infty} \frac{Re^{j\omega}\left(1+\frac{2}{Re^{j\omega}}\right)}{Re^{j\omega}\left(1+\frac{1}{R\cdot e^{j\omega}}\right)Re^{j\omega}\left(1+\right.}$$

$$= \lim_{R\to\infty} \frac{1+\frac{2}{Re^{j\omega}}}{Re^{j\omega}\left(1+\frac{1}{Re^{j\omega}}\right)\left(-1+\frac{1}{Re^{j\omega}}\right)} = 0$$

$$= \lim_{R\to\infty} \frac{Re^{j\omega}}{Re^{j\omega}\cdot R\cdot e^{j\omega}} = 0 \quad \left| -90^\circ \to 0 \to 90^\circ \right.$$

Hence infinite semicircle ABC on the s-plane maps onto the origin of L(s) plane.

Now the Nyquist plot encircles $-1+j0$ pt once in counter clockwise direction

$\therefore N = -1$.

$Z$ = no. of RH poles of CLTF

For stability $Z$ should be "0", that is there should be no poles in RH-s plane in CLTF.

$P$ = no of Open loop poles in RH-s plane. $= 1$

$$Z = N + P = -1 + 1 = 0.$$

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Hence there are no closed loop poles in RHP.

∴ the closed loop system is stable, even though the open loop is unstable.

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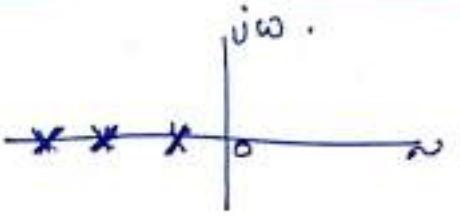
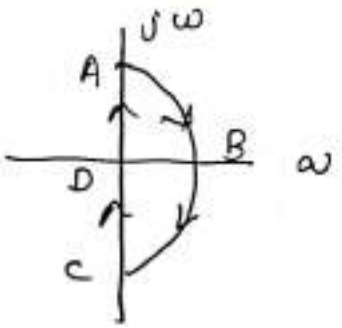
(4) $G(s)H(s) = \dfrac{100}{(s+1)(s+2)(s+3)}$.




Zero's = nil  Poles = −1, −2, −3  ∴ $\boxed{P=0}$




(1) for Section OA

$s \to j\omega$.

$$G(j\omega)H(j\omega) = \frac{100}{(j\omega+1)(j\omega+2)(j\omega+3)}$$

$$|G(j\omega)H(j\omega)| = \frac{100}{\sqrt{\omega^2+1}\ \sqrt{\omega^2+4}\ \sqrt{\omega^2+9}}$$

$$\angle G(j\omega)H(j\omega) = -\tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{3}\right)$$

when $\omega = 0$.  $M = \dfrac{100}{1 \times \sqrt{4} \times \sqrt{9}} = \dfrac{100}{6} = 16.66$  $\phi(\omega) = 0^\circ$

when $\omega = \infty$  $M = 0$  $\phi(\omega) = -90 - 90 - 90 = -270^\circ$

Hence the magnitude varies from $16.66\angle 0^\circ$ to $0\angle -270^\circ$ passing through $-90^\circ$ & $-180^\circ$.

Note: $\tan^{-1}(A) \pm \tan^{-1}(B) = \tan^{-1}\left(\dfrac{A \pm B}{1 \mp AB}\right)$

∴ The point lies in 4th, 3rd & 2nd quadrants.

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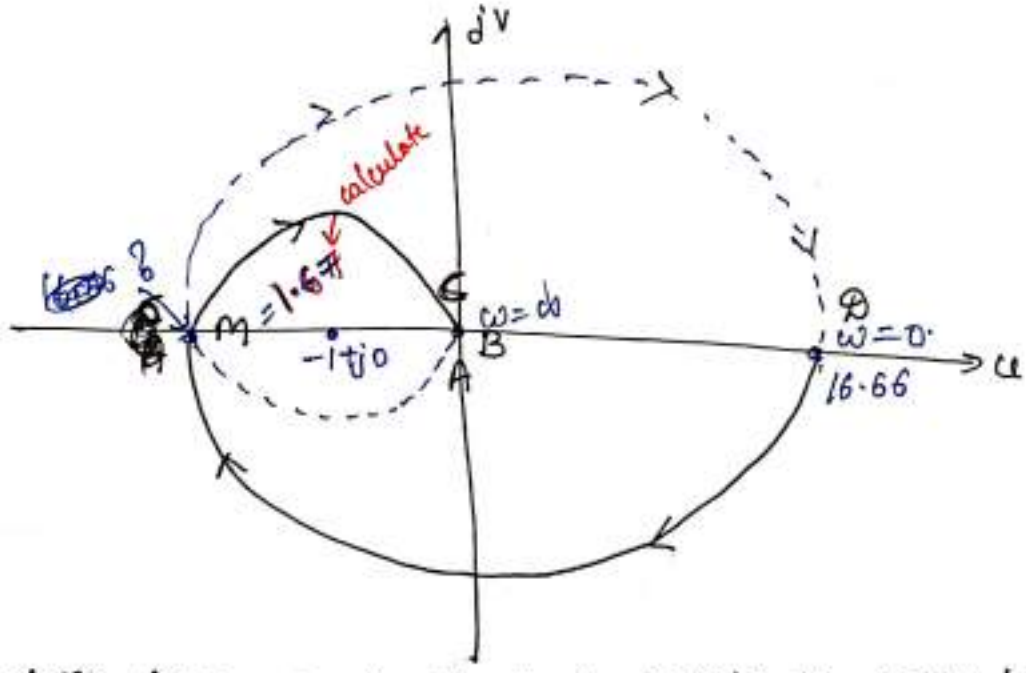


Section ABC ⇒ mapped to origin in F(s) plane.

Path "C-D" is the mirror image of "DA"

We now need to find the point 'm', where the Nyquist plot intercepts -ve real axis in F(s) plane (i.e G(s)H(s) plane)

at m, the Imag part = 0. or another method.

at m phase angle = $-180^\circ$

$$-\tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{3}\right) = -180^\circ$$

$$+\tan^{-1}\left(\frac{\omega + \frac{\omega}{2}}{1 - \omega\cdot\frac{\omega}{2}}\right) + \tan^{-1}\left(\frac{\omega}{3}\right) = 180^\circ.$$

$$\tan^{-1}\left(\frac{3\omega}{2-\omega^2}\right) + \tan^{-1}\left(\frac{\omega}{3}\right) = 180^\circ$$

$$\tan^{-1}\left[\frac{\frac{3\omega}{2-\omega^2} + \frac{\omega}{3}}{1 - \frac{3\omega^2}{3(2-\omega^2)}}\right] = 180^\circ.$$

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$9\omega + \omega^2(2-\omega^2) = 0$ $\quad\quad$ $\tan(180^\circ) = 0$

$11\omega - \omega^3 = 0$

Since $\omega \neq 0$.

$11 - \omega^2 = 0$

$\omega^2 = 11$

or $\omega = \sqrt{11}$ rad/sec.

Magnitude of $G(j\omega)H(j\omega)$ at $\omega = \sqrt{11}$ is

Pf $\quad M = \dfrac{100K}{\sqrt{\omega^2+1}\sqrt{\omega^2+4}\sqrt{\omega^2+9}} \Big|_{\omega=\sqrt{11}}$

$$= \frac{100}{\sqrt{11+1}\sqrt{11+4}\sqrt{11+9}} = \frac{100}{\sqrt{12}\sqrt{15}\sqrt{20}}$$

$$= \frac{100}{60} = 1.67 > 1.$$

Since $M > 1$, means $-1 + j0$ has been encircled twice in the clockwise direction

$\therefore N = 2.$

$Z = N + P = 2 + 0 = 2.$

$\therefore$ the closed loop system is unstable with 2 closed loop poles in RHP of s-plane.

Soln To make it stable reduce 100 to value of K, such that M is less than 1

$M = \dfrac{K}{60}$ $\quad$ for $M = 1$ $\quad K = 60$

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For marginal stability $K=60$ | for stable $0 < K < 60$

Eg: if $K=50$ then $m=0.833$.

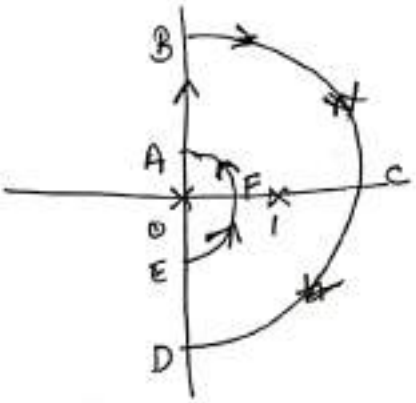
$N=0$

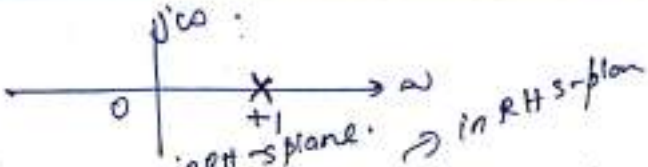
& $Z = N+P = 0+0 = 0$

System will be stable.

---

(5) $G(s)H(s) = \frac{5}{s(s-1)}$



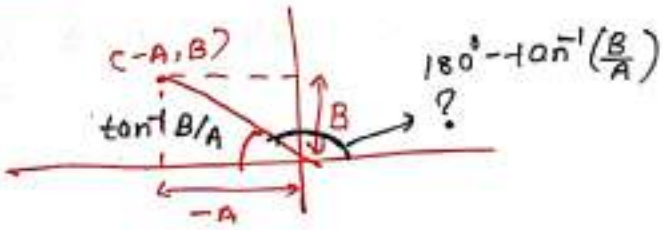

$P = 1$ in RH s-plane. & $Z = 0$ in RH s-plan

Note

$-A+jB = \sqrt{A^2+B^2} \angle 180^\circ - \tan^{-1}\left(\frac{B}{A}\right)$

2nd Quadrant

$jB = B \angle 90^\circ$




(a) for section AB

$s \to j\omega$.

$$G(j\omega)H(j\omega) = \frac{5}{\omega \cdot \sqrt{\omega^2+1}}$$

$$\angle G(j\omega)H(j\omega) = -90^\circ + \tan^{-1}\left(\frac{\omega}{-1}\right)$$

$$= -90^\circ + (180^\circ - \tan^{-1}(\omega))$$

$$= -90^\circ - 180^\circ + \tan^{-1}(\omega)$$

$$= -270^\circ + \tan^{-1}(\omega)$$

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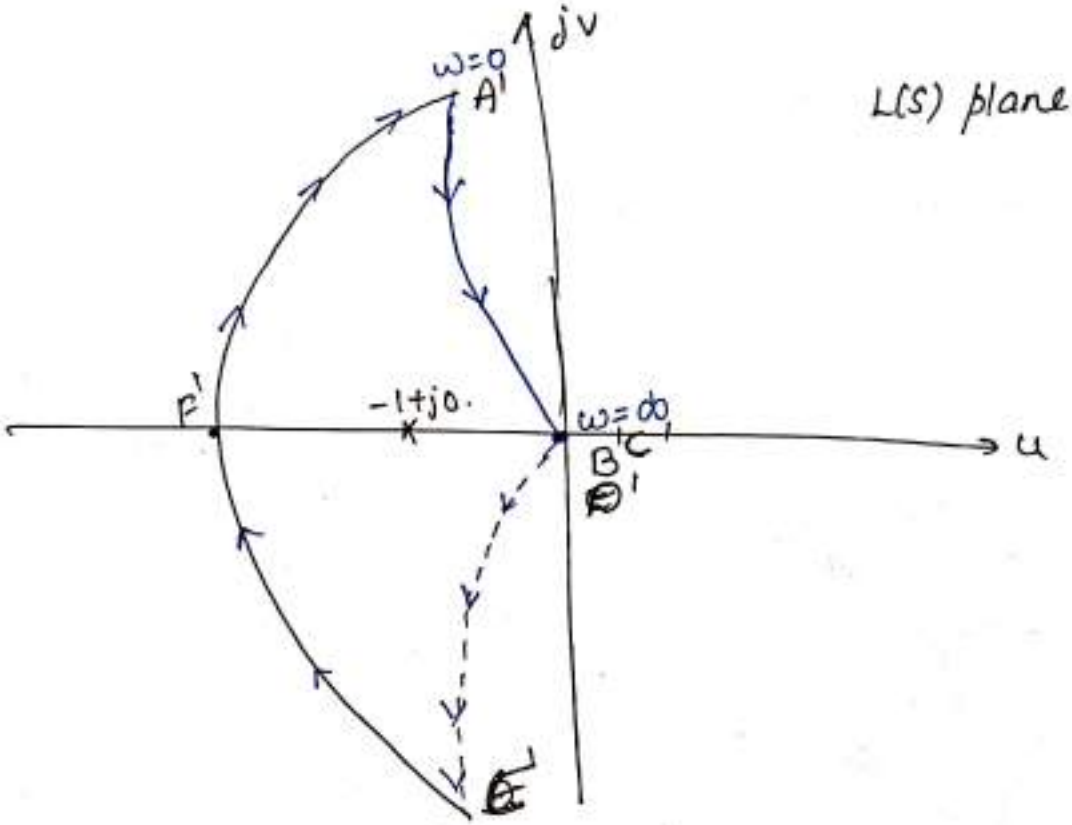
When $\omega = 0$ $\quad M = \infty$ $\quad \phi = -270^\circ$. $\Rightarrow$ $\infty \angle -270^\circ$.

$\omega = \infty$ $\quad M = 0$ $\quad \phi = -270^\circ + 90^\circ \Rightarrow 0 \angle -180^\circ$

(b) Section ED : mirror image of AB.




(c) For the path BCD $\Rightarrow$ The mapping is to origin.

(d) For path EFA $\Rightarrow$ $\omega = R \cdot e^{j\theta}$. $\quad R \to 0$

$\theta \to -90^\circ \to 0^\circ \to 90^\circ$.

$$\lim_{R\to 0} L(s) = \lim_{R\to 0} \frac{5}{R e^{j\theta}(Re^{j\theta} - 1)}$$

$$= \lim_{R\to 0} \frac{5}{R e^{j\theta}.(-1)} = \frac{5}{R \angle \theta + 180^\circ} = -\angle \theta + 180^\circ$$

Angle

As $\theta \to -90^\circ \to 0^\circ \to +90^\circ$

$\phi \Rightarrow -90^\circ \to -180^\circ \to -270^\circ$

$\lim_{R\to 0} |L(s)| = \infty$ $\quad \therefore$

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$$\angle G(j\omega)H(j\omega)\Big|_{\omega=\omega_{pc}} = -180^\circ$$

$$-90^\circ - \tan^{-1}(-\omega) = -90^\circ - (180^\circ - \tan^{-1}(\omega))$$

$$= -270^\circ + \tan^{-1}(\omega) = -180^\circ$$

$$\therefore \tan^{-1}(\omega) = 270^\circ - 180^\circ$$

$$\tan^{-1}(\omega) = 90^\circ$$

$$\omega = \infty$$

Now the nyquist plot encircles $-1+j0$, once in the clockwise direction $\therefore N = 1$ $P = 1$ (RHS Pole)

$$Z = N + P = 1 + 1 = 2$$

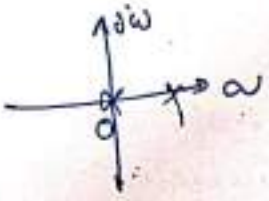
$\therefore$ Closed loop system is unstable with 2 poles in RHP

$$1 + \frac{5}{s(s-1)} = 0.$$

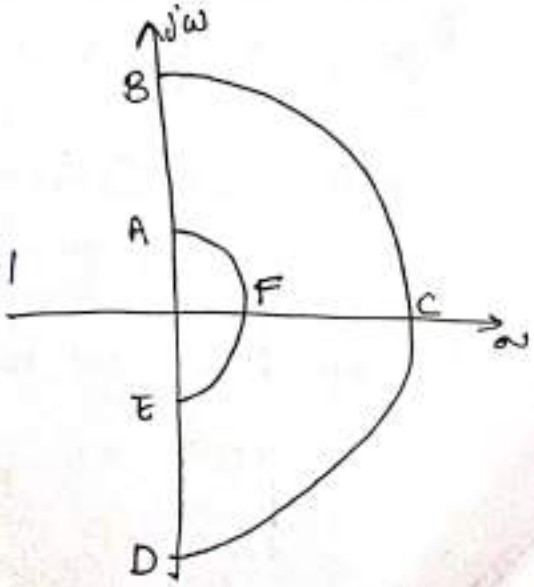
$$\boxed{s^2 - s + 5 = 0}$$

---

(6) $G(s)H(s) = \frac{5}{2(1-s)}$ poles 0, $1-s=0$ or $s=+1$




$P = 1$




ie. for stability $N = -1$

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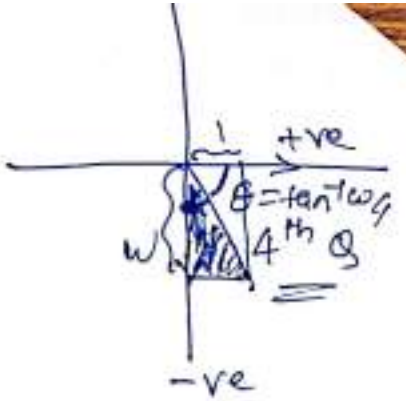
① PATH AB $s = j\omega$




$$G(j\omega)H(j\omega) = \frac{5}{j\omega(1-j\omega)}$$

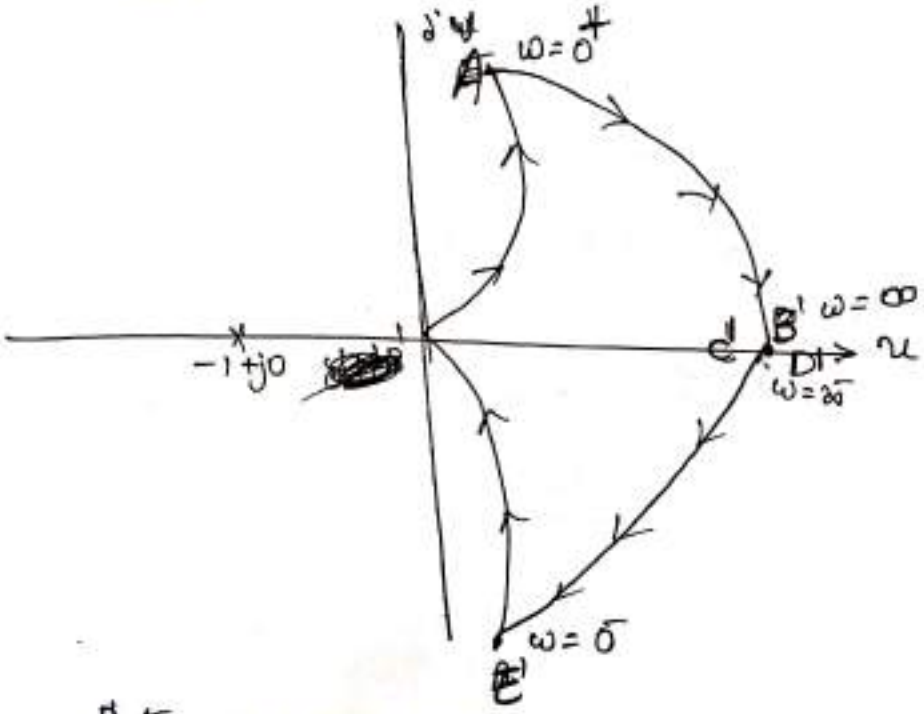
Recall $A - jB = \sqrt{A^2+B^2} \angle -\tan^{-1}(B/A)$

$$\therefore |G(j\omega)H(j\omega)| = \frac{5}{\omega\sqrt{1+\omega^2}}$$

$$\angle G(j\omega)H(j\omega) = -90^\circ + \tan^{-1}\left(\frac{\omega}{1}\right)$$

For $\omega = 0$, $M = \infty$ $\quad \angle = -90^\circ$

$\omega = \infty$, $M = 0$ $\quad \angle = -90^\circ + 90^\circ = 0$




② Path $\overline{DE}$ ⇒ mirror image of A'B'

③ Path BCD: Infinite semicircle in s-plane, maps into origin.

④ Path EFA: $\omega$ vary from $-90 \to 0^\circ \to +90^\circ$

let $s = Re^{j\theta}$ $\quad R \to 0$

$$\lim_{R\to 0} L(s) = \lim_{R\to 0} \frac{5}{Re^{j\theta}(1-Re^{j\theta})} = \lim_{R\to 0} \frac{5}{Re^{j\theta}}$$

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$\infty \angle -\theta$

when $\theta \Rightarrow -90 \to 0 \to 90^\circ$

$\infty \angle +90^\circ \to 0 \to -90^\circ$

From the Nyquist plot $\Gamma_L$ doesnot encircle $-1+j0$ ; $N=0$

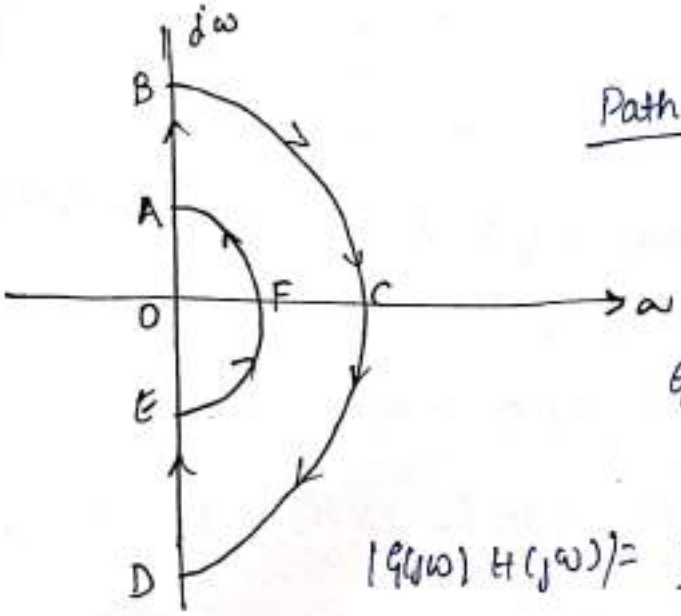
$\therefore Z = N+P = 0+1 = 1$

Therefore the closed loop system is unstable with one pole in the RHS of s plane.

---

(7) $G(s)H(s) = \dfrac{K(s\tau_a+1)}{s^3}$ ; $K\tau_a > 0$.

Poles = 0 (3 at origin) ; $s_{\infty} = -\frac{1}{\tau_a}$ ; $\boxed{P=0}$




Path AB ω varies from $0^+$ to $\infty$

$s = j\omega$.

$$G(j\omega)H(j\omega) = \frac{K(j\omega\tau_a+1)}{(j\omega)^3}$$

$$|G(j\omega)H(j\omega)| = \frac{K\sqrt{\omega^2\tau^2+1}}{\omega^3}$$

$$\angle G(j\omega)H(j\omega) = -270^\circ + \tan^{-1}\omega\tau_a$$

when $\omega = 0^+$ ; $G(j\omega)H(j\omega) = \infty \angle -270^\circ$

when $\omega = \infty$ ; $G(j\omega)H(j\omega) = 0 \angle -270^\circ + 90^\circ = 0 \angle -180^\circ$.

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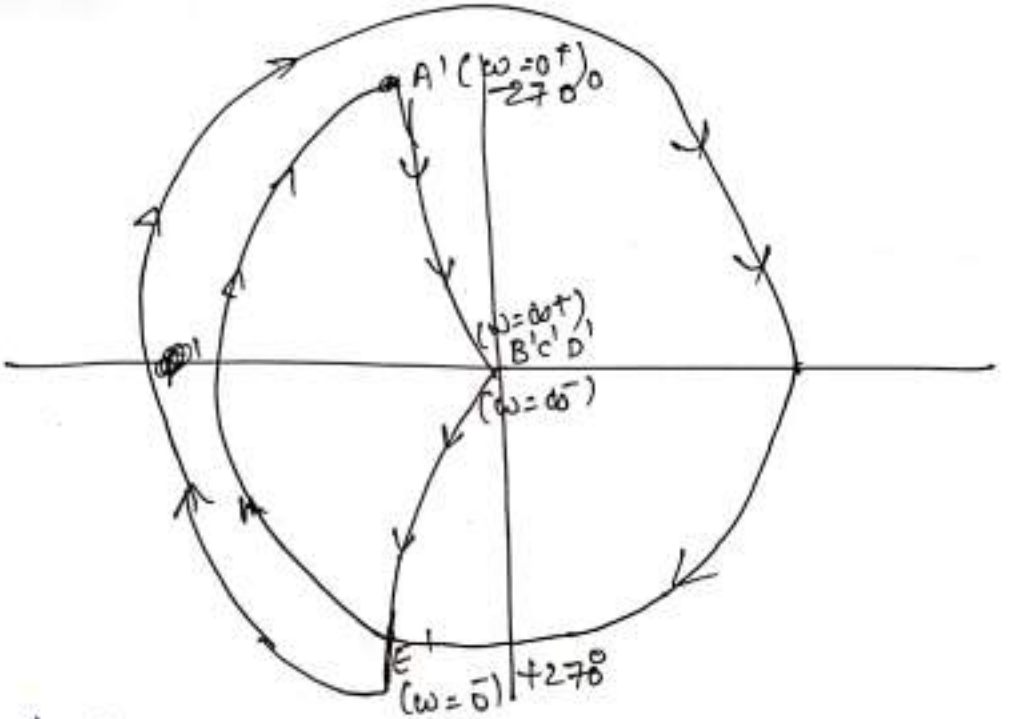


(2) Path DE is the mirror image of A'B'

(3) Pat EFA, $s = Re^{j\theta}$ $\theta$ varies from $-90° \rightarrow 0° \rightarrow +90°$

$$\lim_{R\to 0} \frac{k(R\cdot e^{j\theta}\cdot T_a + 1)}{(R\cdot e^{j\theta})^3} = \lim_{R\to 0} \frac{k}{R^3 e^{j3\theta}}$$

$$= \infty \angle -3\theta \quad \text{as } \theta \Rightarrow -90° \rightarrow 0° \rightarrow 90°$$

$$= \infty \angle 270° \rightarrow \infty \angle 0° \rightarrow \infty \angle -270°$$

The Nyquist plot encircles $-1+j0$ point twice in clockwise

Hence $N = 2$.

$$\therefore Z = N + P = 2 + 0 = 2.$$

The closed loop system is unstable with 2 RHS-poles.

(4) BCD maps to origin.

Stability Analysis

The Nyquist plot encircles $-1+j0$ point twice in clockwise

Hence $N = 2$. $\therefore Z = N + P = 2 + 0 = 2$

$\therefore$ The CL-System is unstable with 2 RHP poles.

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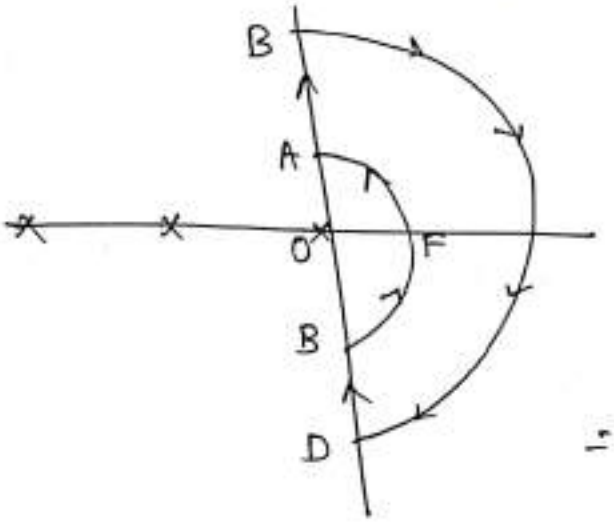
## (8) System with 3 poles

$$G(s)H(s) = \frac{K}{s(\tau_1 s+1)(\tau_2 s+1)}$$

O/L poles $= 0, -1/\tau_1, -1/\tau_2$ $\therefore$ RHS $P = 0$




① PATH AB

$\omega$ $0^+ \to \infty$

$s \to j\omega$.

$$L(j\omega) = \frac{K}{(j\omega)(j\omega\tau_1+1)(j\omega\tau_2+1)}$$

$$= \frac{K}{\omega^2\sqrt{1+\omega^2\tau_1^2}\sqrt{1+\omega_2^2\tau^2}}$$

$$\angle L(j\omega) = -90^\circ - \tan^{-1}\omega\tau_1 - \tan^{-1}\omega\tau_2.$$

① Seg AB

When $\omega = 0^+$ $M = \infty$ $\phi = -90^\circ$.

$\omega = \infty$ $M = 0$ $\phi = -270^\circ$ (i.e $-90^\circ - 90^\circ - 90$)

As $\omega$ varies from $0^+$ to $\infty$, the angle varies from $-90^\circ$ to $-270^\circ$ passing $-180^\circ$. So the plot has to be in the 3rd & 2nd Quadrant.

→ The plot A'B' would intersect the negative real axis at some point $M_1$ which would be $\geq -1$

→ To find the point $M_1$, we will set the imag part of $u+jv$ to zero. & phase angle of $L(j\omega)$ to $-180^\circ$

$$-90^\circ - \tan^{-1}\omega\tau_1 - \tan^{-1}\omega\tau_2 = -180^\circ$$

$$\tan^{-1}\omega\tau_1 + \tan^{-1}\omega\tau_2 = 90^\circ$$

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$$\tan^{-1}\left(\frac{\omega T_1 + \omega T_2}{1 - \omega^2 T_1 T_2}\right) = 90^\circ$$

$$\frac{\omega T_1 + \omega T_2}{1 - \omega^2 T_1 T_2} = \tan(90^\circ) = \infty$$

$$1 - \omega^2 T_1 T_2 = 0$$

$$\omega^2 T_1 T_2 = 1 \quad \text{or} \quad \omega = \frac{1}{\sqrt{T_1 T_2}}$$

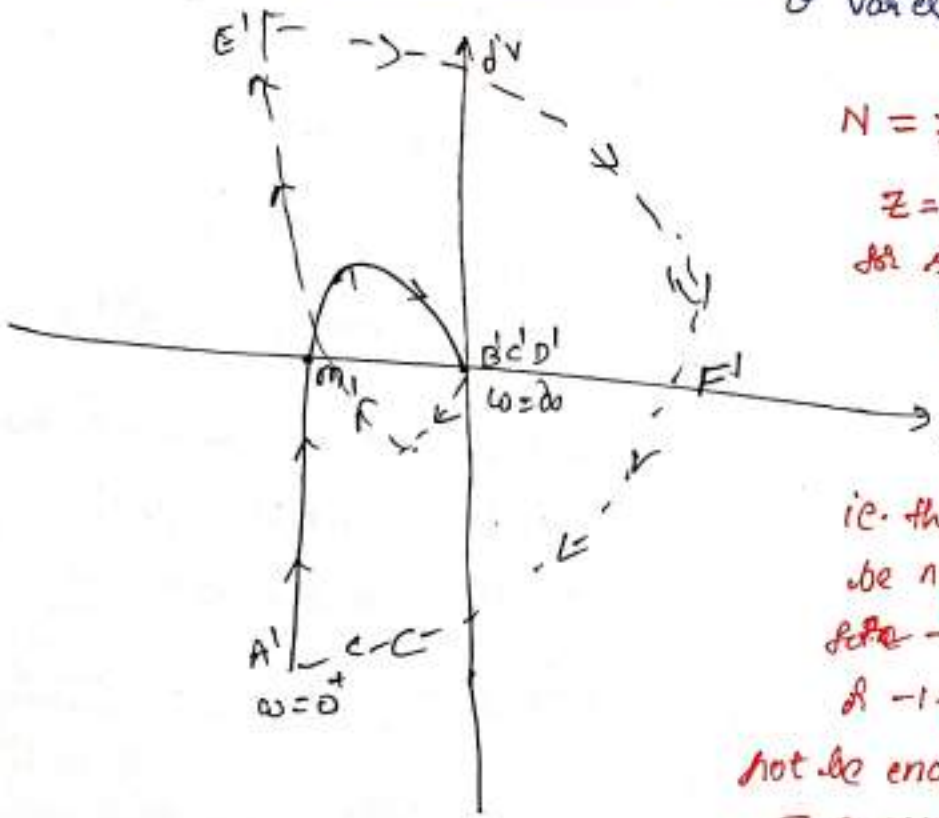
Substituting $\omega = \frac{1}{\sqrt{T_1 T_2}}$ in the expression for the magnitude Eqn, and for stability $M_1 \leq 1$

if $M_1 = 1$ (marginally Stable)

→ path BCD maps to origin.

→ Path DE is the Mirror image of A'B'

→ Path EFA → $\angle G$ $+90^\circ \to 0^\circ \to -90^\circ$ as $\angle\theta$ varies $-90^\circ \to 0 \to 90^\circ$




$N = Z - P$

$Z = N + P$

for stability

$Z = 0$

Here $P = 0$

$\therefore N = 0$ should

i.e. there should be no encirclement for $-1+j0$ pt.

& $-1+j0$ pt should not be encircled

This will happen only if

$m_1 \leq 1$

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$m_1$ is mag at $\omega = \frac{1}{\sqrt{\tau_1 \tau_2}}$

$$\frac{K}{\frac{1}{\sqrt{\tau_1\tau_2}}\sqrt{\frac{1}{\tau_1\tau_2}\cdot\tau_1^2+1}\sqrt{\frac{1}{\tau_1\tau_2}\cdot\tau_2^2+1}} \leq 1$$

$$\Rightarrow \frac{K}{\sqrt{\frac{1}{\tau_1\tau_2}}\left[\sqrt{\frac{\tau_1}{\tau_2}+1}\right]\left[\sqrt{\frac{\tau_2}{\tau_1}+1}\right)} \leq 1$$

$$= \frac{K}{\sqrt{\frac{1}{\tau_1\tau_2}}\sqrt{\frac{\tau_1+\tau_2}{\tau_2}}\sqrt{\frac{\tau_1+\tau_2}{\tau_1}}}$$

$$= \left(\frac{K}{(\sqrt{\tau_1+\tau_2})(\sqrt{\tau_1+\tau_2})}\Big/\sqrt{(\tau_1\tau_2)(\tau_1\tau_2)}\right) \leq 1$$

$$= \frac{K\cdot\tau_1\tau_2}{\tau_1+\tau_2} \leq 1$$

$$\text{or } \boxed{K \leq \frac{\tau_1+\tau_2}{\tau_1\tau_2}} \quad \therefore$$

Satisfying this condition would ensure the point $-1+j0$ is not encircled by the Nyquist plot so.

$N=0 \quad P=0 \;\&\; Z=0$.

Eg: If we consider $\tau_1 = \tau_2 = 1$ then for Stability $\boxed{K \leq 2}$

$K<2 \rightarrow$ Stable ; $K=2$ marginally stable, $K>2$ = unstable

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## method 2

- Consider $L(j\omega)$
- Separate them into real & imag parts.
- Find $m_1$, Equate the imag part to 0

In this case we get $\omega = \frac{1}{\sqrt{Z_1 Z_2}}$.

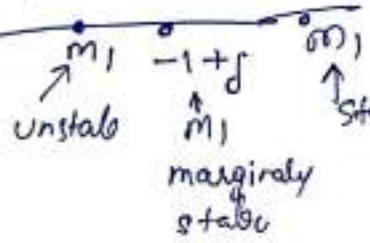
→ Find $u$ at this freq $\omega = \frac{1}{\sqrt{Z_1 Z_2}}$, we get

$$u = \frac{-k Z_1 Z_2}{Z_1 + Z_2}.$$

for stability

$$\frac{-k Z_1 Z_2}{Z_1 + Z_2} \geq -1$$

(ie $m_1$ pt should come




$$\text{if } k \leq \frac{Z_1 + Z_2}{Z_1 Z_2}.$$

---

(9) $G(s)H(s) = \frac{K}{s(s+1)(s+2)}$ ; find the range for stability.

Read this question!

$s = j\omega$.

$$G(j\omega)H(j\omega) = \frac{K}{\omega\sqrt{\omega^2+1}\sqrt{\omega^2+4}} \angle -90^\circ - \tan^{-1}\omega - \tan^{-1}\omega/2$$

To find $m_1$

$$-90^\circ - \tan^{-1}\omega - \tan^{-1}\omega/2 = -180^\circ$$

$$\tan^{-1}\omega + \tan^{-1}\omega/2 = 90^\circ$$

$$\tan^{-1}\left(\frac{\omega + \frac{\omega}{2}}{1 - \frac{\omega^2}{2}}\right) = 90$$

$$\Rightarrow \frac{3\omega/2}{(2-\omega^2)/2} = \tan 90^\circ = \infty \quad \text{or} \quad \frac{2-\omega^2}{2} = 0$$

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or $\frac{2-\omega^2}{2} = 0$ or $2-\omega^2 = 0$

$\omega = \sqrt{2}$ r/sec.

Mag at $\omega = \sqrt{2}$.

$$= \frac{K}{\sqrt{2}\sqrt{2+1}\sqrt{2+4}} = \frac{k}{\sqrt{2}\sqrt{3}\sqrt{6}} = \frac{K}{6}$$

∴ for stability

Here N needs to be "0", since P=0. Then point M should be to the right of -1+j0 or

$$\frac{k}{6} < 1 \quad \text{or} \quad \boxed{k < 6.}$$

Since $K > 0$. The range of K for stability is

$$\boxed{0 < K < 6}$$

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# Relative Stability & Nyquist Criterion

→ **Gain Margin (GM)**

→ GM is the reciprocal of the magnitude $|G(j\omega)|$ at the freq at which angle is $-180^\circ$

→ $GM\ (\text{in dB}) = 20\log\frac{1}{|G(j\omega)|} = -20\log|G(j\omega)|$

GM = +ve if $K_g > 1$

GM = −ve if $K_g < 1$

→ **Phase crossover frequency (PCF) ($\omega_{pc}$)**

→ Frequency at which phase angle of O.L.T.F = −180

→ **Phase Margin**

→ PM is amount of additional phase lag at GCF to bring system to verge of instability

→ $PM = 180 + \phi$

$\phi = \angle G(j\omega)\big|_{\omega=\omega_{ge}}$

→ **Gain crossover frequency (GCF) ($\omega_{ge}$)**

→ Frequency at which magnitude of O.L.T.F = 1

→ When GM & PM are +ve ⇒ System is stable

When GM & PM are −ve ⇒ System is unstable

Q. $G(s) = \frac{1}{s(s+1)(s+0.5)}$. Find a) $\omega_{ge}$ b) $\omega_{pc}$ c) GM d) PM

A. $G(j\omega) = \frac{1}{j\omega(j\omega+1)(j\omega+0.5)} \Rightarrow |G(j\omega)| = \frac{1}{\omega\sqrt{1+\omega^2}\sqrt{\omega^2+\frac{1}{4}}}$ & $\angle G(j\omega) = -90 - \tan^{-1}\omega - \tan^{-1}2\omega$

$= \frac{(-j\omega)(1-j\omega)(0.5-j\omega)}{\omega^2(1+\omega^2)(0.25+\omega^2)} = \frac{-1.5\omega^2 + j(\omega^3-0.5\omega)}{\omega^2(1+\omega^2)(0.25+\omega^2)}$

| $\omega$ | $\lvert G(j\omega)\rvert$ | $\angle G(j\omega)$ | $Re(G(j\omega))$ | $Im(G(j\omega))$ |
|---|---|---|---|---|
| 0 | ∞ | $-90^\circ$ | ∞ | ∞ |
| 1/2 | 2.53 | $-161.56^\circ$ | −2.4 | −0.8 |
| 1 | 0.63 | $-198.43^\circ$ | −0.6 | 0.2 |
| ∞ | 0 | −270 | 0 | 0 |

For path bcd,

$s = Re^{j\phi}$ $\quad \phi = 90$ to $-90$

$\lim R \to \infty$

$$G(s) = \frac{1}{Re^{j\phi}(1+Re^{j\phi})(0.5+Re^{j\phi})} = \frac{1}{Re^{j3\phi}\left(\frac{1}{Re^{j\phi}}+1\right)\left(\frac{0.5}{Re^{j\phi}}+1\right)}$$

$$\lim_{R\to\infty} \frac{1}{Re^{j3\phi}\left(\frac{1}{Re^{j\phi}}+1\right)\left(\frac{0.5}{Re^{j\phi}}+1\right)} = 0.e^{-j3\phi} \Rightarrow G(s) \text{ varies from } -270 \text{ to } 270$$

To find real axis crossing,

$\text{Im}(G(j\omega)) = 0 \Rightarrow \omega^3 - 0.5\omega = 0 \Rightarrow \omega_{pc} = \frac{1}{\sqrt{2}}$ rad/s ⟶ Phase crossover frequency ($\omega_{pc}$)

$$\text{Re}(G(j\omega))\Big|_{\omega=\omega_{pc}} = \frac{-1.5\left(\frac{1}{2}\right)}{\frac{1}{2}\left(1+\frac{1}{2}\right)\left(\frac{1}{4}+\frac{1}{2}\right)} = \frac{-1.5}{1.5\times 0.75} = -1.33$$

$K_g = |-1.33| = 1.33$

$GM = \frac{1}{K_g} = 0.752$

In dB, $GM = 20\log\left(\frac{1}{1.33}\right) = -2.5\text{dB}$

To find $\omega_{gc}$,

$$|G(j\omega)|_{\omega=\omega_{gc}} = 1$$

$$\frac{1}{\omega_{gc}\sqrt{1+\omega_{gc}^2}\sqrt{0.25+\omega_{gc}^2}} = 1 \Rightarrow \omega_{gc}^2(1+\omega_{gc}^2)(0.25+\omega_{gc}^2)$$

$$\omega_{gc}^6 + \frac{5}{4}\omega_{gc}^4 + \frac{1}{4}\omega_{gc}^2 = 1 \Rightarrow \text{Solving eq}^n$$

$\omega_{gc}^2 = t \Rightarrow t^3 + \frac{5}{4}t^2 + \frac{1}{4}t^2 - 1 = 0$

$\omega_{gc}^2 = 0.6 \Rightarrow \omega_{gc} = 0.813$ rad/s

$$\phi = \angle G(j\omega)\Big|_{\omega=\omega_{gc}} = -90 - \tan^{-1}0.813 - \tan^{-1}(2\times 0.813) = -187.5^\circ$$

$PM = 180 + \phi = -7.5^\circ$

GM & PM are -ve ⇒ System Unstable

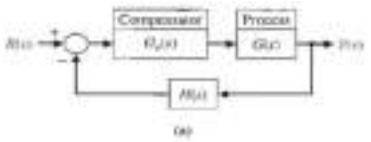
## Design of Feedback Control System

→ Types of Compensators

i) Lag Compensator - Reduces $e_{ss}$

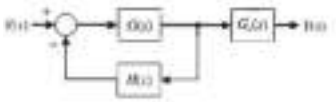
ii) Lead Compensator - Improves transient response

iii) Lead-lag Compensator - Used when both $e_{ss}$ & transient response aren't satisfactory



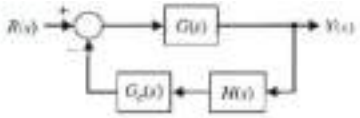

(a)

CASCADE COMPENSATION



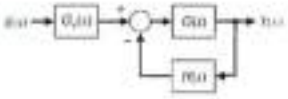
(b)

OUTPUT/LOAD COMPENSATION



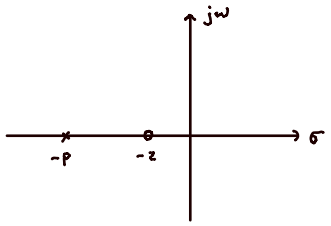
(b)

FEEDBACK COMPENSATION



(b)

INPUT COMPENSATION




## Phase Lead Compensator (Differentiating Network)

→ System with 1 Pole & 1 Dominating Zero (Lead Network)
↳ closer to origin

→ When the system is absolutely unstable, we must stabilize & meet req. performance
ex: System with Type-2 & above, require lead compensation
When the system is stable, we must obtain req. performance
ex: Type 0 & 1, stability can be acheived by adjusting the gain

→ Compensator T.F, $G_c(s) = K \frac{\prod_{i=1}^{m}(s+z_i)}{\prod_{j=1}^{n}(s+p_j)}$

First order compensator, $G_c(s) = K\frac{(s+z)}{(s+p)}$

→ When $|z| < |p| \Rightarrow$ Phase lead network

If $G_c(s) = K\frac{s+z}{s+p}$

$$G_c(j\omega) = K\frac{j\omega+z}{j\omega+p} = \frac{Kz}{p}\left(\frac{j\left(\frac{\omega}{z}\right)+1}{j\left(\frac{\omega}{p}\right)+1}\right) = \frac{K_1(1+j\omega\alpha\tau)}{1+j\omega\tau} \qquad \left(\tau=\frac{1}{p},\ p=\alpha z,\ K_1=\frac{K}{\alpha}\right)$$

$$\phi(\omega) = \tan^{-1}(\alpha\omega\tau) - \tan^{-1}(\omega\tau) \qquad \left(\frac{d}{dx}(\tan^{-1}ax) = \frac{a}{a^2x^2+1}\right)$$

→ Max $\phi$ at $\omega = \omega_m$

$$\frac{d\phi}{d\omega} = 0 \Rightarrow \omega_m = \sqrt{zp} = \frac{1}{\tau\sqrt{\alpha}} \Rightarrow |G(j\omega)| = \frac{1}{\sqrt{\alpha}}$$

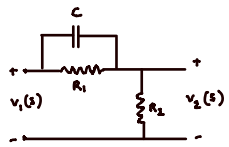
### Effects of Phase Lead

→ $K_v \uparrow$, PM $\uparrow$, $e_{ss} \downarrow$
→ Response is faster

### Advantages

→ speed of system ↑ because GCF ↑
→ $M_p \downarrow$

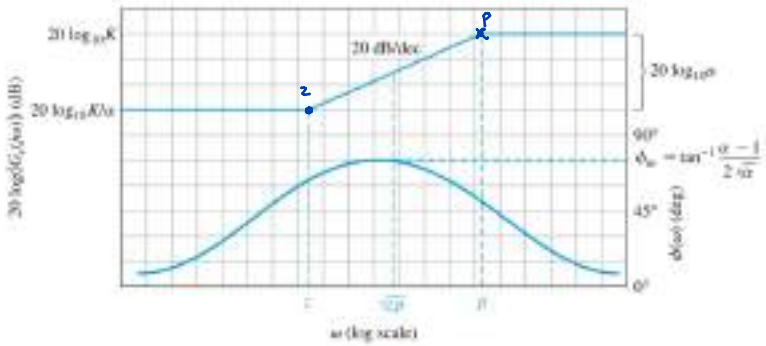
### Phase lead electrical network

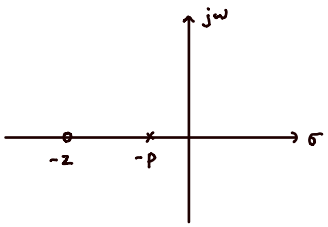



→ $G_c(s) = \frac{V_2(s)}{V_i(s)} = \frac{R_2}{R_2 + \frac{R_1/Cs}{R_1+\frac{1}{Cs}}} = \frac{R_2\left(R_1+\frac{1}{Cs}\right)}{R_1R_2 + \frac{R_2}{Cs} + \frac{R_1}{Cs}} = \frac{R_2\left(R_1+\frac{1}{Cs}\right)}{R_1R_2Cs + R_1 + R_2} = \frac{R_2}{R_1+R_2}\left(\frac{R_1+\frac{1}{Cs}}{\frac{R_1R_2Cs}{R_1+R_2}+1}\right)$

Let $\tau = \left(\frac{R_1R_2}{R_1+R_2}\right)C$ & $\alpha = \frac{R_1+R_2}{R_2}$ Then $G_c(s) = \frac{\tau s + 1/\alpha}{(1+\tau s)} = \frac{\alpha\tau s+1}{\alpha(1+\tau s)}$

## Bode Plot of Phase lead compensators






## Phase Lag Compensation (Integrating Network)

→ $|p| < |z|$

1 Dominating Pole & 1 zero

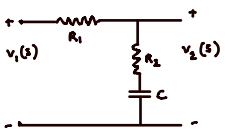
First order compensator $G_c(s) = \frac{K(s+z)}{(s+p)}$

$$G_c(j\omega) = \frac{K(j\omega+z)}{(j\omega+p)} = \frac{Kz}{p}\frac{\left(j\frac{\omega}{z}+1\right)}{\left(j\frac{\omega}{p}+1\right)} = K_1\left(\frac{1+j\omega T}{1+j\omega\alpha T}\right) \qquad \left(T=\frac{1}{z} \quad z=\alpha p \quad K_1=K\alpha\right)$$

$$\phi(\omega) = \tan^{-1}(\omega T) - \tan^{-1}(\alpha\omega T)$$

Max $\phi$ at $\omega_m$,

$$\frac{d\phi}{d\omega} = 0 \Rightarrow \omega_m = \frac{1}{T\sqrt{\alpha}}$$



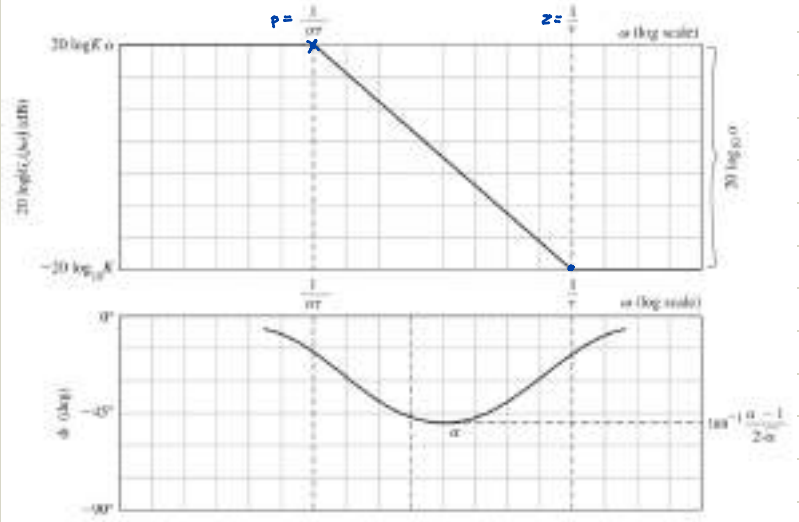

## Phase lag electrical network

→ $$G_c(s) = \frac{V_2(s)}{V_1(s)} = \frac{R_2+\frac{1}{Cs}}{R_1+R_2+\frac{1}{Cs}} = \frac{R_2Cs+1}{(R_1+R_2)Cs+1}$$

Let $T = R_2C$ & $\alpha = \frac{R_1+R}{R_2}$

$$G_c(s) = \frac{1+Ts}{1+\alpha Ts} = \frac{1}{\alpha}\left(\frac{s+z}{s+p}\right) \qquad \left(z=\frac{1}{T} \quad \& \quad p=\frac{1}{\alpha T}\right)$$

## Bode Plot of Phase lag compensators




For Phase Lead/Lag using bode Plot refer slides from below ↓

And for reference click this 🥰 (YT video)

But best example : CLICK THIS 🥰🥰

# Unit 5: Design of Feedback Control Systems

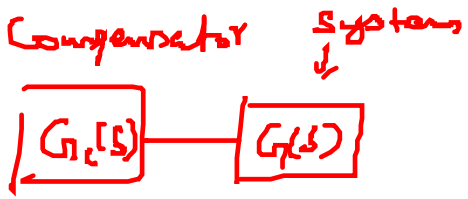
## Phase-Lead design using Bode plot

**Dr. Anuradha M**
Department of Electronics and Communication Engineering

# Design of Feedback Control Systems

## Phase-Lead design using Bode Plot

- Design the Lead compensator such that system meets desired specifications.
- How? Frequency response of cascade compensation network is added to frequency response of the uncompensated system
- There are steps to design the Lead compensator



# Design of Feedback Control Systems

## Phase-Lead design using Bode Plot

- Step 1: Evaluate the uncompensated system phase margin($\emptyset_{un}$) with error constants satisfied.
- Step 2: determine the necessary additional phase lead $\emptyset_m$ with a small amount of safety margin
  - $\emptyset_1 = desired\ PM\ - (\emptyset_{un})$
  - $\emptyset_m = \emptyset_1 + safety\ margin$
  - Safety margin = 5% of $\emptyset_1$ , where % can be varied until you get the desired PM

# Design of Feedback Control Systems
## Phase-Lead design using Bode Plot

- Step 3: Evaluate $\propto$ using the following equation
  - $sin\,\emptyset_m = \frac{\propto -1}{\propto +1}$ or $\propto = \frac{1+\sin(\emptyset_m)}{1-\sin(\emptyset_m)}$
- Step 4: Evaluate $10\log_{10}\propto$ and determine the frequency where the uncompensated magnitude curve is equal to $-10\log_{10}\propto$ dB. (Because the compensation network provides a gain of $10\log_{10}\propto$ at $\omega_m$ and this is the new 0dB cross over frequency)

# Design of Feedback Control Systems

## Phase-Lead design using Bode Plot

- Step 5: Calculate the pole $p = \omega_m\sqrt{\alpha}, z = \frac{p}{\alpha}$
- Step 6: Draw the compensated frequency response, check the resulting phase margin and repeat the steps if necessary

$\omega_m = \frac{1}{z\sqrt{\alpha}}$

$z = \frac{p}{\sqrt{\alpha}} \Rightarrow p = \omega_m\sqrt{\alpha}$

$z = \frac{1}{\alpha z} = \frac{p}{\alpha}$

$G_c(s) = \frac{K(s+z)}{(s+p)}$, $L(s) = G_c(s)\,G(s)$

freq. response of $L(s)$, check PM of $L(s)$

# Design of Feedback Control Systems

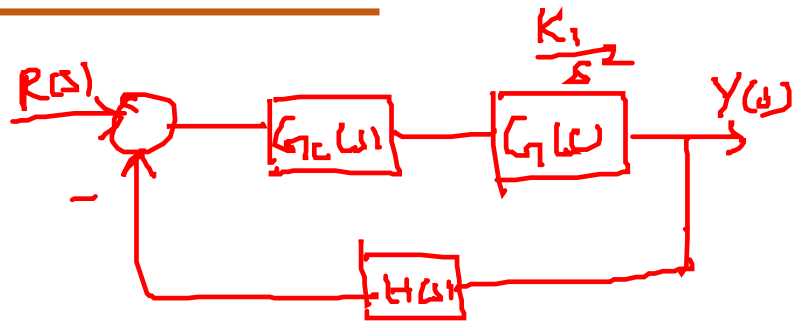
## Example 1

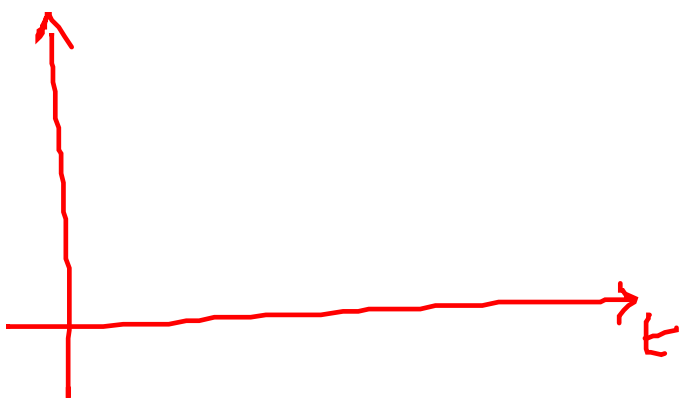
- $G(s) = \frac{K_1}{s^2}$ and H(s) = 1, $T(s) = \frac{K_1}{s^2+K_1}$
- Desired specifications:
  - Settling time, $t_s \leq 4\ s$
  - System damping constant, $\zeta \geq 0.45$

$$=> t_s = \frac{4}{\zeta \omega_n} = 4$$

$$=> \omega_n = \frac{1}{\zeta} = 2.22$$

$$=> K_1 = \omega_n^2 \approx 5$$



# Design of Feedback Control Systems

## Example 1

- But we need specification in terms of phase margin
  - $\emptyset_{pm} = \frac{\zeta}{0.01} = 45\ degree$ ⇒ desired PM
- Step 1: Draw Bode plot for $G(j\omega) = \frac{K_1}{(j\omega)^2}$
- $K_1$=10 with safety margin as 5 or add in the step 2

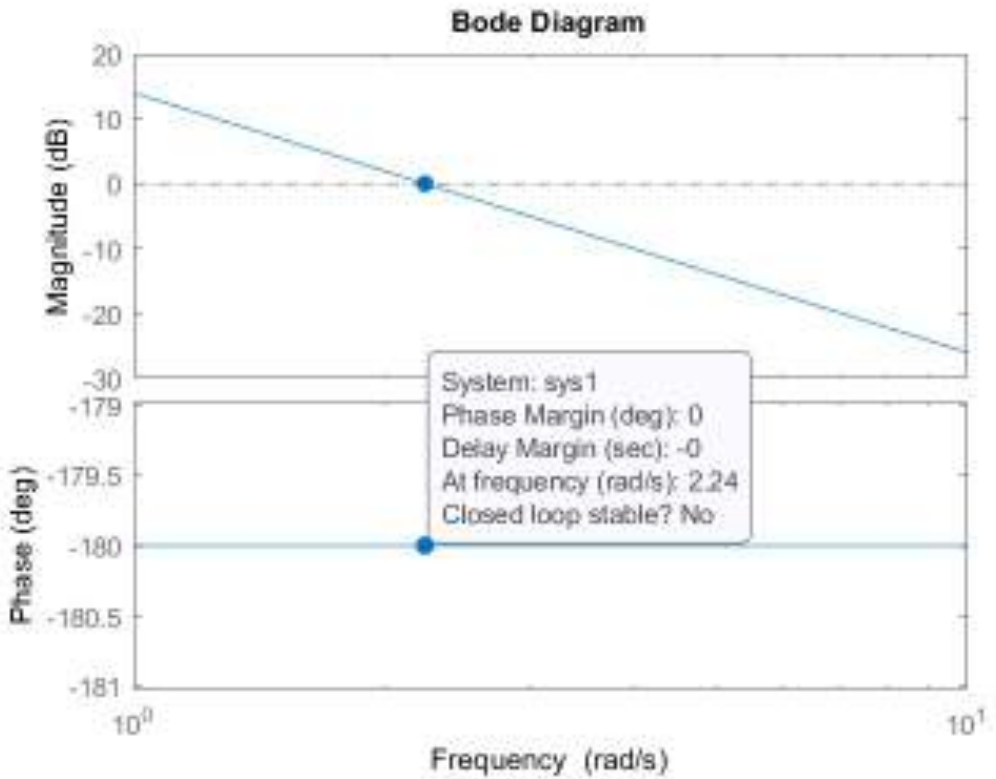
, $G_1(j\omega) = \frac{10}{(j\omega)^2}$

$|G(j\omega)| = 20\log 10 - 20\log \omega^2$ , $\angle G(j\omega) = -\pi$

$\omega = 0.1$ , $A = 60$ dB

# Design of Feedback Control Systems

## Example 1




$PM = 180 + (-180^\circ) = 0^\circ$

# Design of Feedback Control Systems

## Example 1

- Step 2:
  - $\emptyset_1 = \boldsymbol{desired\ PM} - (\emptyset_{un})$
  - $\emptyset_m = \emptyset_1 + \boldsymbol{safety\ margin}$
  - **Desired PM , $\emptyset_{pm}$= 45 and $\emptyset_{un} = 0$**
  - $\emptyset_1 = 45$
  - $\emptyset_m = 45 + 0\ as\ safety\ margin = 45$
- Step 3: Evaluate $\propto$ using the following equation
  - $sin\ \emptyset_m = \frac{\propto -1}{\propto +1}$ or $\propto = \frac{1+\sin(\emptyset_m)}{1-\sin(\emptyset_m)} = 5.8 \approx 6$

# Design of Feedback Control Systems

## Example 1

- Step 4: Evaluate $10\log_{10}\propto$ = 7.78 dB
  - frequency at which the uncompensated magnitude curve is equal to $-7.78$ dB is $\omega_m = 4.95\,rad/sec$ (Because the compensation network provides a gain of $10\log_{10}\propto$ at $\omega_m$ and this is the new 0dB cross over frequency)
- Step 5: Calculate the pole $p = \omega_m\sqrt{\propto} = 4.95\sqrt{6} = 12,\ z = \frac{p}{\propto} = \frac{12}{6} = 2$

# Design of Feedback Control Systems

## Example 1

- Step 6: Draw the compensated frequency response,
  - $L(j\omega) = G_c(j\omega)G(j\omega) = \frac{10[\frac{j\omega}{2}+1]}{(j\omega)^2[\frac{j\omega}{12}+1]}$
  - Total DC loop gain must be raised by a factor of α in order to account for the factor $\frac{1}{\alpha}$ then the

$$G_c(s) = \frac{\frac{s}{z}+1}{\frac{s}{p}+1}$$

  - $L(s) = G_c(s)G(s) = \frac{10[\frac{s}{2}+1]}{(s)^2[\frac{s}{12}+1]} = \frac{60(s+2)}{s^2(s+12)}$
  - $T(s) = \frac{60(s+2)}{s^3+12s^2+60s+120}$
  - check the resulting phase margin and repeat the steps if necessary

# Design of Feedback Control Systems

## Example 1

| factor | C.F | slope (dB/dec) | Net slope (dB/dec) | freq. range (rad/sec) |
|---|---|---|---|---|
| $10/(jw)^2$ | none | -40 | -40 | $w = 0.1$ to $w = 2$ |
| $\left(\frac{jw}{2}+1\right)$ | 2 | 20 | -20 | $w = 2$ to $w = 12$ |
| $\frac{1}{\frac{jw}{12}+1}$ | 12 | -20 | -40 | $w = 12$ onwards |

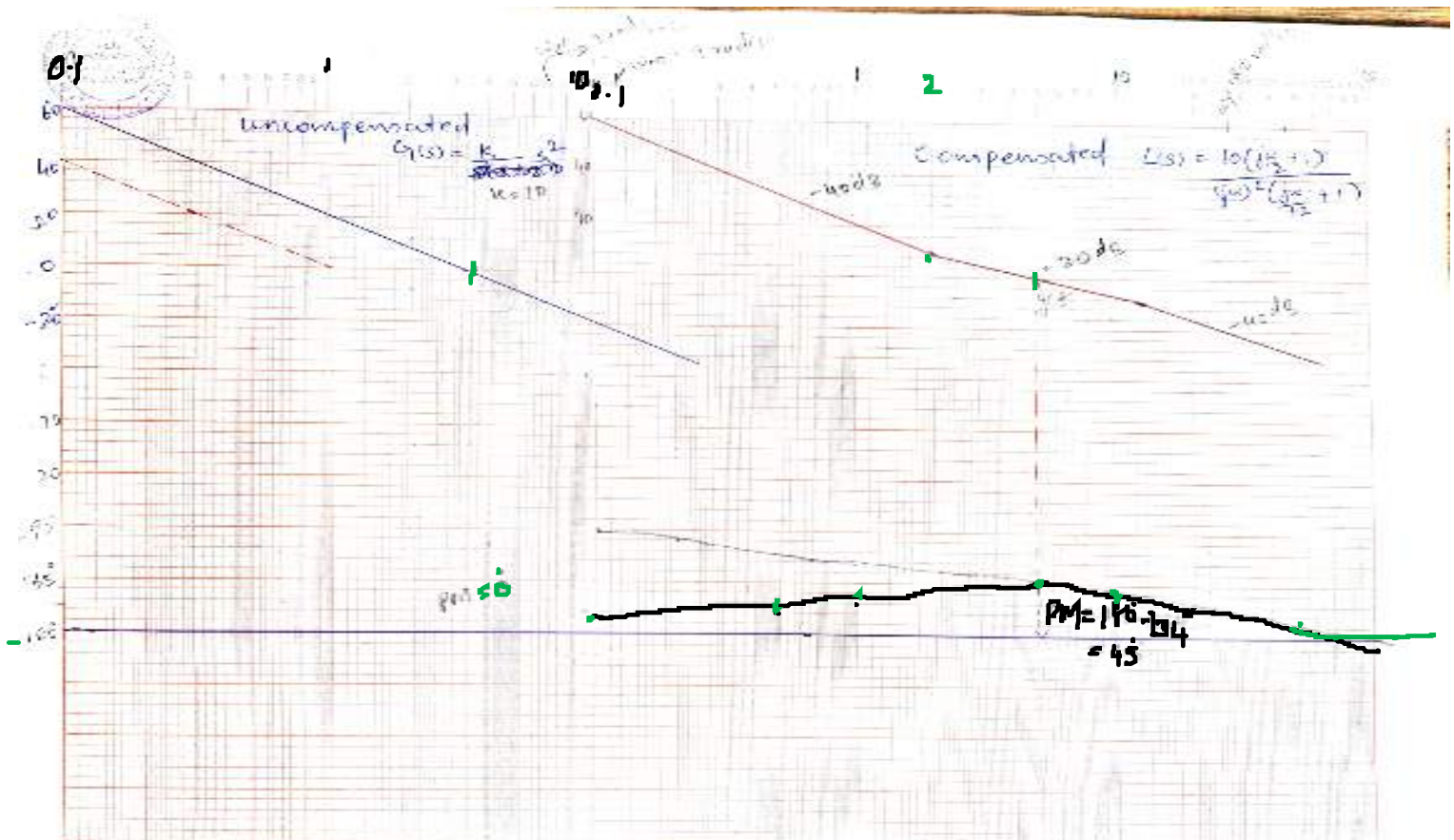
$A = 20 \log \frac{10}{w^2} = 20 \log 10 - 40 \log w$

$A$ at $w = 0.1 = 60$ dB

$\angle G(jw) = \tan^{-1}\frac{w}{2} - 180^\circ - \tan^{-1}\frac{w}{12}$

# Design of Feedback Control Systems

## Example 1




GCF, $\omega_{gc}$ = 5 rad/sec

desired PM = 45°

| $\omega$ | 0.1 | 0.5 | 1 | 5 | 10 |
|---|---|---|---|---|---|
| $\angle G(j\omega)$ | -177 | -168 | -158 | -134 | -141 |

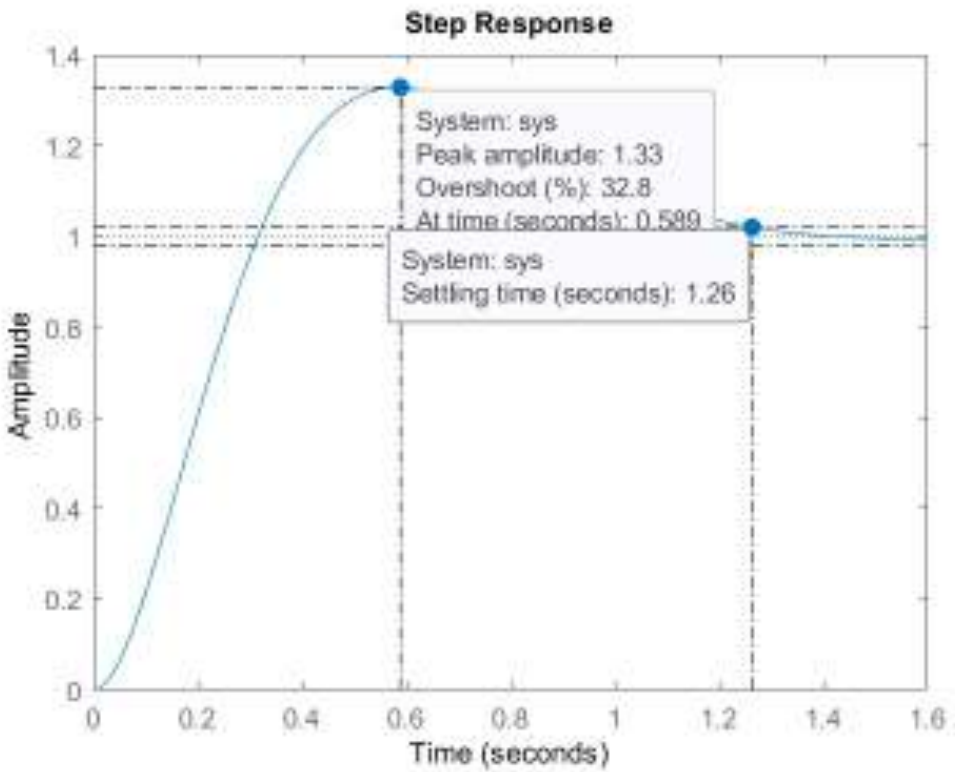
# Design of Feedback Control Systems

## Example 1

- But the desired specifications are in terms of settling time damping constant
- Therefore, overall transfer function after compensating with lead compensator is
  - $T(s) = \frac{60(s+2)}{s^3+12s^2+60s+120}$
  - Step response of compensated system is used to check the specifications

# Design of Feedback Control Systems

## Example 1



# Design of Feedback Control Systems

## Example 2

- $G(s) = \frac{K}{s(s+2)}$ and H(s) = 1
- Desired specification
  - Phase margin = 45 degree
  - Steady state error for a ramp input equal to 5% of the velocity of the ramp
    - $e_{ss} = \frac{A}{K_v} \Rightarrow 0.05A = \frac{A}{K_v} \Rightarrow K_v = 20$

$K_v = \lim_{s \to 0} s\,G(s) = \lim_{s \to 0} s \cdot \frac{K}{s(s+2)}$

$K_v = \frac{K}{2}$

$2K_v = K$

$G(s) = \frac{2K_v}{s(s+2)} = \frac{2K_v}{2s(\frac{1}{2}s+1)} = \frac{K_v}{s(\frac{1}{2}s+1)}$

| factor | C.F | slope | net slope | freq. range |
|---|---|---|---|---|
| $\frac{20}{j\omega}$ | none | -20 | -20 | $0.1 \le 2$ |
| $\frac{1}{0.5j\omega+1}$ | 2 | -20 | -40 | 2 onwards |

- Step 1: Draw Bode plot for $G(j\omega) = \frac{K_v}{j\omega(0.5j\omega+1)}$

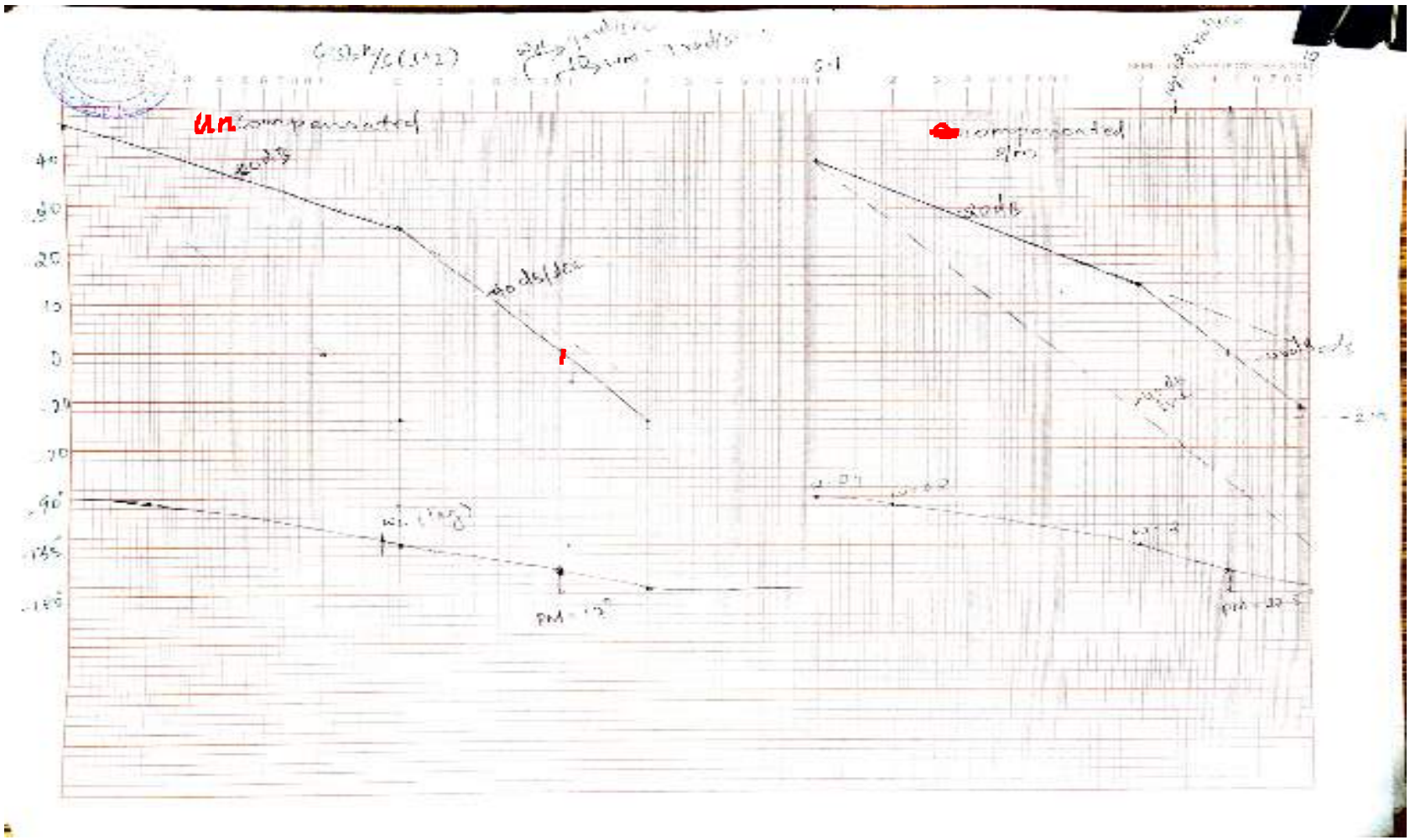
$\angle G(j\omega) = -90^\circ - \tan^{-1} 0.5\omega$

$A = 20 \log \frac{20}{\omega} = 20 \log 20 - 20 \log \omega$

$= 46.02$ dB

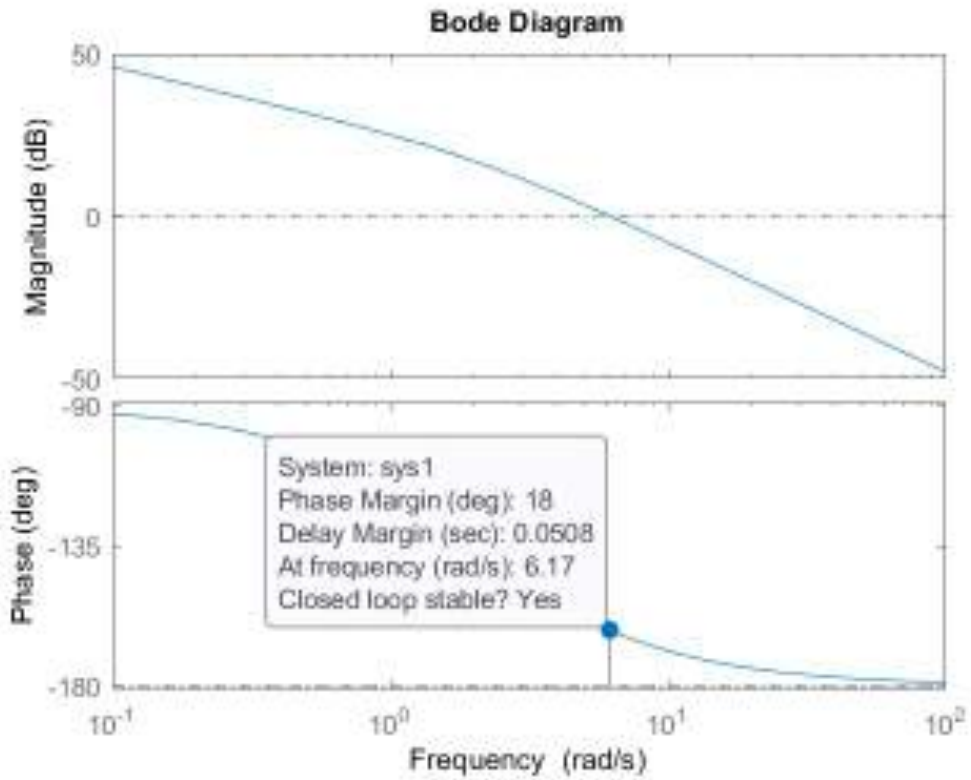
# Design of Feedback Control Systems

## Example 2



# Design of Feedback Control Systems

## Example 2



# Design of Feedback Control Systems

## Example 2

- Step 2:
  - $\emptyset_1 = \boldsymbol{desired\ PM} - (\emptyset_{un})$
  - $\emptyset_m = \emptyset_1 + \boldsymbol{safety\ margin}$
  - **Desired PM , $\emptyset_{pm}$= 45 and** $\emptyset_{un} = 18$
  - $\emptyset_1 = 45\text{-}18 = 27$
  - $\emptyset_m = 27 + 3\ (10\%\ of \emptyset_1 as\ safety\ margin) = 30$
- Step 3: Evaluate $\propto$ using the following equation
  - $\boldsymbol{sin}\ \emptyset_m = \frac{\propto -1}{\propto +1}$ or $\propto = \frac{1+\sin(\emptyset_m)}{1-\sin(\emptyset_m)} = 3$

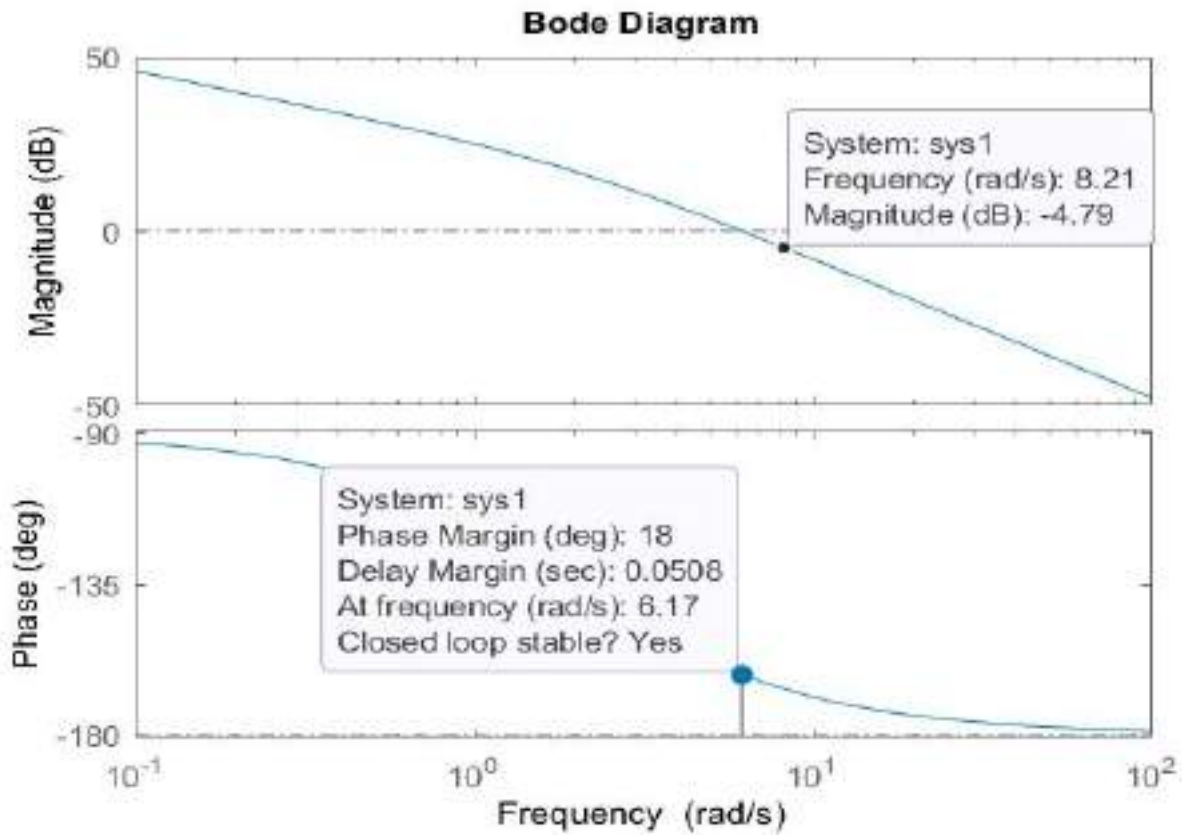
# Design of Feedback Control Systems

## Example 2

- Step 4: Evaluate $10\log_{10}\propto$ = 4.8 dB
  - frequency at which the uncompensated magnitude curve is equal to $-4.8$ dB is $\omega_m = 8.21 rad/sec$ (Because the compensation network provides a gain of $10\log_{10}\propto$ at $\omega_m$ and this is the new 0dB cross over frequency)

# Design of Feedback Control Systems

## Example 2



# Design of Feedback Control Systems
## Example 2

- Step 5: Calculate $p = \omega_m\sqrt{\propto} = 8.21\sqrt{3} = 14.2201,$
  - $z = \frac{p}{\propto} = \frac{14.2201}{3} = 4.74$
- Step 6: Draw the bode plot for compensated system
  - $L(j\omega) = G_c(j\omega)G(j\omega) = \frac{20[\frac{j\omega}{4.74}+1]}{j\omega(0.5j\omega+1)[\frac{j\omega}{14.22}+1]}$
  - $L(s) = G_c(s)G(s) = \frac{20[\frac{s}{4.74}+1]}{s(0.5s+1)[\frac{s}{14.22}+1]}$
  - check the resulting phase margin and repeat the steps if necessary

$A = 20\log\frac{20}{\omega}$

$= 20\log 20 - 20\log\omega$

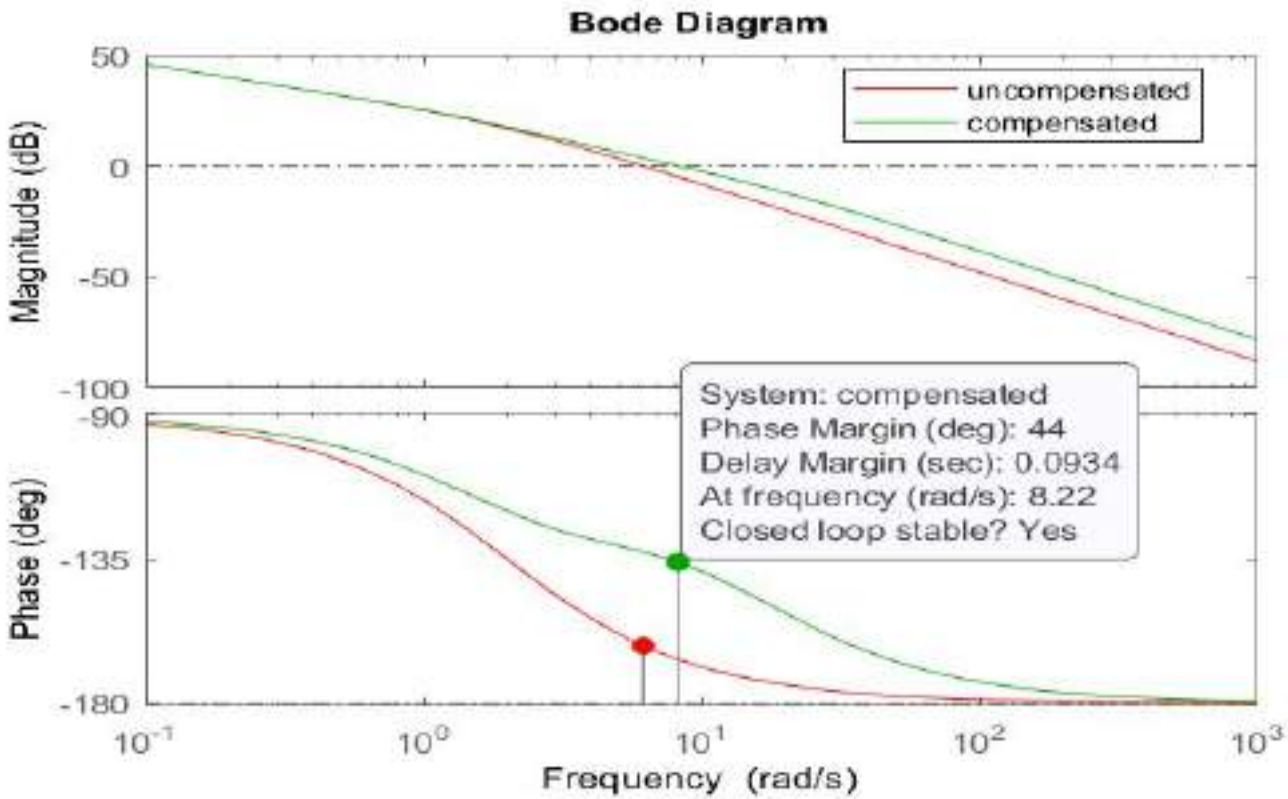
$A(\omega=0.1) = 46\ dB$

| factor | C.F | slope | net slope | freq. range |
|---|---|---|---|---|
| $\frac{20}{j\omega}$ | none | -20 | -20 | $0.1 \le 2$ |
| $\frac{1}{0.5j\omega+1}$ | 2 | -20 | -40 | $2 \le 4.74$ |
| $\frac{j\omega}{4.74}+1$ | 4.74 | 20 | -20 | $4.74 \le 14.22$ |
| $\frac{j\omega}{14.22}+1$ | 14.22 | -20 | -40 | 14.22 onwards |

$\angle L(j\omega) = \tan^{-1}\frac{\omega}{4.74} - 90 - \tan^{-1}0.5\omega - \tan^{-1}\frac{\omega}{14.22}$

# Design of Feedback Control Systems

## Example 2



# Unit 5: Design of Feedback Control Systems

# Phase – Lag using Bode Plot

**Dr. Anuradha M**

Department of Electronics and Communication Engineering

# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot

- $G_c(j\omega) = \frac{1+j\omega\tau}{1+j\omega\alpha\tau}$

$p = \frac{1}{\alpha\tau}$ , $z = \frac{1}{\tau}$

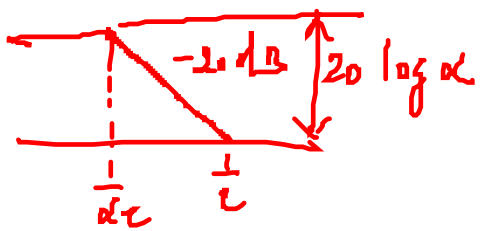
- Step1: Obtain the Bode plot for uncompensated system with the gain adjusted for the desired error constant
- Step2: obtain the phase margin of uncompensated system, if it is insufficient do the following,
- Step3: Determine the frequency where the PM requirement would be satisfied as $\omega_c'$
  - i.e $\emptyset_1$ =desired PM + safety margin(10% or 5 deg of phase lag) then determine frequency at (180- $\emptyset_1$) on phase plot

# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot

- Step4: Place the zero of the compensator one decade below the new crossover frequency $\omega_c'$

  - $z = \frac{\omega_c'}{10}$




- Step5: Measure the necessary attenuation at $\omega_c'$ to ensure that the magnitude curve crosses at this frequency

- Step6: Calculate $\propto$ by noting that the attenuation introduced by the phase lag network is $20\log_{10} \propto$ at $\omega_c'$

- Step7: Calculate $p = \frac{1}{\propto \tau} = \frac{z}{\propto}$

# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 1

$\phi = -90 - \tan^{-1} 0.5\omega/\omega = \omega_{gc}$

$\phi = -162^\circ$

$PM = 180 + \phi$

$= 180^\circ - 162^\circ$

$= 18^\circ$

- $G(s) = \frac{K}{s(s+2)}$ and H(s) = 1
- Desired specification
  - Phase margin = 45 degree
  - Steady state error for a ramp input equal to 5% of the velocity of the ramp
    - $e_{ss} = \frac{A}{K_v} \Rightarrow 0.05A = \frac{A}{K_v} \Rightarrow K_v = 20$

$20 = \omega_{gc}^2 \sqrt{\frac{\omega_{gc}^2}{4} + 1}$

$400 = \omega_{gc}^2 \left(\frac{\omega_{gc}^2}{4} + 1\right)$

$1600 = \omega_{gc}^4 + 4\omega_{gc}^2$

$\omega_{gc}^4 + 4\omega_{gc}^2 - 1600 = 0$

$K_v = \lim_{s \to 0} s\,G(s) = \lim_{s \to 0} s \cdot \frac{K}{s(s+2)}$

$K_v = \frac{K}{2}$

$K = 2K_v$

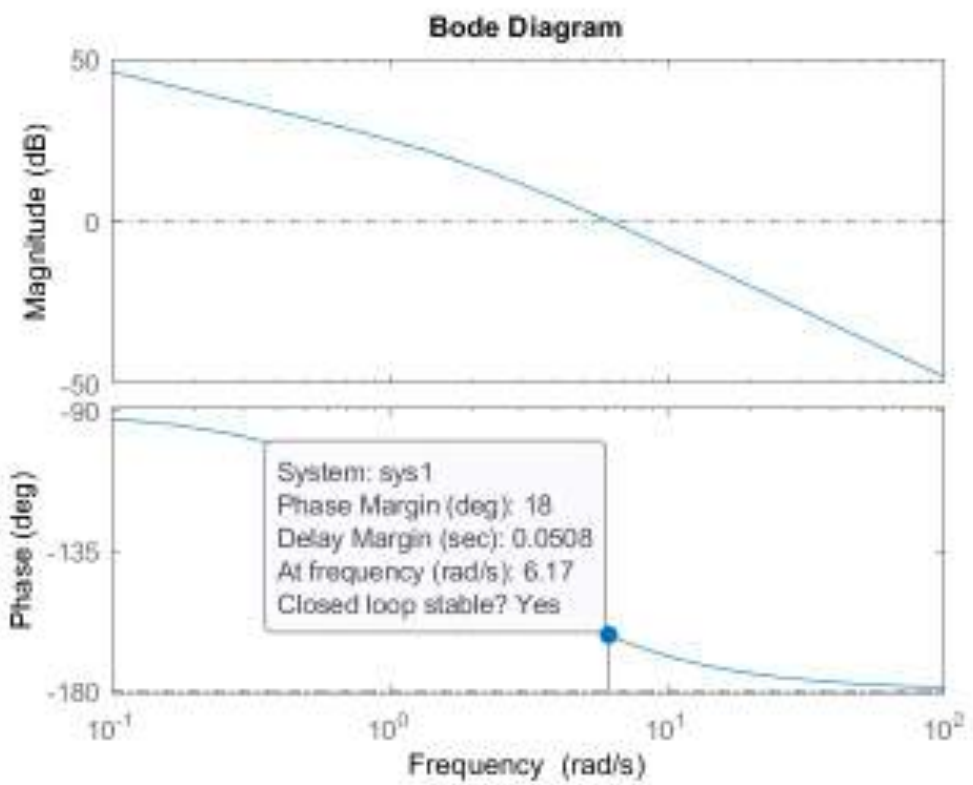
- Step 1: Draw Bode plot for $G(j\omega) = \frac{K_v}{j\omega(0.5j\omega+1)}$

$G(s) = \frac{2K_v}{s(s+2)} = \frac{2K_v}{2s\left(\frac{s}{2}+1\right)} = \frac{20}{s\left(\frac{s}{2}+1\right)}$

$|G(j\omega)| = \left.\frac{20}{\omega\sqrt{\omega^2/4+1}}\right|_{\omega=\omega_{gc}} = 1 \Rightarrow \omega_{gc} = 6.16$ rad/sec

# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 1




$\phi = -90^\circ - \tan^{-1} 0.5\omega$

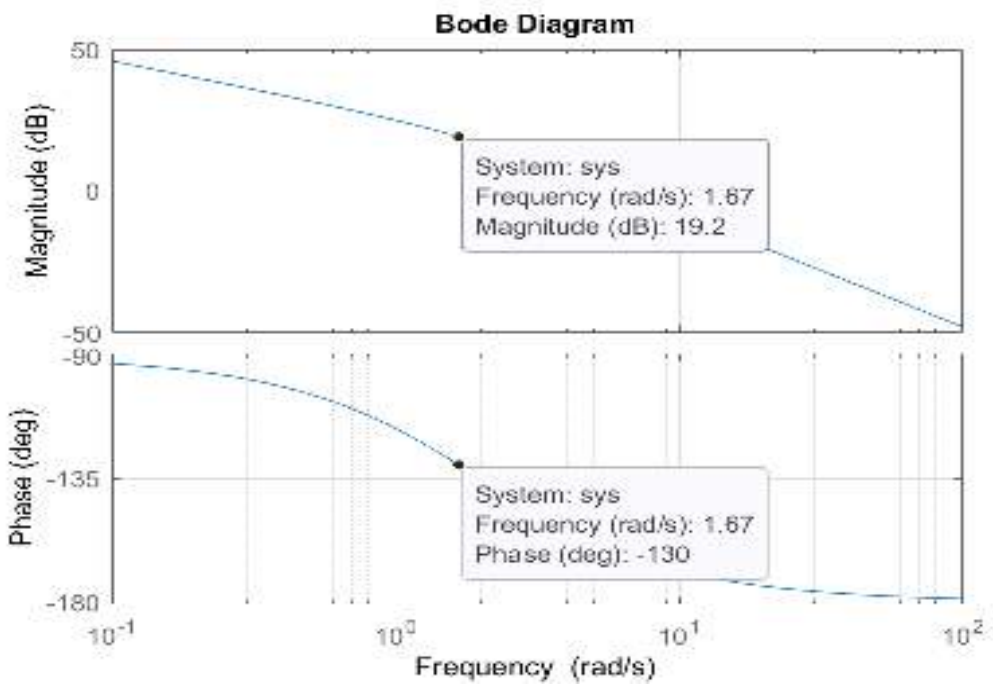
| $\omega$ | 0.1 | 1 | 5 | 10 | 20 |
|---|---|---|---|---|---|
| $\phi$ | $-92.8$ | $-116.5$ | $-158$ | $-168$ | $-180^\circ$ |

- Since PM is 18, it is less than the desired, therefore we need to increase the PM

# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 1

- $\emptyset_1$ =desired PM + safety margin = $45° + 5° = 50°$
- Determine the frequency at $-(180° - 50°) = -130°$ => $\omega_c' = 1.66$



# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 1

- Calculate $\propto$ by noting that the attenuation introduced by the phase lag network is $20\log_{10}\propto = Magnitude$ at $\omega_c'$ $\Rightarrow$ $20\log_{10}\alpha = 19.2$

- $z = \frac{\omega_c'}{10} = \frac{1.66}{10} = 0.166$ $\quad \log_{10}\alpha = \frac{19.2}{20} = 0.96$

- $p = \frac{1}{\propto\tau} = \frac{z}{\propto} = \frac{0.166}{9.2257} = 0.0180$ $\quad \alpha = 10^{0.96} = 9.22$

- Compensated System

$$G_c(s) = \frac{\frac{s}{z}+1}{\frac{s}{p}+1}$$

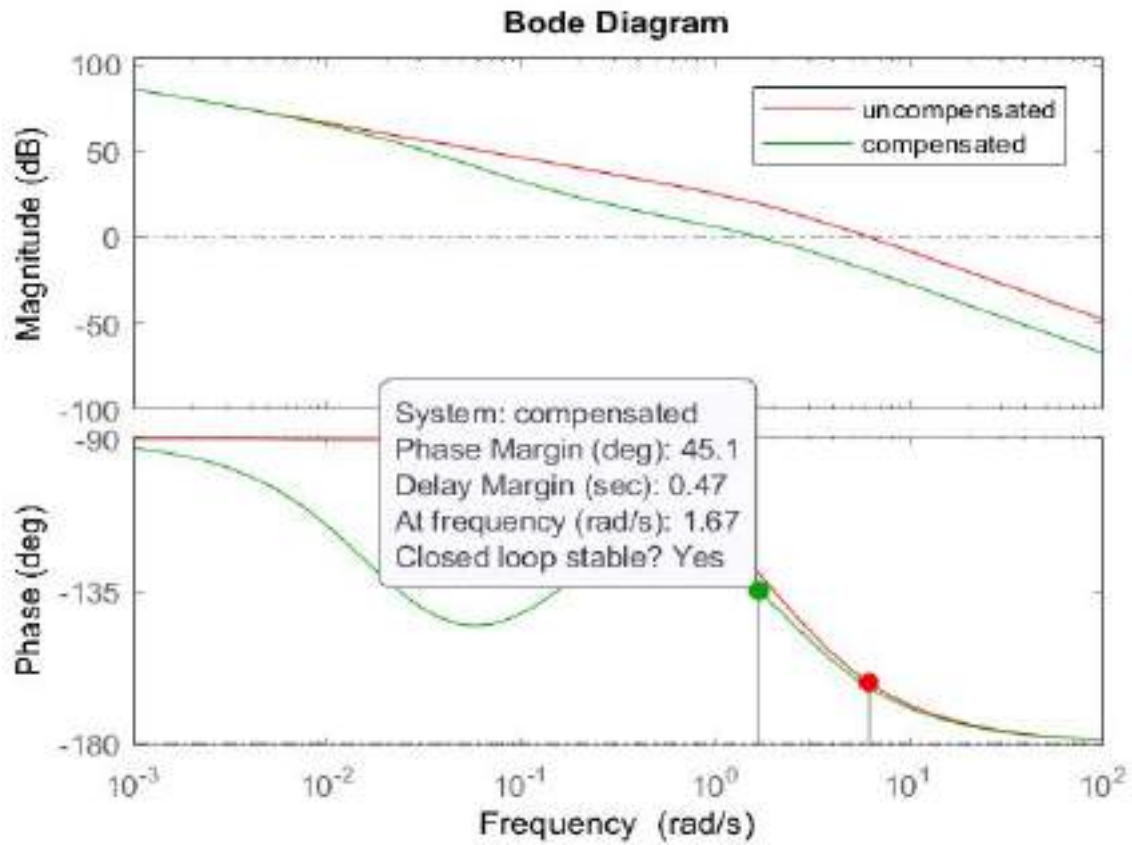
- $L(j\omega) = G_c(j\omega)G(j\omega) = \frac{20\left[\frac{j\omega}{0.166}+1\right]}{j\omega(0.5j\omega+1)\left[\frac{j\omega}{0.0180}+1\right]}$

- $L(s) = G_c(s)G(s) = \frac{20[6.0241s+1]}{s(0.5s+1)[55.5556\,s+1]}$

| | C.F | Slope | Net | freq. |
|---|---|---|---|---|
| $\frac{20}{j\omega}$ | none | -20 | -20 | 0.001 to 0.01 |
| $\frac{1}{\frac{j\omega}{0.01}+1}$ | 0.01 | -20 | -40 | 0.01 to 0.166 |
| $\frac{j\omega}{0.166}+1$ | 0.166 | 20 | -20 | 0.166 to 2 |
| $\frac{1}{0.5j\omega+1}$ | 2 | -20 | -40 | 2 onwards |

# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 1




$A = 20 \log \frac{20}{\omega}$

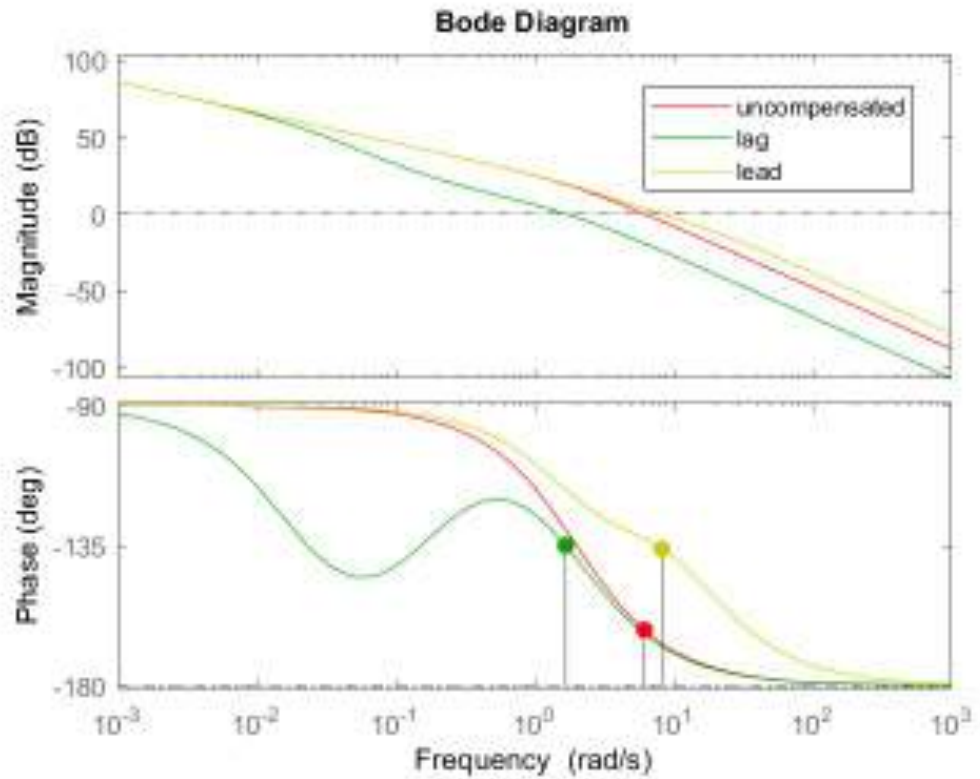
$= 20 \log 20 - 20 \log \omega$

at $\omega = 0.001$, $A = 86.02$ dB

$\angle L(j\omega) = \tan^{-1} \frac{\omega}{0.126} - 90 - \tan^{-1} 0.5\omega - \tan^{-1} \frac{\omega}{0.018}$

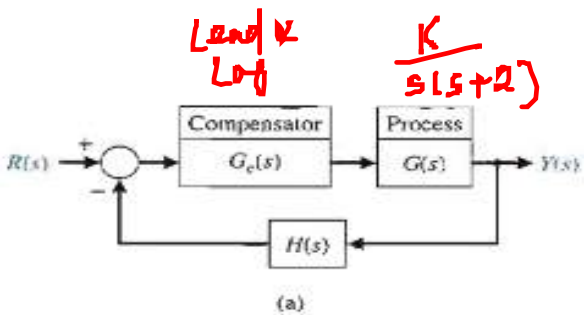
# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 1

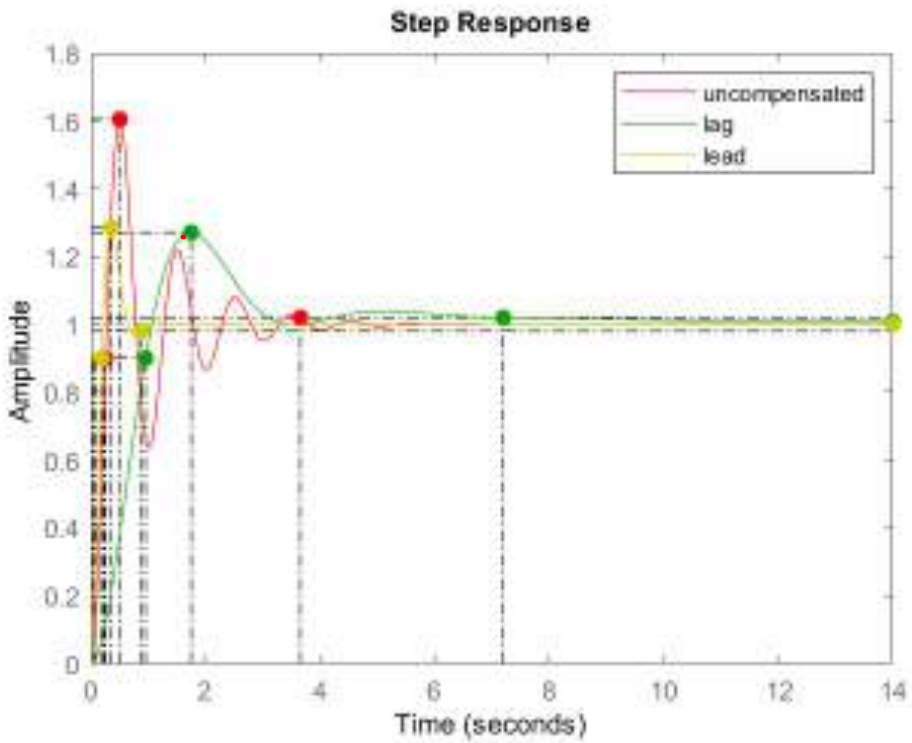


# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 1




$G_c(s) = K\frac{s+2}{s+p}$




CLTF of uncompensated

( T(s) = G(s)/(1+G(s)) and

compensated system

( T(s) = L(s)/(1+L(s))

where, L(s) = Gc(s)G(s)

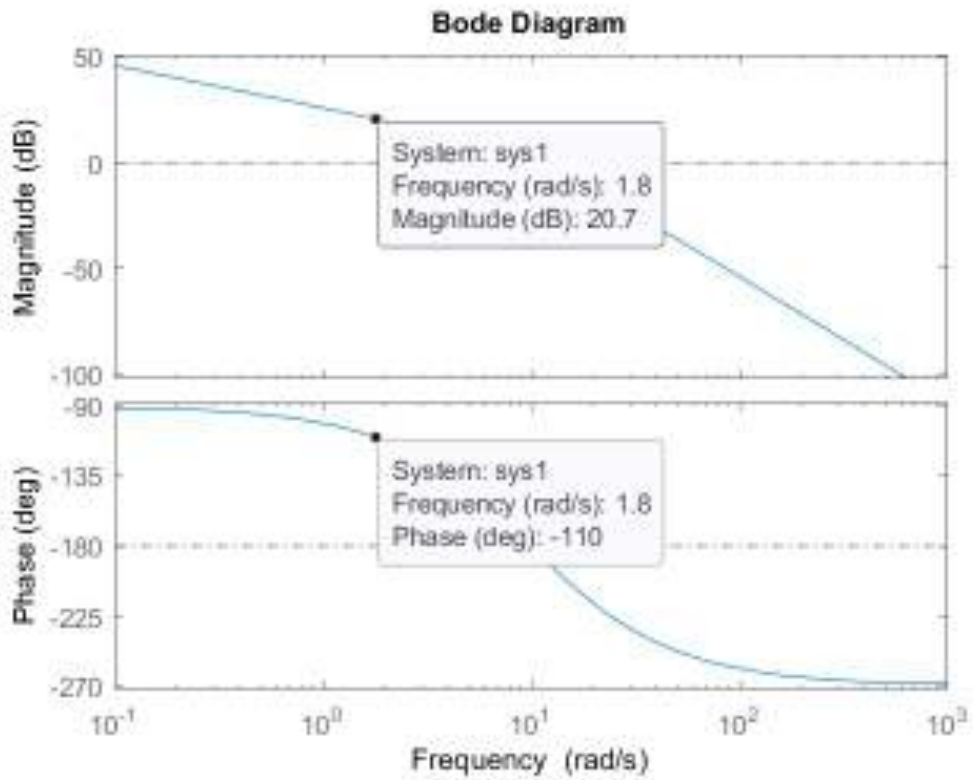
# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 2

- $G(s) = \frac{K}{s(s+10)^2}$ and H(s) = 1
- Desired specification
  - Velocity constant of $K_v = 20$
  - $\zeta = 0.707$
    - $\emptyset_{pm} = \frac{\zeta}{0.01} = 65°$(since 65 is max)
- $G(j\omega) = \frac{K}{j\omega(j\omega+10)^2} = \frac{K_v}{j\omega(0.1j\omega+1)^2}$
- $K_v = \frac{K}{100}$
- Draw Bode plot of $G(j\omega) = \frac{20}{j\omega(0.1j\omega+1)^2}$

# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 2



# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 2

- $\emptyset_1$ =desired PM + safety margin = 65° + 5° = 70°
- Determine the frequency at $-(180° - 70°) = -110°$ =>

  $\omega_c' = 1.8$
- $-20\log_{10} \propto = Magnitude$ at $\omega_c'$
- $20\log_{10} \propto = 20.7$
- => $\propto = 10.8393$

# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 2

- $z = \frac{\omega_c'}{10} = \frac{1.8}{10} = 0.18$
- $p = \frac{1}{\propto\tau} = \frac{z}{\propto} = \frac{0.18}{10.8393} = 0.0166$
- Compensated System
  - $L(j\omega) = G_c(j\omega)G(j\omega) = \frac{20[\frac{j\omega}{0.18}+1]}{j\omega(0.1j\omega+1)^2(\frac{j\omega}{0.0166}+1)}$
  - $L(s) = G_c(s)G(s) = \frac{20[5.5556s+1]}{s(0.1j\omega+1)^2(60.2410\,s+1]}$

$G_c(j\omega)$

from Matlab

num = [$\frac{20}{0.18}$ 20]

den = [ $d_2$ ]

bode ( num , den )

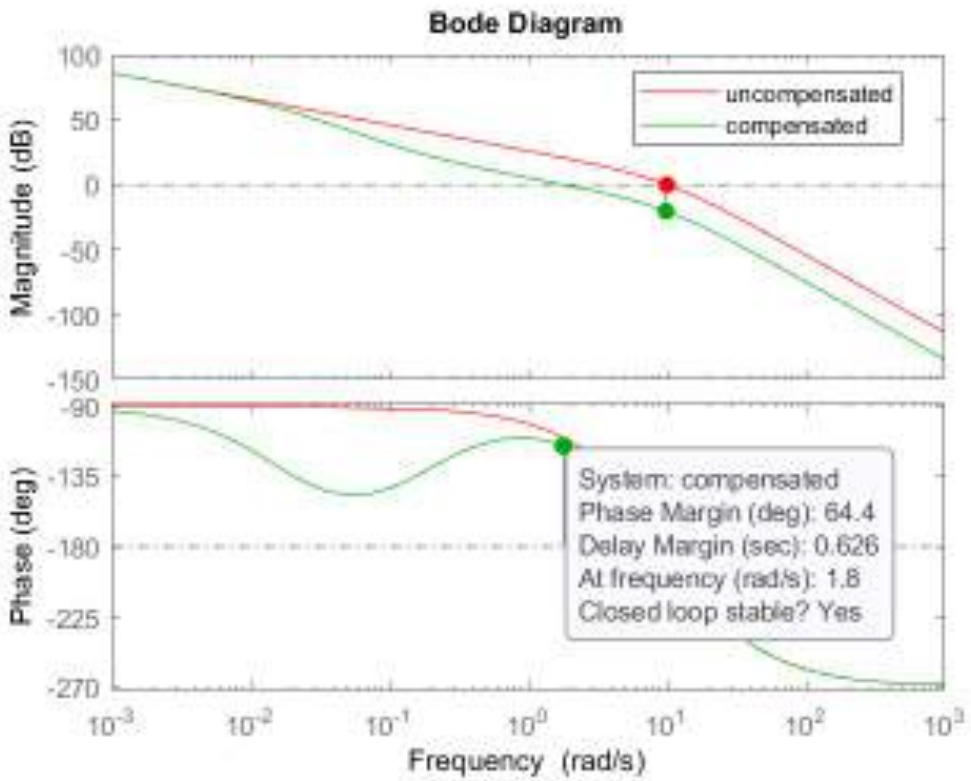
to multiply 2 poly

$d_1$ = conv [(1 0) ($0.1)^2$ 2x0.1 1)]

$d_2$ = [$d_1$ (16.966 1)]

# Design of Feedback Control Systems

## Phase – Lag design using Bode Plot – Example 2




Bode

# Design of Feedback Control Systems

## How to Find GM and PM manually?

- For the given $G(s) = \frac{10}{s(s+10)}$, determine PM.

$\Rightarrow \angle G(j\omega) = -90^\circ - \tan^{-1}\frac{\omega}{10}$

- Sol: calculate $\omega_{gc}$,

$|G(j\omega)|$ at $\omega = \omega_{gc}$ is equal to one

$$\left|\frac{1}{\omega\sqrt{(\frac{\omega^2}{100}+1)}}\right| = 1 \Rightarrow \omega_{gc} = 1 rad/sec$$

$$\emptyset = -90 - \tan^{-1}\frac{1}{10} = -95.71$$

PM = 180 + Ø

Ans: PM = 84.29deg

$1 = \omega\sqrt{(\frac{\omega^2}{100}+1)}$

$1 = \omega^2(\frac{\omega^2}{100}+1)$

$100 = \omega^4 + 100\omega^2$

$\omega^4 + 100\omega^2 - 100 = 0$

$\omega^2 =$ $\Rightarrow \omega$

# Design of Feedback Control Systems

## How to Find GM and PM manually?

- Given $G(s) = \frac{1}{s(1+2s)(1+s)}$, Determine GM.
- Sol:
  - Find $\omega_{pc}$ => $\angle G(j\omega)$ $at$ $\omega = \omega_{pc}$ is equal to -180
    - -180 = $-90 - \tan^{-1} 2\omega_{pc} - \tan^{-1} \omega_{pc}$
    - $\tan^{-1} \frac{2\omega_{pc}+\omega_{pc}}{1-2\omega_{pc}*\omega_{pc}} = 90 \Rightarrow \omega_{pc} = \sqrt{\frac{1}{2}}$ rad/sec
  - GM(dB) = -20log($|G(j\omega)|$ $at$ $\omega = \omega_{pc}$)
  - GM(dB) = $-20\log \left| \frac{1}{\omega_{pc}\sqrt{(4\omega_{pc}{}^2+1)}\sqrt{(\omega_{pc}{}^2+1)}} \right|$
  - Ans : 3.5dB

$$\frac{2\omega_{pc}+\omega_{pc}}{1-2\omega_{pc}^2} = \frac{1}{0}$$

$$1-2\omega_{pc}^2 = 0$$

$$\omega_{pc}^2 = \frac{1}{2}$$

$$\omega_{pc} = \sqrt{1/2}$$

# Design of Feedback Control Systems

## Ex 3 : Lag Compensators

- $G(s) = \frac{1}{s(s+1)(0.5s+1)}$ and H(s) = 1
- Desired specification
  - Phase margin = 40 degree
  - Velocity error constant, $K_v = 5/sec$
  - Gain margin is at least 10dB

# Design of Feedback Control Systems

## Ex 3 : Lag Compensators

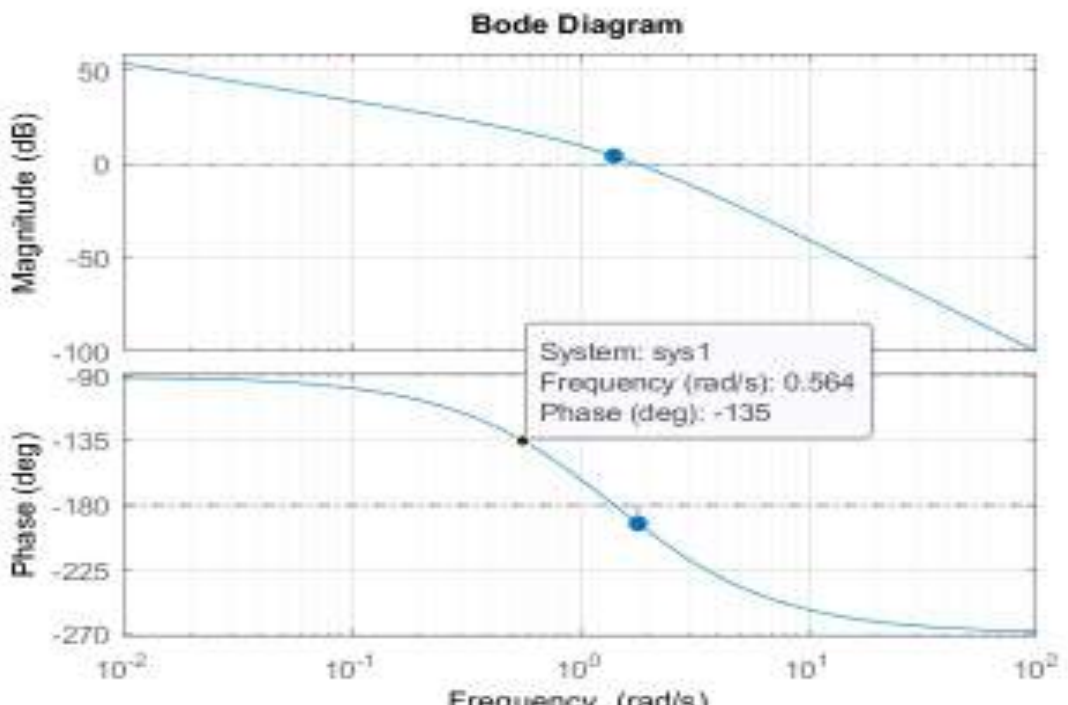
- If K is not given in G(s), include K in the numerator and find K such that velocity error constant, $K_v$ is satisfied.
  - $K_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} s\frac{K}{s(s+1)(0.5s+1)} = 5 \Rightarrow K = 5$
  - => $G(s) = \frac{5}{s(s+1)(0.5s+1)}$ => $G(j\omega) = \frac{5}{j\omega(j\omega+1)(0.5j\omega+1)}$
- Step 1: Draw Bode plot for
- $G(j\omega) = \frac{5}{j\omega(j\omega+1)(0.5j\omega+1)}$

- find K in terms of Kv
- sub. K in G(s)
- convert into time-constant form

# Design of Feedback Control Systems

## Ex 3 : Lag Compensators

- $\emptyset_1$ =desired PM + safety margin = $40° + 5° = 45°$
- Determine the frequency at $-(180° - 45°) = -135°$ => $\omega_c' = 0.562$



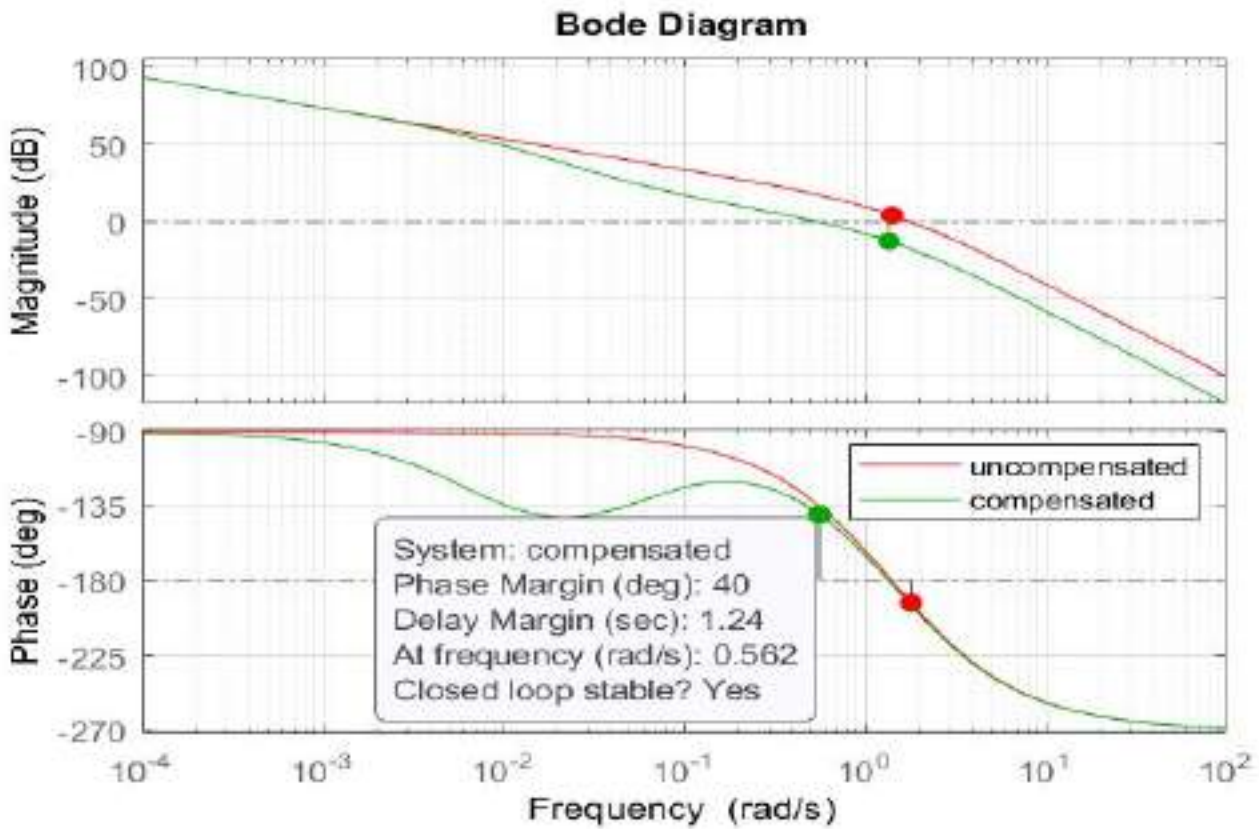
# Design of Feedback Control Systems

## Ex 3 : Lag Compensators

- $-20\log_{10}\propto = Magnitude$ at $\omega_c'$
- $20\log_{10}\propto = 17.5$
- => $\propto = 7.498$
- $z = \frac{\omega_c'}{10} = \frac{0.562}{10} = 0.0562$
- $\boldsymbol{p = \frac{1}{\propto \tau} = \frac{z}{\propto} = \frac{0.0562}{7.498} = 0.0075}$
- Compensated System
  - $L(j\omega) = G_c(j\omega)G(j\omega) = \frac{5[\frac{j\omega}{0.0562}+1]}{j\omega(0.5j\omega+1)[\frac{j\omega}{0.0075}+1](j\omega+1)}$
  - $L(s) = G_c(s)G(s) = \frac{5[17.7936s+1]}{s(0.5s+1)[133.4331s+1](1+s)}$

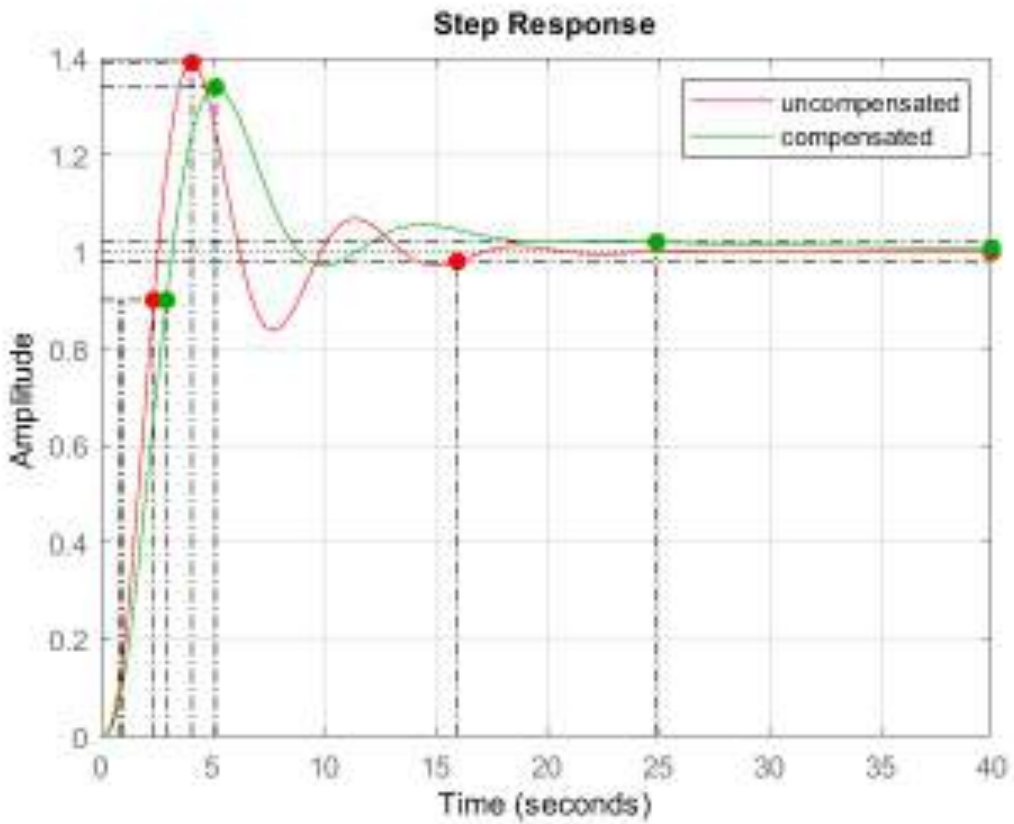
# Design of Feedback Control Systems

## Ex 3 : Lag Compensators



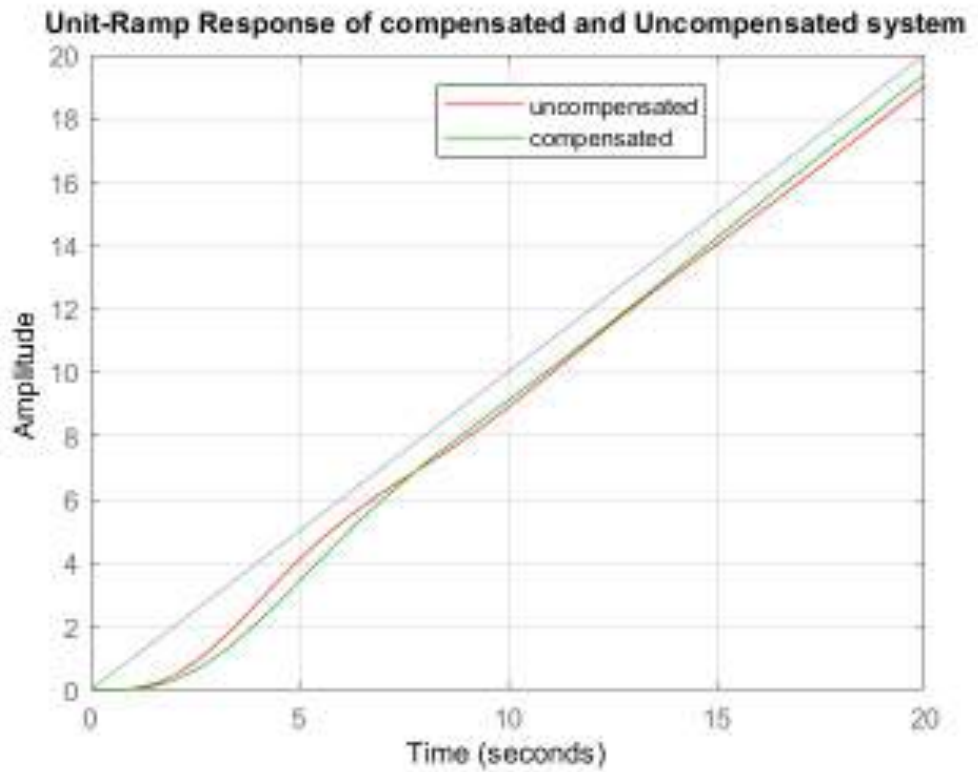
# Design of Feedback Control Systems

## Ex 3 : Lag Compensators



# Design of Feedback Control Systems

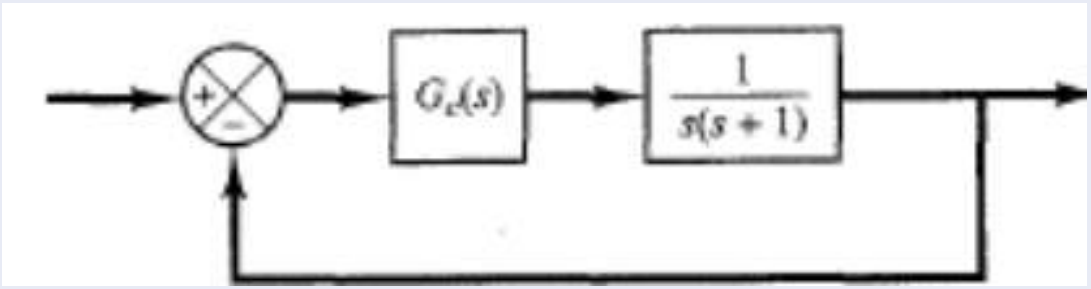
## Ex 3 : Lag Compensators



# Design of Feedback Control Systems

## Ex 4 : Lead Compensators

Consider the system shown in figure.




Design a compensator such that the static velocity error constant $K_v$ is 50/sec, phase margin is 50 degree, and gain margin not less than 8 dB.

# Design of Feedback Control Systems

## Ex 4 : Lead Compensators

- $K_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} s\frac{K}{s(s+1)} = 50 \Rightarrow \frac{K}{1} = 50 \Rightarrow K = 50$
- $\Rightarrow G(s) = \frac{50}{s(s+1)} \Rightarrow G(j\omega) = \frac{50}{j\omega(j\omega+1)}$
  - Draw Bode plot for $G(j\omega) = \frac{50}{j\omega(j\omega+1)} \Rightarrow \emptyset_{un} = 8$

$\frac{50}{j\omega}$ none slope $-20$ Net slope $-20$ $\omega = 0.1$ to $1$

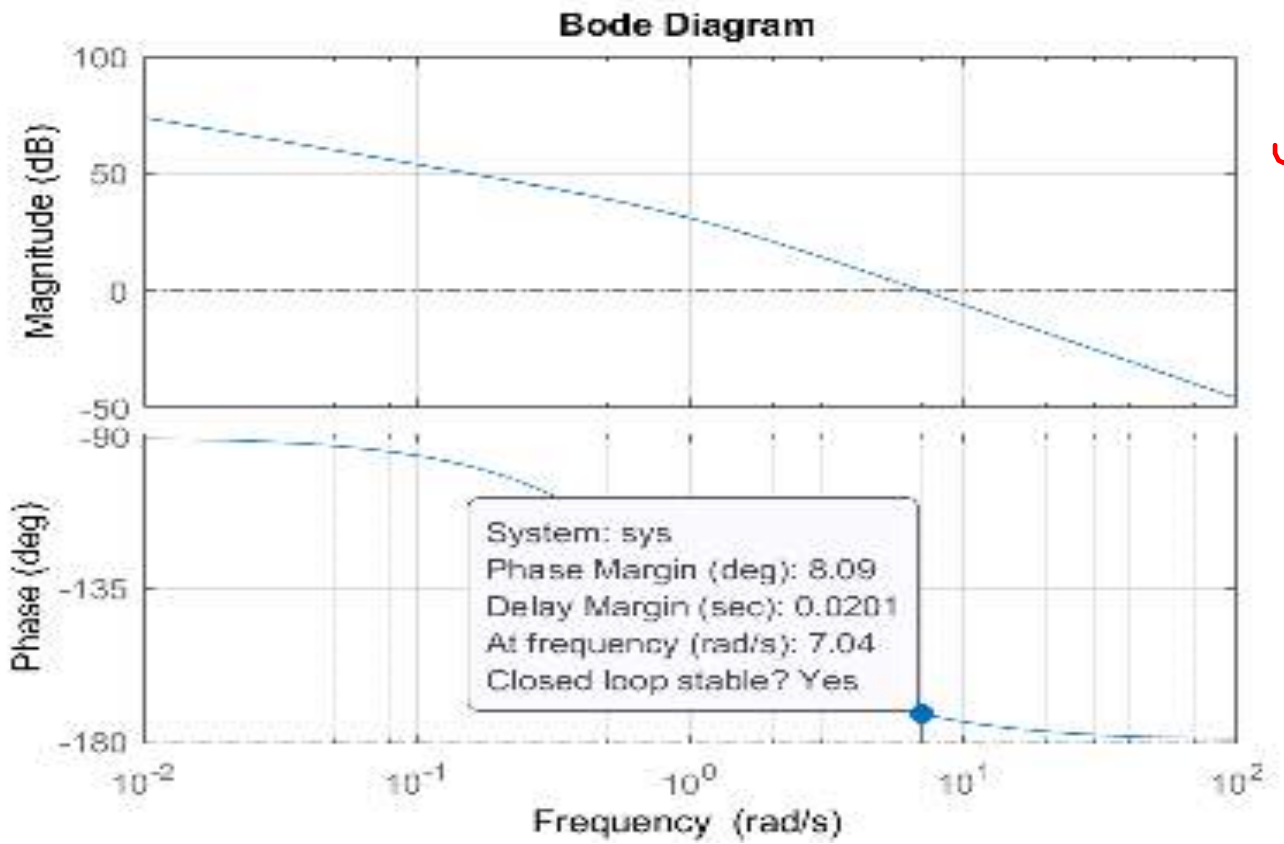
$\frac{1}{j\omega+1}$ $1$ $-20$ $-40$ $\omega = 1$ onwards

$A = 20\log\frac{50}{\omega}$

$= 20\log 50 - 20\log\omega$

# Design of Feedback Control Systems

## Ex 4 : Lead Compensators




$\phi = \angle G(j\omega) = -90 - \tan^{-1}\omega$

$\omega = 0.1 \quad 0.5 \quad 1 \quad 10$

$\phi =$

$|G(j\omega)| = \dfrac{50}{\omega=\omega_{gc}\ \omega\sqrt{\omega^2+1}} = 1$

$\omega_{gc}\sqrt{\omega_{gc}^2+1} = 50$

$\omega_{gc}^2(\omega_{gc}^2+1) = 2500$

$\omega_{gc}^4 + \omega_{gc}^2 - 2500 = 0$

# Design of Feedback Control Systems

## Ex 4 : Lead Compensators

- $\emptyset_1 = \boldsymbol{desired\ PM} - (\emptyset_{un})$
- $\emptyset_m = \emptyset_1 + \boldsymbol{safety\ margin}$
- **Desired PM ,** $\emptyset_{pm}$**= 50 and** $\emptyset_{un} = 8$
- $\emptyset_1 = 50\text{-}8 = 42$
- $\emptyset_m = 42 + 3\ (10\%\ of \emptyset_1 as\ safety\ margin) = 45$
- $\boldsymbol{sin}\ \emptyset_m = \frac{\alpha - 1}{\alpha + 1}$ or $\alpha = \frac{1+\sin(\emptyset_m)}{1-\sin(\emptyset_m)} = 5.828$

# Design of Feedback Control Systems

## Ex 4 : Lead Compensators

- Evaluate $10\log_{10}\propto$ = 7.66 dB
- frequency at which the uncompensated magnitude curve is equal to $-7.66$ dB is $\boldsymbol{\omega_m = 11\,rad/sec}$ (Because the compensation network provides a gain of $10\log_{10}\propto$ at $\boldsymbol{\omega_m}$ and this is the new 0dB cross over frequency)
- $\boldsymbol{p = \omega_m\sqrt{\propto} = 11\sqrt{5.828} = 26.556,}$
- $\boldsymbol{z = \frac{p}{\propto} = 4.556}$

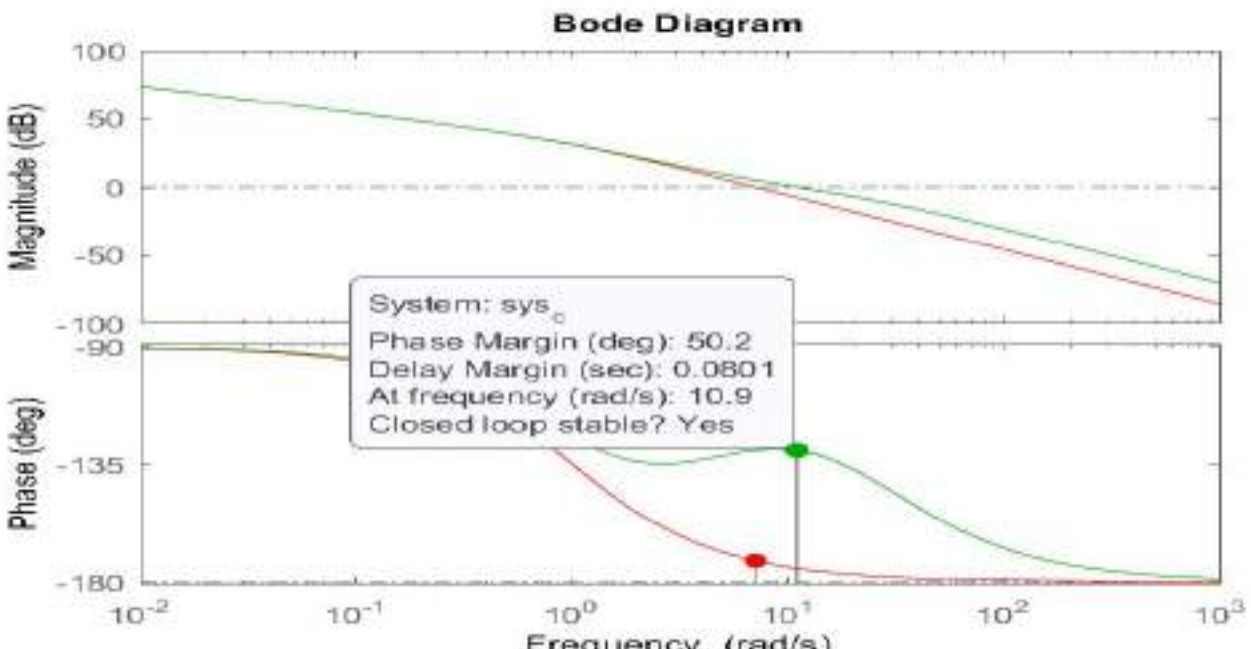
# Design of Feedback Control Systems

## Ex 4 : Lead Compensators

- Draw the bode plot for compensated system
- $L(j\omega) = G_c(j\omega)G(j\omega) = \dfrac{K_0[\frac{j\omega}{z}+1]}{j\omega(j\omega+1)\ (\frac{j\omega}{p}+1)}$

$z = 4.556$

$p = 26.556$



# Design of Feedback Control Systems

## Ex 5 : Lead Compensators

- Consider the system transfer function

$$G(s) = \frac{K}{s^2(0.2s + 1)}$$

$K_a = 10$

- Design a lead compensator such that the static acceleration error constant is greater than equal to 10/sec^2, and desired phase margin is 35 degree.

$$K_a = \lim_{s \to 0} s^2 G(s) = \lim_{s \to 0} s^2 \frac{K}{s^2(0.2s+1)} = K$$

$K = 10$

# Design of Feedback Control Systems

## Ex 5 : Lead Compensators

Draw Bode plot for $G(j\omega) = \dfrac{10}{(j\omega)^2(0.2j\omega+1)}$

| | | slope | net slope | |
|---|---|---|---|---|
| $\frac{10}{(j\omega)^2}$ | none | $-40$ | $-40$ | $\omega = 0.1$ to $\omega = 5$ |
| $\frac{1}{0.2j\omega+1}$ | $\frac{1}{0.2} = 5$ | $-20$ | $-60$ | $\omega = 5$ onwards |

$$A = 20 \log \frac{10}{\omega^2}$$

$$= 20 \log 10 - 20 \log_{10} \omega^2$$

$$\angle G(j\omega) = -180^\circ - \tan^{-1}(0.2\omega)$$

# Design of Feedback Control Systems

## Ex 5 : Lead Compensators

PM = $-33^\circ$ , desired PM = $35^\circ$

$\phi_1 = 35^\circ - (-33^\circ) = 68^\circ$

$\phi_m$ = 68° + Safety margin

= $68^\circ + 10^\circ = 78^\circ$ ( $\phi_m$ max $65^\circ$ )

Since $\phi_m > 65^\circ$ , $\frac{\phi_m}{2} = \frac{78}{2} = 39^\circ$

$$\alpha = \frac{1 + \sin 39^\circ}{1 - \sin 39^\circ} = 4.395$$

$$-10 \log_{10} \alpha = 10 \log_{10} (4.395) = -6.43 \text{ dB}$$

# Design of Feedback Control Systems

## Ex 5 : Lead Compensators

Since we are designing 2 compensators in cascade

we have the total gain $= -(6.43 + 6.43)$ dB

$= -12.8$ dB

freq. at $-12.8$ dB is $\omega_m = 5.44$ rad/sec

$P = \omega_m \sqrt{\alpha} = 11.4046$

$Z = \frac{P}{\alpha} = 2.5949$

# Design of Feedback Control Systems

## Ex 5 : Lead Compensators

$$L(s) = \frac{10}{s^2(0.2s+1)} \left(\frac{\frac{s}{2.5949}+1}{\frac{s}{11.4046}+1}\right)\left(\frac{\frac{s}{2.5949}+1}{\frac{s}{11.4046}+1}\right)$$

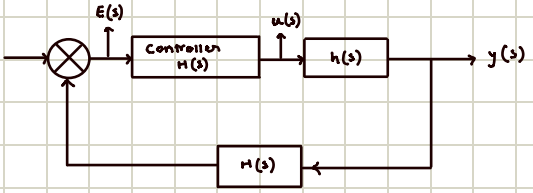
$$PM = 30.5$$

## Controller

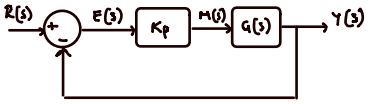
→ Take an open loop


Assuming system is unstable, we take -ve feedback to make system stable

We add a comparator & controller




### Types of controller




1) **Proportional Controller (P-controller)**

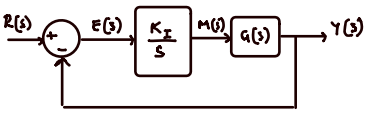
With proportional control, Actuator applies a corrective force proportional to amount of error

$u(t) = K_p e(t)$ $\quad K_p$ : proportional gain (real no. b/w $-\infty$ & $\infty$)

$U(s) = K_p E(s) \implies \frac{U(s)}{E(s)} = K_p$

Advantage : Simple to design, Reduces overshoot

Disadvantage : Results in non-zero steady state error




2) **Integral Controller (I- Controller)**

Adds a pole at origin for OLTF & TYPE of the system increases

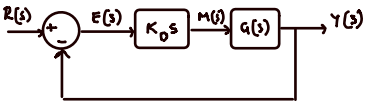
$u(t) = K_I \int_0^t e(t)\,dt$ $\quad K_I$ : Integral gain (real no. b/w $-\infty$ & $\infty$)

$u(s) = K_I \frac{E(s)}{s} \implies \frac{U(s)}{E(s)} = \frac{K_I}{s}$

Advantage : steady state error can be reduced to 0

Disadvantage : Introduces pole at origin

Actuator saturation




3) **Derivative Controller (D - Controller)**

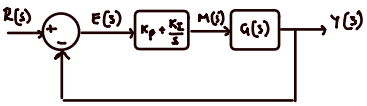
$u(t) = K_d \frac{de(t)}{dt}$

$U(s) = K_d s E(s) \implies \frac{u(s)}{E(s)} = K_d s$

Advantage : Anticipatory Controller

Disadvantage : Introduces a zero

| | Rise Time | Over-shoot | Settling Time | S-S Error |
|---|---|---|---|---|
| $K_p$ | ↓ | ↑ | Small Change | ↓ |
| $K_i$ | ↓ | ↑ | ↑ | ↓ |
| $K_d$ | Small change | ↓ | ↓ | - |




4) Proportional - Integral Controller (PI Controller)

$$u(t) = K_p e(t) + K_I \int_0^t e(t)\,dt$$

$$U(s) = K_p E(s) + K_I \frac{E(s)}{s} \Rightarrow \frac{U(s)}{E(s)} = K_p + \frac{K_I}{s} = \frac{K_p s + K_I}{s} = G_c(s) \Rightarrow \text{zero} = \frac{-K_I}{K_p}$$

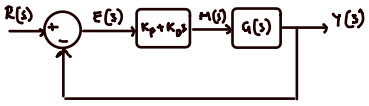
pole = origin

→ Increases rise time

→ Decreases bandwidth

→ Improves GM, PM & resonant peak

→ Improved damping, zero offset, no steady state error




5) Proportional - Derivative Controller (PD Controller)

Adds zero on real axis for OLTF

→ Root Locus gets pulled to the left, system is more stable & response faster

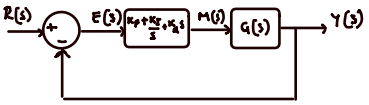
→ Differentiation makes system sensitive to noise

→ It is high pass filter

→ Reduces overshoot, rise time & settling time

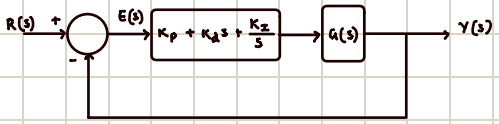
→ Increases Bandwidth, GM, PM & Resonant Peak

$$G_c(s) = K_p + K_d s \Rightarrow \text{zero} = \frac{-K_p}{K_d}$$




6) Proportional - Integral - Derivative Controller (PID Controller)

$$G_c(s) = K_p + \frac{K_I}{s} + K_d s \Rightarrow \begin{matrix}2 \text{ zeros} \\ 1 \text{ pole}\end{matrix}$$




→ It is band-pass or band-stop filter

→ $K_p$ decreases rise time

→ $K_i$ eliminates $e_{ss}$

→ $K_d$ decreases $M_p$ & $t_s$

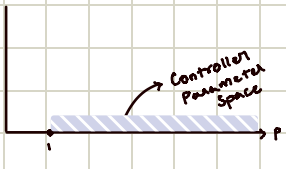
Q. $G(s) = \frac{1}{s+1}$

A. $H(s) = 1$

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{K_p}{s-1}}{1 + \frac{K_p}{s-1}} = \frac{K_p}{s - 1 + K_p} = \frac{K_p}{s + (K_p - 1)}$$

C.E $= s + K_p - 1 = 0$

If we use $K_p > 1$, we get poles on LHS of s-plane



## Controllability

→ A system is completely state controllable if it is possible to transfer the system state from an initial state $x(t_0)$ to any desired state $x(t)$ in specified finite time by control vector $u(t)$

→ Condition : No cancellation of poles & zeros should occur

→ Testing : Take $\dot{x} = Ax + Bu$ and form a composite matrix as follows,

$$Q_c = [B \quad AB \quad A^2B \ \dots \ A^{n-1}B] \Rightarrow \text{Controllability matrix}$$

$n$ : order of system

## Observability

→ A system is observable if every state $x(t)$ can be completely identified by measurements of output $y(t)$ over finite time interval

→ Condition : No cancellation of poles & zeros should occur

→ Testing : Take $\dot{x} = Ax + Bu$

$$y = Cx + Du$$

$$Q_o = [C \quad CA \quad CA^2 \dots C^{n-1}]$$

$n$ : order of system

**Q.** A single input single output system is given as

$$\dot{x}(t) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u$$

$$y(t) = \begin{bmatrix} 1 & 0 & 2 \end{bmatrix} x(t)$$

Test for controllability & observability

A. $n = 3$

$$B = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad AB = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix} \quad A^2B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}$$

$$\Rightarrow Q_c = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$|Q_c| = 0$, System is not controllable

$$C = \begin{bmatrix} 1 & 0 & 2 \end{bmatrix} \quad CA = \begin{bmatrix} 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & -6 \end{bmatrix} \quad CA^2 = \begin{bmatrix} -1 & 0 & -6 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 18 \end{bmatrix}$$

$$Q_o = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 0 & -6 \\ 1 & 0 & 18 \end{bmatrix} \Rightarrow |Q_o| = 0, \text{ System is not observable}$$